Master's Thesis

Exploratory modelling of collective effects in the FCC-ee High-Energy Booster using machine learning

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List of Acronyms

AE	autoencoder.
AI	artificial intelligence.
BO	Bayesian optimization.
CERN	European Organization for Nuclear Research.
CNN	convolutional neural network.
CPU	central processing unit.
CUDA	Compute Unified Device Architecture.
CVAE	conditional variational autoencoder.
DL	deep learning.
DOI	digital object identifier.
ELBO	evidence lower bound.
FCC	Future Circular Collider.
FCC-ee	Future Circular Collider electron–positron collider.
FFT	fast Fourier transform.
FNO	Fourier Neural Operator.
GANIL	Grand Accélérateur National d'Ions Lourds.
GELU	Gaussian error linear unit.
GNO	Graph Neural Operator.
GPU	graphics processing unit.
HEB	High-Energy Booster.
IPAC	International Particle Accelerator Conference.
JACoW	Joint Accelerator Conferences Website.
KL	Kullback–Leibler.
MAD-X	Methodical Accelerator Design, version X.
MHA	multi-head attention.
ML	machine learning.
MLP	multilayer perceptron.
MSE	mean squared error.
NEG	non-evaporable getter.
OOD	out of distribution.
PDE	partial differential equation.
RF	radio frequency.
RMS	root mean square.
SWD	sliced Wasserstein distance.
TF	Transformer.
UQ	uncertainty quantification.
VAE	variational autoencoder.
VFP	Vlasov–Fokker–Planck.

Glossary

accelerator lattice	Ordered sequence of magnets, radiofrequency systems, drifts, and other elements that determines charged-particle transport through an accelerator.
attention	Neural-network mechanism that forms weighted combinations of value representations using learned similarities between queries and keys.
autoencoder	Neural network trained to encode data into a lower-dimensional representation and decode that representation back to the input space.
beam centroid	First statistical moment of a beam-coordinate distribution; for coordinate ξ_i, it is the ensemble average $\langle \xi_i \rangle$.
beam coupling impedance	Frequency-domain representation of the electromagnetic response of the accelerator environment to the beam.
beam size	Transverse spatial extent of a beam, conventionally expressed through an RMS coordinate spread and related to emittance, optical beta function, dispersion, and momentum spread.
betatron motion	Oscillatory transverse motion of a particle around the reference orbit under the action of focusing elements.
bunch	Localized ensemble of charged particles transported together through an accelerator.
bunch charge	Total electric charge carried by one particle bunch.
causal attention	Attention mask that permits a token to receive information only from itself and preceding tokens in an ordered sequence.
collective effects	Beam-dynamics phenomena in which particles interact through fields generated by the beam itself or by its electromagnetic interaction with the accelerator environment.
conditional latent uncertainty	Variability obtained by repeatedly sampling the conditional latent distribution while holding model parameters and conditioning variables fixed.
conditional variational autoencoder	Variational autoencoder whose posterior and generation process are conditioned on auxiliary physical or machine variables.

convolution	Integral operation combining a function with a translated kernel; in beam dynamics it relates a longitudinal density to its induced wake potential.
coordinate-wise RMS spread	Square root of the second central moment of one beam coordinate; it remains defined for non-Gaussian distributions with finite variance but does not fully characterize their shape.
coupled-bunch instability	Coherent instability in which wake-mediated interactions couple the motion of different bunches circulating in an accelerator.
`cpymad`	Python interface to MAD-X used to load and manipulate accelerator lattice descriptions.
decoder	Model component that maps a latent representation back to the data or observable space.
digital twin	Computational representation of a physical system that combines models and data to support prediction, monitoring, calibration, or control.
discretization invariance	Property of an operator-learning architecture whose parameters do not depend on one fixed input or output discretization and whose discrete realizations converge under refinement.
dispersion	Dependence of the closed orbit or particle position on relative momentum deviation; it contributes to the measured transverse beam size.
emittance	Phase-space area or volume characterizing the spread of a beam in a conjugate coordinate– momentum plane.
empirical measure	Discrete probability measure formed by assigning equal point mass to each simulated macroparticle.
encoder	Model component that maps high-dimensional observations into features or a latent representation.
Fourier Neural Operator	Neural-operator architecture that parameterizes non-local integral transformations through learned multiplication of Fourier modes.
Fourier transform	Transformation between a function in physical space and its representation in frequency or wavenumber space.
graph message passing	Update mechanism in which graph nodes aggregate learned messages from neighbouring nodes connected by edges.

Graph Neural Operator	Neural-operator architecture that discretizes a learned kernel through graph-based message passing on arbitrary or physically structured nodes.
Haïssinski distribution	Stationary longitudinal bunch distribution obtained from the self-consistent balance of RF confinement, wakefields, synchrotron-radiation damping, and quantum excitation below the microwave-instability threshold.
inductive bias	Architectural assumption that favours particular solution structures, such as locality, causality, translation structure, or graph connectivity.
intrabeam scattering	Multiple small-angle Coulomb scattering among particles within the same bunch, which can increase beam emittances and energy spread.
kernel	Function defining weighted interactions or an integral operator between input and output locations.
Kullback–Leibler divergence	Asymmetric divergence measuring how one probability distribution differs from another; in a VAE it regularizes the approximate posterior toward the prior.
latent space	Learned lower-dimensional representation in which complex observations are encoded and manipulated.
Liouville's theorem	Statement that Hamiltonian phase-space flow preserves phase-space volume and transports densities incompressibly.
local-window attention	Attention mechanism restricted to a finite neighbourhood of each token, imposing local support on information transfer.
longitudinal line density	Particle or charge density projected onto the longitudinal coordinate.
macroparticle	Numerical simulation particle representing many physical particles while carrying effective charge and phase-space coordinates.
microwave instability	Single-bunch longitudinal instability associated with short-range wakefields and highfrequency impedance.
Monte Carlo estimator	Approximation of an expectation or integral by an average over random samples.
neural operator	Learned parametric mapping between function spaces rather than only between fixeddimensional vectors.

operator learning	Machine-learning framework for approximating mappings that transform input functions, fields, or distributions into output functions, fields, or distributions.
Panofsky–Wenzel theorem	Relation connecting longitudinal and transverse wakefields through derivatives of the integrated electromagnetic force.
phase space	Space of particle positions and conjugate momenta; the beam state in this thesis is represented in six dimensions.
phase-space marginal	Lower-dimensional distribution obtained by integrating the full phase-space density over unobserved coordinates.
posterior distribution	Probability distribution updated after conditioning on observed data or model inputs.
prior distribution	Probability distribution specified before conditioning on observations; the standard VAE prior is commonly a unit Gaussian.
pushforward measure	Distribution obtained by transporting a measure through a deterministic map.
PyColleff	Python module developed by LNLS for impedance modelling and collectiveinstability analysis in electron storage rings.
PyWIT	Python Wake and Impedance Toolbox used to represent and manipulate accelerator wake and impedance models.
quantile	Value below which a specified fraction of a probability distribution lies; useful for describing non-Gaussian beam tails and asymmetric distributions.
reparameterization trick	Differentiable sampling construction used in variational autoencoders, typically written as a deterministic transformation of Gaussian noise.
resistive-wall impedance	Beam coupling impedance produced by finite electrical conductivity of the vacuum chamber walls.
self-consistent force	Force that depends on the instantaneous beam distribution and therefore couples the evolution of individual particles to the collective state.
sliced Wasserstein distance	Distributional distance obtained by averaging one-dimensional Wasserstein distances over projected directions.

space charge	Collective electromagnetic interaction arising directly from the charge and current distribution of the beam.
surrogate model	Computationally inexpensive approximation of a more costly simulator, experiment, or physical model.
symplectic map	Phase-space transformation that preserves the symplectic structure of Hamiltonian dynamics and therefore respects the associated geometric invariants.
synchrotron radiation	Electromagnetic radiation emitted by relativistic charged particles undergoing transverse acceleration, producing damping and quantum excitation in electron rings.
Transformer	Sequence-model architecture based on attention layers and token-wise transformations.
Twiss parameters	Optical functions α, β, and γ used to describe linear transverse beam envelopes and phase-space ellipses.
uncertainty calibration	Agreement between predicted uncertainty intervals or probabilities and their empirical coverage or observed error frequencies.
under-dispersion	Condition in which the variability predicted by an ensemble or probabilistic model is systematically smaller than the observed prediction error.
variational autoencoder	Probabilistic autoencoder that learns an approximate latent posterior and is trained using reconstruction and prior-regularization terms.
Vlasov equation	Kinetic equation describing collisionless evolution of a phase-space density under external and self-consistent forces.
Vlasov–Fokker–Planck equation	Kinetic equation combining Vlasov transport with damping and diffusion terms.
wake function	Green-function response of the accelerator environment to a point-like source charge.
wake potential	Integrated electromagnetic effect experienced by a beam distribution, obtained by convolving the wake function with the source distribution.
Wasserstein distance	Optimal-transport distance between probability distributions that accounts for the cost of moving probability mass.

Xsuite	Open-source Python suite for multiparticle accelerator simulations, including tracking, optics manipulation, collective-effects interfaces, and accelerator-model workflows.
Xtrack	Xsuite component used to construct accelerator lines and track macroparticles through lattice elements.

List of Symbols

α	Under-relaxation coefficient in the longitudinal fixed-point iteration.
$\beta_{x,y}$	Horizontal or vertical Twiss beta function.
$\mathbf{c}$	Conditioning vector supplied to a conditional model.
$D_{x,y}$	Horizontal or vertical dispersion function.
δ	Relative momentum or energy deviation.
$\varepsilon_{x,y}$	Horizontal or vertical geometric emittance.
η	Phase-slip factor in longitudinal dynamics.
$\mathcal{G}$	Target evolution or solution operator acting on a function or beam distribution.
H_{ij}	Two-dimensional histogram or probability-mass representation of coordinates (ξ_i, ξ_j).
$\mathcal{L}$	Generic training objective or loss function.
λ	Normalized longitudinal line density.
m_i	Centroid of coordinate ξ_i.
μ	Vector of machine, lattice, material, or collective-effect conditioning parameters.
N_b	Number of histogram bins per coordinate.
N_p	Number of macroparticles per beam cloud.
N_s	Number of simulated input–output samples or trajectories.
N_ζ	Number of points in the longitudinal discretization grid.
Φ_μ	Single-particle or macroparticle-cloud transport map for parameters μ.
$p_\psi(X \mid z, c)$	Conditional decoder distribution parameterized by ψ.
p_x	Horizontal normalized canonical momentum.
p_y	Vertical normalized canonical momentum.
Q	Quality factor of a resonator model.
$q_\phi(z \mid X, c)$	Approximate conditional latent posterior parameterized by encoder parameters ϕ.
R	Number of stochastic posterior samples used in a Monte Carlo ensemble.

ρ	Beam probability density in six-dimensional phase space.
ρ^N	Empirical beam measure formed from a finite macroparticle cloud.
σ_i	Coordinate-wise RMS spread of coordinate ξ_i.
W_1	Wasserstein-1 distance.
$W_\parallel$	Longitudinal wake function.
$W_\perp$	Transverse wake function.
x	Horizontal transverse position.
ξ	Six-dimensional phase-space coordinate, typically $\xi = (x, p_x, y, p_y, \zeta, \delta)$.
y	Vertical transverse position.
ζ	Longitudinal position relative to the synchronous particle.
$\mathbf{z}$	Latent random variable or encoded latent state.
$Z_\parallel$	Longitudinal beam coupling impedance.
$Z_\perp$	Transverse beam coupling impedance.

Abstract Collective effects in modern particle accelerators generate nonlinear correlations in the six-dimensional beam phase space and require computationally expensive multiparticle simulations. This work investigates exploratory surrogate models for beam-distribution evolution in the High-Energy Booster of the Future Circular Collider (FCC-ee), with the objective of accelerating numerical studies while retaining physically relevant distributional information. Two complementary modelling strategies are developed. Reduced longitudinal dynamics are learned directly as mappings between density functions using Fourier Neural Operators. For the six-dimensional problem, beam states are represented by the fifteen pairwise two-dimensional phase-space marginals, supplemented by six centroids and six coordinate-wise RMS spreads. A conditional variational autoencoder compresses these projected distributions into a probabilistic latent representation, whose full-lattice input-to-output evolution is modelled using a causal Transformer, a local-window Transformer, or a Graph Neural Operator. Only the final downstream state is supervised and validated; the internal element-indexed latent states are not claimed to be physically validated intermediate beam states. The comparison of the latent propagators shows that restricting the support of information transfer provides a beneficial inductive bias for the investigated endpoint-transport task. Relative to global causal attention, local-window attention reduces distributional reconstruction errors and systematic artefacts. Among the selected checkpoints, the Graph Neural Operator has the lowest observed relative histogram error, decreasing it from 0.681 for the global Transformer to 0.511, while requiring approximately 2.34 ms per sample under the reported neural-model evaluation conditions, about eight times faster than the global Transformer. These values come from selected single-seed runs and are not accompanied by seed-to-seed uncertainty estimates. The conclusions remain restricted to the sampled simulation domain, the fixed full-lattice endpoint map, and the fifteen-marginal representation, which does not uniquely determine the complete six-dimensional joint distribution. Conditional latent sampling produces structured variability but is strongly under-dispersed relative to the observed prediction error and must not be interpreted as calibrated predictive coverage. Within these limitations, the study identifies structured causal support, explicit physical-statistics prediction, and graph-based latent propagation as promising components for future accelerator surrogate models. Preliminary results from this work were published in the JACoW proceedings of the International Particle Accelerator Conference (IPAC) [1].
Keywords *Machine learning, accelerator physics, FCC-ee, collective effects, beam dynamics, neural oper- ators, Fourier Neural Operator, conditional variational autoencoder, Graph Neural Operator, Transformer, surrogate modelling.*

1 Introduction

1.1 Motivation

Collective effects are among the main performance-limiting phenomena in modern circular accelerators. As the beam intensity increases, the beam can no longer be described as a collection of independent particles evolving only under externally applied electromagnetic fields. Instead, the beam interacts with its environment and with itself through wake fields, impedance, space charge, radiation effects, and other collective mechanisms. These interactions may lead to emittance growth, coherent tune shifts, bunch lengthening, coupled-bunch motion, transverse mode coupling, and single-bunch instabilities. The wakefield and impedance formalisms therefore provide the standard language for describing beam–environment interactions and for assessing the stability margins of a machine [2].

These questions are particularly relevant in the context of the Future Circular Collider for electrons and positrons (FCC-ee), a proposed high-luminosity machine designed to operate over a broad energy range and to deliver precision measurements of the Z, W, and Higgs bosons, as well as of the top quark. Within this complex, the High-Energy Booster (HEB) is a key element of the injector chain, responsible for accumulating and ramping beams to collider operating energies while preserving beam quality. Even if the booster bunch population is lower than that of the collider rings, collective effects at injection remain a significant concern, especially when resistive-wall effects, transverse coupled-bunch motion, wakefields, space charge, intrabeam scattering, and synchrotron-radiation effects interact within the same dynamical system [3].

A reliable study of these effects generally requires multi-particle tracking over many turns, together with detailed impedance and wakefield models. Such simulations are indispensable because collective instabilities arise from the coupling of many degrees of freedom and from the non-trivial interplay between machine optics, beam parameters, feedback systems, and material properties. However, this level of fidelity comes at substantial computational cost. Parameter scans over intensity, RF settings, vacuum chamber properties, optics choices, and impedance scaling are therefore difficult to perform exhaustively. This creates a natural motivation for surrogate modelling: one seeks a learned approximation that is much cheaper to evaluate than the original simulator while preserving the physical information required for exploratory studies, stability scans, and design-space reduction.

Machine learning has recently become an important tool in accelerator physics, with applications including diagnostics, optics correction, anomaly detection, online optimization, and surrogate modelling of numerical simulations [4, 5]. In the present work, the objective is not to replace high-fidelity tracking codes. Rather, the objective is to investigate whether machine-learning models can learn reduced representations of collective beam dynamics accurately enough to accelerate exploration and to reveal which architectural biases are appropriate for high-dimensional beam-distribution evolution.

Preliminary results from the associated FCC-ee HEB impedance and surrogate-modelling study were disseminated in the JACoW proceedings of IPAC 2026 [1]. The present the-

sis provides the complete methodological development, validation, and limitations of that exploratory contribution.

1.2 Problem statement

The central object of this report is the evolution of a beam distribution in six-dimensional phase space. Let

$$\xi = (x, p_x, y, p_y, \zeta, \delta) \in \mathbb{R}^6 \tag{1}$$

denote the transverse and longitudinal phase-space coordinates, and let $\rho_s(\xi)$ be the beam density at longitudinal position or turn index s. For fixed machine and collective-effect parameters μ, the ideal target is a nonlinear evolution operator

$$\mathcal{G}_{\Delta s,\mu} : \rho_s \mapsto \rho_{s+\Delta s}. \tag{2}$$

When collective effects are absent, this operator reduces to the pushforward of the distribution under a single-particle lattice map. When collective effects are present, the force depends on the distribution itself and the operator becomes nonlinear. Equation (2) defines the ideal operator family, but the empirical six-dimensional study carried out here is narrower. The available tracking data contain only an upstream cloud and one downstream cloud after the complete modelled lattice section. The validated target is therefore the fixed endpoint map

$$\mathcal{G}_{L,\mu} : \rho_{\text{in}} \mapsto \rho_{\text{out}}, \tag{3}$$

where L is the full propagation length represented in the dataset. No ground-truth snapshots are available at intermediate lattice positions, so accuracy for arbitrary Δs and physical validity of internal latent states are not established. Within this scope, the research question can be stated as follows:

> Can the evolution of high-dimensional beam distributions, generated by expensive tracking and collective-effects simulations, be approximated by learned surrogate operators that preserve the physically relevant structure of the beam dynamics?

The exploratory nature of the project is important. Several architectures were investigated as different inductive biases for approximating related endpoint operators. In particular, the work compares spectral operator learning for reduced longitudinal profiles with latent sequence modelling, local-window attention, and graph-kernel message passing for the six-dimensional input-to-output task. The main conclusion is limited to the evaluated configurations: structured local support is a useful architectural prior for the endpoint latent-propagation problem, not proof that the underlying beam dynamics are strictly local or that the learned internal rollout is a physically resolved trajectory.

2 Beam dynamics

2.1 Linear and Nonlinear beam dynamics

Considering the theoretical description of a charged particle through a series of lattice elements, in the absence of collective forces and strong nonlinearities, its coordinates ξ_j at the entrance of element j are mapped to the coordinates ξ_{j+1} at the exit of the element by

$$\xi_{j+1} = \Phi_{j,\mu_j}(\xi_j), \tag{4}$$

where μ_j contains the local element parameters. In the linear approximation, $\Phi_{j,\mu_j}(\xi) = M_j(\mu_j)\xi$ and a sequence of N elements gives

$$\xi_N = M_{N-1} \cdots M_1 M_0 \xi_0. \tag{5}$$

At the distribution level, the corresponding evolution is the pushforward

$$\rho_{j+1} = (M_j)_{\#}\rho_j, \tag{6}$$

defined weakly by the following the Liouville's theorem condition for all suitable test functions

$$\int \varphi(\xi)\, \mathrm{d}(M_j)_{\#}\rho_j(\xi) = \int \varphi(M_j\xi)\, \mathrm{d}\rho_j(\xi). \tag{7}$$

The dynamics of intense charged-particle beams in circular accelerators cannot generally be described by independent single-particle motion alone. The beam interacts with the surrounding vacuum chamber and accelerator components, generating electromagnetic fields that act back on trailing particles. The time-domain response is described by wake functions, while the frequency-domain response is described by the beam coupling impedance. Together, wakefields and impedance determine energy loss, tune shifts, emittance growth, bunch lengthening, and the onset of collective instabilities [2].

If $\lambda(z)$ denotes the normalized longitudinal line density of a bunch, the induced longitudinal wake potential can be written as the convolution

$$V_{\parallel}(z) = \int_{-\infty}^{\infty} W_{\parallel}(z - z')\lambda(z')\, \mathrm{d}z'. \tag{8}$$

Similarly, the transverse wake potential is

$$V_{\perp}(z) = \int_{-\infty}^{\infty} W_{\perp}(z - z')\lambda(z')\, \mathrm{d}z'. \tag{9}$$

These equations show why collective effects naturally suggest an operator-learning viewpoint: the bunch profile is mapped to an induced field through an integral kernel.

The wake formalism is naturally expressed as a Green-function description of the beamenvironment response. In the rigid-beam and impulse approximations, one focuses on the integrated Lorentz force experienced by a trailing test particle rather than on the instantaneous force during the traversal of the structure. The longitudinal and transverse wake

functions, denoted respectively by $W_\parallel$ and $W_\perp$, represent the response of the environment to a point-like source charge. In that sense, the wake function is the shock response, or Green kernel, of the beam pipe and its discontinuities. The wake function is associated with the integrated force, and the wake potential is obtained by convolving that Green function with the beam distribution [2].

It is often more convenient to work in the frequency domain because material properties and resonant structures are more naturally characterized spectrally. The longitudinal impedance is the Fourier transform of the longitudinal wake, and similarly for the transverse case. A common convention is

$$Z_\parallel(\omega) = \frac{1}{c}\int_{-\infty}^{\infty} W_\parallel(z) e^{i\omega z/c}\,\mathrm{d}z, \tag{10}$$

$$Z_\perp(\omega) = \frac{i}{c}\int_{-\infty}^{\infty} W_\perp(z) e^{i\omega z/c}\,\mathrm{d}z, \tag{11}$$

up to sign conventions. The inverse transforms recover the wake functions. Because the wake is real-valued, the impedance satisfies the Hermitian symmetry relation

$$Z^*(\omega) = Z(-\omega). \tag{12}$$

This is one of the basic structural properties emphasized in Métral's discussion on wakefields and impedance [2]. A related formulation, written explicitly as inverse Fourier transform relations for wake functions, is [6]:

$$W_\parallel(\tau_d) = \frac{L}{2\pi}\int_{-\infty}^{\infty} Z_\parallel(\omega) e^{-i\omega\tau_d}\,\mathrm{d}\omega, \tag{13}$$

$$W_\perp(\tau_d) = \frac{iL}{2\pi}\int_{-\infty}^{\infty} Z_\perp(\omega) e^{-i\omega\tau_d}\,\mathrm{d}\omega, \tag{14}$$

with L the effective interaction length and τ_d the longitudinal delay variable.

A crucial theoretical relation is the Panofsky-Wenzel theorem, connecting longitudinal and transverse wakefields. In one standard form,

$$\nabla_\perp W_\parallel(z, \mathbf{r}_\perp) = \frac{\partial}{\partial z} W_\perp(z, \mathbf{r}_\perp), \tag{15}$$

or equivalently, in terms of the delay variable τ_d,

$$\frac{1}{c}\frac{\partial}{\partial \tau_d} W_\perp = \nabla_\perp W_\parallel. \tag{16}$$

This theorem follows from Maxwell's equations under the rigid-beam and impulse approximations and does not require a specific boundary condition. It is therefore one of the most general structural constraints in wakefield theory. In practice, it is used to relate transverse and longitudinal impedances and to check the consistency of numerical or experimental impedance data [2]. Kim [6] gives the same relation explicitly as Eq. (3.11) and emphasizes that it is one of the bases of beam-instability analysis.

2.2 Longitudinal Beam Dynamics

Considering the simplified case for longitudinal beam dynamics, we write the system as a Hamiltonian system. With z the longitudinal position relative to the synchronous particle and δ the relative energy deviation, a simplified dynamical formulation is the following

$$z' = -\eta\delta, \qquad \delta' = K(z), \tag{17}$$

where η is the phase-slip factor and $K(z)$ is the longitudinal restoring force. In the conservative limit, the distribution $\psi(z, \delta, s)$ satisfies the Vlasov equation

$$\frac{\partial\psi}{\partial s} - \eta\delta\frac{\partial\psi}{\partial z} + K(z)\frac{\partial\psi}{\partial\delta} = 0, \tag{18}$$

or, equivalently,

$$\frac{\partial\psi}{\partial s} + \{\psi, H\} = 0. \tag{19}$$

This formulation describes explicitly Liouville's theorem and the incompressibility of phase-space flow. By following this approach, the Vlasov equation is the starting point for many analytical and numerical studies of collective effects, including linear stability analysis, mode decomposition, and perturbative treatments of wakefield-induced instabilities.

When considering a more complete description including synchrotron-radiation damping and quantum excitation, the resulting kinetic model has Vlasov–Fokker–Planck form,

$$\frac{\partial\psi}{\partial s} - \eta\delta\frac{\partial\psi}{\partial z} + K(z)\frac{\partial\psi}{\partial\delta} = \frac{2}{c\tau_z}\frac{\partial}{\partial\delta}\left(\delta\psi + \sigma_\delta^2\frac{\partial\psi}{\partial\delta}\right), \tag{20}$$

where τ_z is the longitudinal damping time and σ_δ the equilibrium energy spread. The righthand side introduces relaxation and diffusion, making stationary distributions possible. Below the microwave-instability threshold, the stationary solution of the Vlasov-Fokker-Planck equation is given by the Haïssinski equation. In the formulation given by Zhou et al.[7], the equilibrium line density $\lambda_h(z)$ satisfies

$$\frac{\mathrm{d}\lambda_h(z)}{\mathrm{d}z} + \left[\frac{z}{\sigma_{z0}^2} - \frac{1}{\eta\sigma_\delta^2}F_h(z)\right]\lambda_h(z) = 0, \tag{21}$$

where $F_h(z)$ is the longitudinal wake force generated self-consistently by the bunch itself. Its explicit solution can be written

$$\lambda_h(z) = A\exp\left[-\frac{z^2}{2\sigma_{z0}^2} - \frac{I}{\sigma_{z0}}\int_z^\infty W_k(z')\,\mathrm{d}z'\right], \tag{22}$$

with normalization constant A such that

$$\int_{-\infty}^{\infty}\lambda_h(z)\,\mathrm{d}z = 1, \tag{23}$$

and bunch wake potential

$$W_k(z) = \int_{-\infty}^{\infty}W_k(z - z')\lambda_h(z')\,\mathrm{d}z'. \tag{24}$$

With this formulation, it is clear that the natural RF Gaussian confinement is modified by the self-induced wake potential, and the stationary bunch profile is the Boltzmann-like density corresponding to that deformed effective potential.

2.3 Distribution-level Beam Dynamics

The full dimensionality study of the dynamics can be considered by studying macroparticles in six-dimensional phase space, derived from the canonical coordinates $(x, p_x, y, p_y, \zeta, \delta)$, where x and y are the transverse positions, p_x and p_y are the corresponding momenta, ζ is the longitudinal position relative to the synchronous particle, and δ is the relative energy deviation. The state vector is

$$\xi = (x, p_x, y, p_y, \zeta, \delta) \in \mathbb{R}^6. \tag{25}$$

2.3.1 Statistical descriptors and beam-size convention

For a cloud of N_p macroparticles, the coordinate-wise centroid and RMS spread used throughout this work are the empirical first and centred second moments,

$$m_i = \frac{1}{N_p}\sum_{n=1}^{N_p}\xi_{n,i}, \qquad \sigma_i = \left[\frac{1}{N_p}\sum_{n=1}^{N_p}(\xi_{n,i} - m_i)^2\right]^{1/2}. \tag{26}$$

The symbol σ_i therefore denotes the RMS spread of the statistical distribution of coordinate ξ_i. This definition requires a finite second moment but does not assume a Gaussian distribution. For non-Gaussian beams, the centroid and RMS spread do not fully characterize tails, asymmetry, multimodality, or higher-order correlations. Those features are retained partially by the pairwise histograms introduced in Chapter 3, quantiles are not supplied as additional model inputs in the present study.

The term beam size is used more narrowly for a transverse position coordinate. In uncoupled linear optics, and neglecting correlations between betatron motion and relative momentum deviation, the projected horizontal RMS beam size satisfies

$$\sigma_x^2 = \beta_x \varepsilon_x + D_x^2 \sigma_\delta^2, \tag{27}$$

where ε_x is the geometric projected emittance, β_x is the Twiss beta function, D_x is the horizontal dispersion, and σ_δ is the RMS relative momentum spread. If the betatron coordinate x_β is correlated with δ, the more general decomposition is

$$\sigma_x^2 = \beta_x \varepsilon_x + 2D_x \operatorname{Cov}(x_\beta, \delta) + D_x^2 \sigma_\delta^2. \tag{28}$$

An analogous expression applies in the vertical plane. By contrast, $\sigma_{p_x}, \sigma_{p_y}, \sigma_\zeta$, and σ_δ are referred to as coordinate-wise RMS spreads, not as transverse beam sizes. This terminology is used consistently in the data description, model architecture, and results.

For a lattice section with parameters μ, the microscopic dynamics are described by a map

$$\Phi_\mu : \mathbb{R}^6 \to \mathbb{R}^6, \qquad \xi_{\text{out}} = \Phi_\mu(\xi_{\text{in}}). \tag{29}$$

At the distribution level, the corresponding density evolution is the pushforward

$$\mathcal{T}_\mu[\rho] = (\Phi_\mu)_\# \rho, \tag{30}$$

characterized weakly by

$$\int \varphi(\xi)\, \mathrm{d}\mathcal{T}_\mu[\rho](\xi) = \int \varphi(\Phi_\mu(\xi))\, \mathrm{d}\rho(\xi) \tag{31}$$

for all suitable test functions φ. When collective effects are included, the map becomes density dependent. A generic continuum description is

$$\partial_s \rho + \nabla_\xi \cdot (\mathbf{F}_{\text{ext}}(\xi, s, \mu)\rho) + \nabla_\xi \cdot (\mathbf{F}_{\text{coll}}[\rho](\xi, s, \mu)\rho) = \mathcal{D}[\rho], \tag{32}$$

where F_{ext} is the external lattice force, $F_{\text{coll}}[\rho]$ is the self-consistent collective force, and $\mathcal{D}$ denotes damping and diffusion.

A useful conceptual splitting for one lattice section is

$$\rho_{j+1} \approx \exp(\Delta s\, \mathcal{D}_j) \exp(\Delta s\, \mathcal{C}_j[\rho_j])(M_{j,\mu_j})_\# \rho_j, \tag{33}$$

where M_{j,μ_j} denotes external transport and $\mathcal{C}_j$ the collective generator. By considering this splitting, it is possible to consider the nonlinear operator as the main object of interest for surrogates and to study the collective effects in isolation from the external transport. This is particularly useful for the longitudinal dynamics, where the collective effects are often the dominant source of nonlinearity.

The theoretical relations introduced in this chapter enter the experiments at different levels. The wake convolution and the Haïssinski fixed-point structure motivate the PyColleff longitudinal benchmark and the FNO density-to-density map. The Panofsky–Wenzel theorem constrains the consistency of longitudinal and transverse wake models, but it is not imposed as a loss or architectural identity in the six-dimensional latent surrogate. Likewise, the Haïssinski profile is not used as a checkpoint constraint for the full-lattice Transformer or GNO rollout. The present six-dimensional validation is restricted to the final tracked cloud; a multiturn equilibrium test or an intermediate-position consistency test would require additional simulator outputs that are not contained in the reported dataset.

3 Simulation workflow

Two independently generated simulation datasets are used and are not pooled into a single benchmark. The longitudinal dataset provides one-dimensional density-to-density examples for the Fourier Neural Operator (FNO). The six-dimensional dataset provides paired input–output particle clouds for the CVAE and latent-propagation models. Their numerical errors are task-specific and must not be compared directly.

3.1 Six-dimensional Xsuite tracking dataset

The six-dimensional dataset was generated with Xsuite version 0.48.4, the latest public release available on 1 April 2026 [8, 9]. Xsuite is an open-source Python suite for multiparticle accelerator simulations. In the present workflow, its Xtrack component performs macroparticle transport after the MAD-X lattice description has been imported through cpymad. The complete generation workflow is available in the public `ML4CollEffects` repository [10], serving as the six-dimensional data-generation entry point.

For each sample s, a parameter vector $\mu^{(s)}$ and a sampled input beam family define the macroparticle cloud

$$Z_{\text{in}}^{(s)} = \left\{\xi_n^{(s)}\right\}_{n=1}^{N_p}, \tag{34}$$

which is tracked through the selected lattice realization to obtain

$$Z_{\text{out}}^{(s)} = \Phi_{\mu^{(s)}}\left(Z_{\text{in}}^{(s)}\right). \tag{35}$$

The resulting supervised dataset is

$$\mathcal{D}_{\text{track}} = \left\{\left(Z_{\text{in}}^{(s)}, \mu^{(s)}, Z_{\text{out}}^{(s)}\right)\right\}_{s=1}^{2048}. \tag{36}$$

The working archive used for the reported models contains 2048 paired trajectories, with $N_p = 4096$ retained macroparticles in each input and output cloud. The fixed data-partition indices are stored in the arrays train, val, and test. The split contains 1640 training samples, 204 validation samples, and 204 test samples. The reference generator used a larger macroparticle population before the stored clouds were reduced to the fixed array size used by the machine-learning pipeline. The coordinate order used in the archive is

$$(x, y, \zeta, p_x, p_y, \delta), \tag{37}$$

which differs from the canonical plane-grouped notation $(x, p_x, y, p_y, \zeta, \delta)$ used in parts of Chapter 2. Array indices and pairwise-histogram channels follow the ordering in Eq. (37); this convention is stated explicitly to avoid ambiguity in the physical interpretation of the stored channels.

The optics input corresponds to the FCC-ee HEB lattice release, obtained from the public FCC-ee HEB optics archive [11]. The generation script imports the MAD-X driver, selects the FCC-ee HEB sequence, and converts the selected lattice section into an Xtrack line. The FCC-ee HEB wake and impedance inputs were obtained from the public FCC-ee Booster PyWIT model repository [12]. Since both the optics archive and the impedance model remain subject to updates, the archival dataset release will preserve the exact input files used in this work, together with their cryptographic checksums.

For component μ_k, the nominal in-distribution range is defined from the training subset only,

$$I_k = \left[\min_{s\in\mathcal{D}_{\text{train}}} \mu_k^{(s)}, \max_{s\in\mathcal{D}_{\text{train}}} \mu_k^{(s)}\right]. \tag{38}$$

No validity claim is made outside $\mathcal{J}^{Q}_{kk}$. Following the element-conditioned latent-dynamics formulation of [13], each lattice element is described by the seven-dimensional parameter vector

$$\mu_i = (L_i, K_{1,i}, K_{2,i}, \phi_i, V_{\mathrm{rf},i}, f_{\mathrm{rf},i}, \varphi_{\mathrm{rf},i}) \in \mathbb{R}^7, \tag{39}$$

where L_i is the element length, $K_{1,i}$ and $K_{2,i}$ are the quadrupole and sextupole strengths, ϕ_i is the dipole bending angle, and $V_{\mathrm{rf},i}$, $f_{\mathrm{rf},i}$, and $\varphi_{\mathrm{rf},i}$ are the RF-cavity voltage, frequency, and phase. For non-RF elements, $V_{\mathrm{rf},i} = f_{\mathrm{rf},i} = 0$.

Before embedding, each component is normalized according to its numerical scale. The RF voltage and frequency are compressed using the transformation $x \mapsto \log(1 + x)$, which reduces their large dynamic range while preserving the zero value assigned to non-RF elements. The remaining parameters are normalized as

$$\begin{gathered} \widetilde{L}_i = \frac{L_i}{1.0\,\mathrm{m}}, \quad \widetilde{K}_{1,i} = \frac{K_{1,i}}{10.0\,\mathrm{m}^{-2}}, \quad \widetilde{K}_{2,i} = \frac{K_{2,i}}{10.0\,\mathrm{m}^{-3}}, \\ \widetilde{\phi}_i = \frac{\phi_i}{2\pi}, \qquad \widetilde{\varphi}_{\mathrm{rf},i} = \frac{\varphi_{\mathrm{rf},i}}{2\pi}. \end{gathered} \tag{40}$$

3.2 PyColleff longitudinal dataset

The longitudinal benchmark was generated using the `pycolleff` module distributed in the LNLS collective effects repository [14]. PyColleff is a Python package for impedance modelling and instability analysis in electron storage rings. In this work, it is used as a controlled operator-learning benchmark rather than as a validated end-to-end model of the FCC-ee HEB. The implemented configuration uses a 3, GeV ring model, a uniform longitudinal grid with $N_\zeta = 2048$, a resonator contribution with quality factor $Q = 104$, three fixed-point relaxation steps, and an under-relaxation coefficient $\alpha = 0.7$. The sampled conditioning variables are the bunch current, chamber radius, and non-evaporable-getter coating thickness. Their uniform ranges are $I_b \in [200, 350]$ A, $R \in [5, 12]$ mm, and $t_{\mathrm{NEG}} \in [1, 3]$ μm. These variables modify the longitudinal response and the resistive-wall and material contributions represented by the benchmark configuration.

$$\mu^{(s)} = \left(I_b^{(s)}, R^{(s)}, t_{\mathrm{NEG}}^{(s)}\right). \tag{41}$$

The induced voltage is evaluated spectrally as

$$V_{\mathrm{ind}}(z) = -\mathcal{F}^{-1}\left[Z_{\parallel}(\omega;\mu)\hat{\lambda}(\omega)\right](z), \tag{42}$$

and the total voltage is

$$V_{\mathrm{tot}}(z) = V_{\mathrm{RF}}(z) + V_{\mathrm{ind}}[\lambda](z). \tag{43}$$

The truncated fixed-point update is

$$\lambda^{(k+1)} = (1-\alpha)\lambda^{(k)} + \alpha\mathcal{T}_{\mu}\left(\lambda^{(k)}\right), \tag{44}$$

with $K = 3$ iterations and $\lambda_{\text{out}} = \lambda^{(K)}$. The resulting data are

$$\mathcal{D}_\lambda = \left\{ \left(\lambda_{\text{in}}^{(s)}, \mu^{(s)}, \lambda_{\text{out}}^{(s)} \right) \right\}_{s=1}^{2048}. \tag{45}$$

The ranges and units of every component of μ are therefore explicit. Reproducibility additionally requires the RF settings, impedance components, material model, boundary conventions, and PyColleff version; these items are preserved with the generation configuration so that the FNO result is not mistaken for a surrogate of a different impedance model.

3.3 Data representation and physical quantities

3.3.1 Empirical measures and reduced profiles

A cloud defines the empirical probability measure

$$\rho^{N_p} = \frac{1}{N_p} \sum_{n=1}^{N_p} \delta_{\xi_n}. \tag{46}$$

For the longitudinal task, this measure is projected onto the uniform grid $\{\zeta_j\}_{j=1}^{\zeta}$ to form

$$\lambda^h = (\lambda(\zeta_1), \dots, \lambda(\zeta_{N_\zeta})) \in \mathbb{R}^{N_\zeta}. \tag{47}$$

The profile is normalized to unit integrated mass before being supplied to the FNO.

3.3.2 Fifteen pairwise phase-space projections

A direct 646 discretization of the full six-dimensional density is infeasible. The tracking state is therefore represented by all fifteen pairwise two-dimensional histograms,

$$\mathcal{P} = \{(i, j) : 1 \le i < j \le 6\}, \qquad H(\rho) = \{H_{ij}\}_{(i,j)\in\mathcal{P}} \in \mathbb{R}^{15\times 64\times 64}. \tag{48}$$

For sample s, coordinate i is histogrammed over the sample-adaptive interval

$$\left[m_i^{(s)} - 5\sigma_i^{(s)},\, m_i^{(s)} + 5\sigma_i^{(s)} \right], \tag{49}$$

where m_i and σ_i are the empirical centroid and coordinate-wise RMS spread defined in Eq. (26). Every pairwise channel is converted to a probability-mass array,

$$\sum_{a=1}^{64} \sum_{b=1}^{64} H_{ij}^{(s)}[a, b] = 1. \tag{50}$$

Thus, the CVAE histogram branch models normalized projected shape. The six centroids and six RMS spreads are retained as a separate 12-component physical vector and provide the location and coordinate scale omitted by independent channel normalization. When a physical probability density is required, the normalized mass is divided by the physical bin area,

$$\rho_{ij}^{(s)}[a,b] = \frac{H_{ij}^{(s)}[a,b]}{\Delta\xi_{i,a}^{(s)}\Delta\xi_{j,b}^{(s)}}. \tag{51}$$

The word *beam size* is therefore not used for all twelve auxiliary quantities. The transverse position spreads σ_x and σ_y may be related to conventional optical beam sizes under the assumptions stated in Section 2.3.1; the remaining entries are coordinate-wise RMS spreads of p_x, p_y, ζ, and δ. The RMS definition remains meaningful for non-Gaussian distributions but is not a complete shape descriptor. Non-Gaussian information is carried by the histograms, while explicit quantiles are left to future work.

Pairwise marginals do not uniquely determine a six-dimensional joint density and predicted channels need not be mutually consistent. For coordinate i, let $h_i^{(ij)}$ be the one-dimensional marginal obtained by summing H_{ij} over coordinate j. A shared-marginal consistency diagnostic is

$$E_{\text{cons}} = \frac{1}{6}\sum_{i=1}^{6}\frac{1}{|\mathcal{C}_i|}\sum_{(ij),(ik)\in\mathcal{C}_i}\left\|h_i^{(ij)} - h_i^{(ik)}\right\|_1, \tag{52}$$

where $\mathcal{C}_i$ contains histogram channels sharing coordinate i.

The information loss can be stated more formally through an ANOVA/Hoeffding-type decomposition of a joint density,

$$\begin{aligned}\rho(\xi_1,\dots,\xi_6) = \rho_0 &+ \sum_i f_i(\xi_i) + \sum_{i<j} f_{ij}(\xi_i,\xi_j) \\ &+ \sum_{i<j<k} f_{ijk}(\xi_i,\xi_j,\xi_k) + \cdots + f_{1\cdots6}(\xi_1,\dots,\xi_6).\end{aligned} \tag{53}$$

The fifteen pairwise marginals constrain one- and two-coordinate information but do not identify the three- through six-coordinate interaction terms. For a multivariate Gaussian distribution, the mean vector and covariance matrix determine the full joint density, so pairwise information is sufficient in that special case. The collective, halo-bearing, and strongly distorted distributions of interest here are not assumed to be Gaussian; omitted higher-order correlations may therefore be dynamically relevant. The reported CVAE reconstruction error measures fidelity to the fifteen-channel representation, not error with respect to the unknown complete six-dimensional density.

A direct estimate of this representation error was not retained in the present experiments. Such a test would require a model that can evaluate or sample a six-dimensional joint density and then compare observables that depend on three or more coordinates. Point-cloud generative models and maximum-entropy normalizing-flow tomography provide natural alternatives because they represent a joint six-dimensional distribution without a 646 grid [15]. A controlled comparison against one of these alternatives, including halo-rich and strongly distorted cases, is therefore a priority for future work rather than a result claimed here.

3.4 Computational cost

Table 4 reports exploratory generation timings recorded for the reference pipeline. They depend on hardware, parallelization, numerical precision, cache state, and software versions. The measurements were not obtained under one final hardware-normalized protocol and are therefore not used as a controlled simulator-versus-network benchmark; they document only the order of magnitude of data generation. The tracking timing corresponds to the reference high-particle-count generator, whereas the stored machine-learning archive contains 4096 retained macroparticles per cloud.

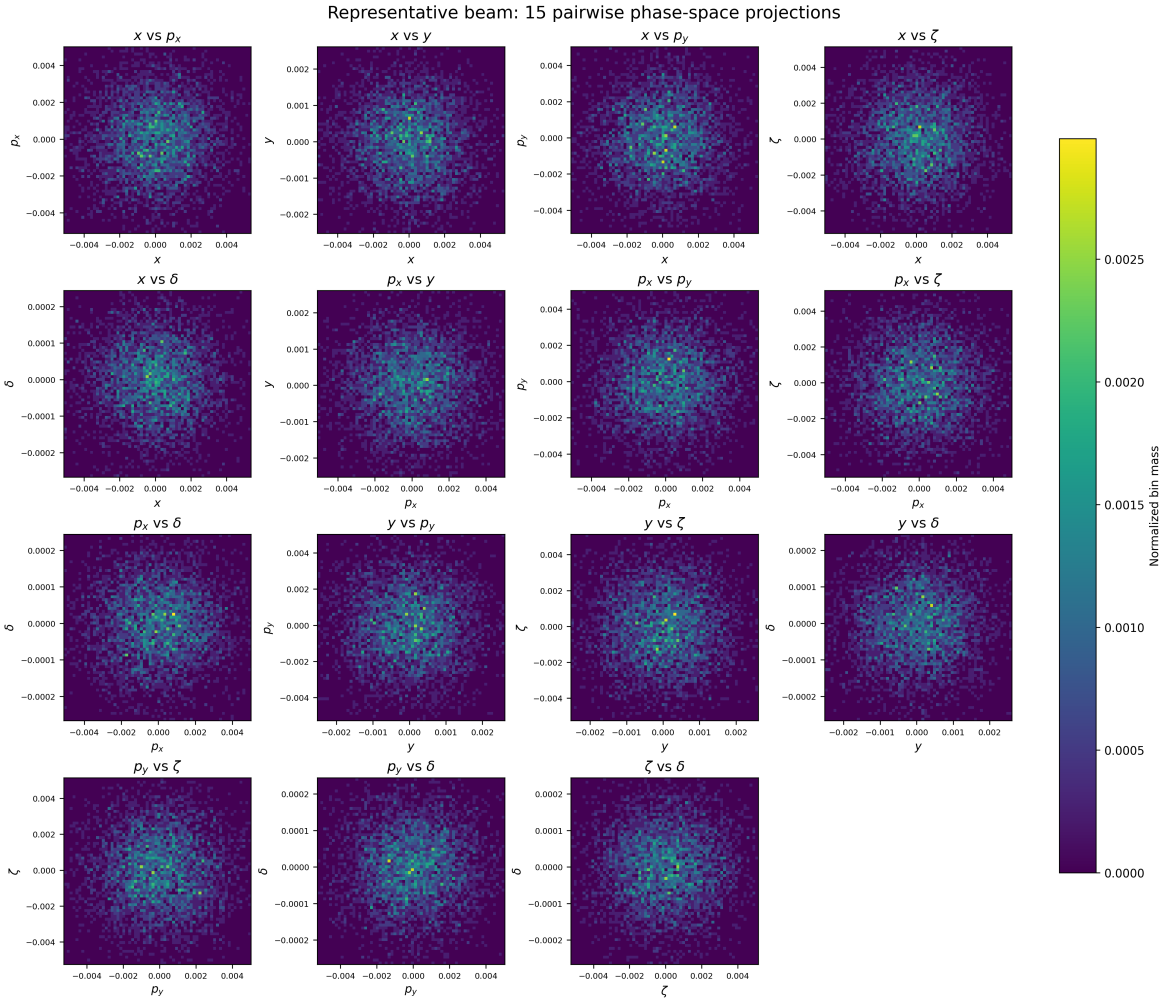


Fig. 1: Representative beam state encoded through its fifteen pairwise two-dimensional phase-space marginals. The associated centroid and coordinate-wise RMS-spread vectors retain complementary information about physical position and extension. The figure represents a reduced description of a six-dimensional state, not a unique reconstruction of the complete joint density.

ence time of 2.338 ms per sample gives the nominal ratio

$$\frac{1.41\,\mathrm{s}}{2.338\times10^{-3}\,\mathrm{s}} \simeq 603. \tag{54}$$

This approximately 600× ratio is useful for indicating the potential order of magnitude,

Tab. 7: Exploratory computational cost recorded for the reference pipeline. "Particles/grid" denotes macroparticles for tracking and histogram generation, and longitudinal grid points for PyColleff.

Data source	Samples	Particles/grid	Time/sample [s]	Total [s]
Xsuite tracking	2048	10^6 particles	1.41	2904.52
PyColleff longitudinal solver	2048	2048 grid points	2.72	5568.32
6D histogram generation	2048	10^6 particles	2.32×10^{-3}	4.75

but it is not reported as a controlled speed-up because the simulator and neural timings were not obtained under a common hardware, batching, and precision protocol.

3.5 Data and code availability

The curated six-dimensional beam-dynamics dataset is publicly available through Zenodo [16]. The archival record preserves the train–validation–test indices, generation and preprocessing scripts, configuration files, software environment, lattice source and checksum, parameter ranges and units, and a machine-readable manifest of the array shapes. The dataset is released subject to the applicable FCC collaboration and institutional procedures and is tagged with both FCC and TwinRISE.

The training, evaluation, figure-generation, and data-generation code developed for this work is publicly available under the MIT licence in the ML4CollEffects GitHub repository [10]. The six-dimensional datasets were generated using the corresponding entry-point script included in that repository. Since the codebase remains under active development, the Zenodo record identifies the exact Git release or commit associated with the results reported in this thesis, thereby ensuring reproducibility independently of subsequent changes to the evolving `main` branch.

4 Machine-learning methodology

4.1 Machine-learning objectives and foundations

The models developed in this work are trained in a supervised setting from simulated input–output pairs. The inputs may include beam distributions, phase-space histograms, latent states, lattice-element descriptors, or machine parameters, while the targets correspond to evolved densities, reconstructed distributions, propagated latent states, or physical beam descriptors. Standard neural-network and variational-inference definitions used throughout this section are collected in Appendix A.1.

4.1.1 From data to prediction

Given a dataset

$$\mathcal{D} = \left\{ \left(x^{(s)}, y^{(s)} \right) \right\}_{s=1}^{N}, \tag{55}$$

a parameterized model $f_\theta : \mathcal{X} \to \mathcal{Y}$ is trained by minimizing an empirical loss,

$$\theta^\star = \arg\min_\theta \frac{1}{N} \sum_{s=1}^{N} \ell\left(f_\theta\left(x^{(s)}\right), y^{(s)}\right). \tag{56}$$

During training, gradients obtained through backpropagation are used to update the model parameters. The complete workflow is summarized in Fig. 2.

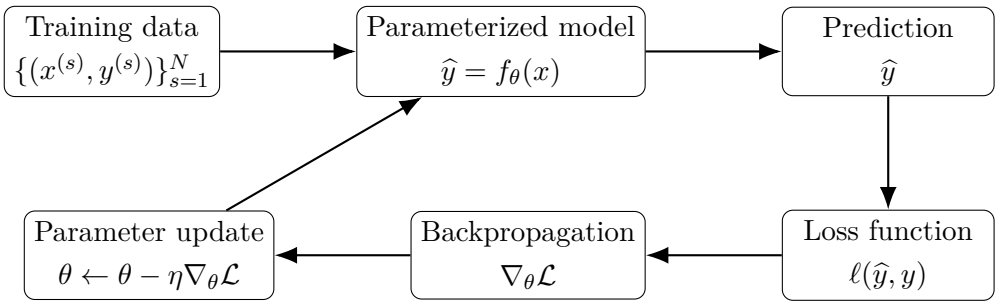


Fig. 2: Basic supervised machine-learning workflow. Input examples are propagated through a parameterized model to obtain predictions. A loss function compares predictions with the ground-truth targets, and backpropagation provides gradients used to update the model parameters. After training, inference consists only of evaluating the learned forward map.

Once training is complete, the optimization loop is removed and inference consists only of evaluating

$$\widehat{y}_{\text{new}} = f_{\theta^\star}(x_{\text{new}}). \tag{57}$$

The purpose of the surrogate models developed in this work is to make this forward evaluation substantially cheaper than rerunning the corresponding reference simulation.

4.1.2 Neural-network building blocks

Dense layers are used throughout the models to transform global variables such as machine parameters, physical beam descriptors, and latent states. They apply learned affine transformations followed by nonlinear activation functions, allowing the networks to represent nonlinear input–output relations.

Convolutional layers are used for the projected beam distributions. Each six-dimensional beam cloud is represented by its fifteen pairwise two-dimensional histograms,

$$X = H(\rho) \in \mathbb{R}^{15 \times N_b \times N_b}. \tag{58}$$

Shared convolutional kernels exploit the local organization of these histograms and can identify structures such as compact cores, asymmetries, tails, filamentation, and halo. Successive convolutional and downsampling layers compress the distributions into learned features, while the decoder reverses this process to reconstruct the projected beam state.

4.1.3 Conditional variational latent representation

Directly propagating all fifteen $N_b \times N_b$ histograms through the accelerator lattice would be computationally expensive. The beam representation is therefore compressed using a conditional variational autoencoder (CVAE). The encoder maps the distributions and conditioning variables to a probabilistic latent state,

$$q_\phi(z \mid X, c), \tag{59}$$

and the decoder reconstructs the distributions through

$$p_\psi(X \mid z, c). \tag{60}$$

The latent state retains a compact representation of the projected beam shape, while variational regularization produces a structured latent space that can be sampled and propagated. The complete CVAE architecture and its training objective are introduced in Section 4.6.

The normalized histograms do not retain all information about the physical position and extension of the beam. The representation therefore also includes six coordinate centroids and six coordinate-wise RMS spreads,

$$\begin{aligned} \mathbf{m} &= (\langle x\rangle, \langle p_x\rangle, \langle y\rangle, \langle p_y\rangle, \langle \zeta\rangle, \langle \delta\rangle), \\ \sigma &= (\sigma_x, \sigma_{p_x}, \sigma_y, \sigma_{p_y}, \sigma_\zeta, \sigma_\delta), \end{aligned} \tag{61}$$

combined into

$$\mathbf{a} = [\mathbf{m}, \sigma] \in \mathbb{R}^{12}. \tag{62}$$

For a projected plane (ξ_i, ξ_j), its physical axes are determined by the shared descriptors $(m_i, \sigma_i, m_j, \sigma_j)$. The model therefore separates normalized distributional shape, represented in latent space, from physical position and extension, represented explicitly by the statistical descriptors.

4.2 Operator-learning formulation

The beam-dynamics problems introduced in the preceding sections are naturally formulated at the level of functions and probability densities. In the absence of collective effects, a phase-space density is transported by the pushforward of a single-particle map. Once self-consistent forces, wakefields, damping, or diffusion are included, however, the evolution becomes a generally nonlinear transformation acting on the distribution itself. The learning problem is therefore more naturally posed as the approximation of an operator rather than as conventional finite-dimensional regression. This would allow to learn different lattice configurations, collective-effect parameters, and initial conditions within a single model.

Let A and U denote suitable spaces of input and output functions, densities, or reduced beam representations, and let

$$\mathcal{M} \subset \mathbb{R}^m \tag{63}$$

denote the admissible machine-parameter domain. The physical target is an operator

$$\mathcal{G}^\dagger : \mathcal{A} \times \mathcal{M} \longrightarrow \mathcal{U}, \qquad (a, \mu) \longmapsto u, \tag{64}$$

where $a \in A$ denotes the input state, $u \in U$ the output state, and $\mu \in \mathcal{M}$ contains machine or collective-effect parameters.

Given $\mathcal{D}$, the set of physical observations corresponding to the operator $\mathcal{G}^\dagger$,

$$\mathcal{D} = \left\{ \left(a^{(s)}, \mu^{(s)}, u^{(s)} \right) \right\}_{s=1}^{N}, \tag{65}$$

the empirical objective is to approximate $\mathcal{G}^\dagger$ by a parameterized family $\mathcal{G}_\theta$,

$$\theta^\star = \arg\min_\theta \frac{1}{N} \sum_{s=1}^{N} \left\| \mathcal{G}_\theta \left(a^{(s)}, \mu^{(s)} \right) - u^{(s)} \right\|_{\mathcal{U}}^2 . \tag{66}$$

The choice of norm depends on the physical object being learned. A discrete L^2 loss is a natural choice for smooth density profiles [20]. Sobolev-type objectives additionally penalize discrepancies in derivatives and are therefore useful when gradient information or fine-scale structure is relevant [21]. Low-order moment penalties can directly constrain physically meaningful quantities such as centroids and coordinate-wise RMS spreads, whereas optimal-transport discrepancies account for the geometric displacement of probability mass and are consequently well suited to comparing spatial probability distributions [22].

This formulation is particularly important for the present work because the different machine-learning models do not necessarily operate on the same discretized representation. The longitudinal FNO acts directly on gridded line densities, whereas the six-dimensional models act on compressed representations of projected phase-space densities. Nevertheless, both can be interpreted as approximations of physical evolution operators at different levels of description.

4.3 Neural operators

4.3.1 From kinetic equations to solution operators

The motivation for neural operators becomes clearer by returning to the governing equations. At the kinetic level, collective beam dynamics are commonly described using the Vlasov equation, or the Vlasov–Fokker–Planck equation when damping and diffusion are included [23–25]. A generic collective beam-density evolution may therefore be written schematically as

$$\partial_s \rho + \nabla_\xi \cdot [\mathbf{F}_{\text{ext}}(\xi, s, \mu)\rho] + \nabla_\xi \cdot [\mathbf{F}_{\text{coll}}[\rho](\xi, s, \mu)\rho] = \mathcal{D}_\mu[\rho], \tag{67}$$

where

$$\xi = (x, p_x, y, p_y, \zeta, \delta) \in \mathbb{R}^6 \tag{68}$$

denotes the six-dimensional phase-space coordinate, F_{ext} is the external lattice force, $F_{\text{coll}}[\rho]$ is a density-dependent collective force, and $\mathcal{D}^{\mu}[\rho]$ represents dissipative or diffusive effects.

Equation (67) may be written abstractly as

$$\partial_s \rho = \mathcal{L}_{\mu}[\rho], \tag{69}$$

where $\mathcal{L}_{\mu}$ denotes the generally nonlinear kinetic generator. For suitable initial data ρ_0, the corresponding evolution defines a parameter-dependent solution operator

$$\mathcal{S}_s^{\mu} : \rho_0 \longmapsto \rho_s. \tag{70}$$

Equivalently, over a finite propagation step Δs, the solution operator may be written as

$$\rho_{s+\Delta s} = \mathcal{S}_{\Delta s}^{\mu}[\rho_s]. \tag{71}$$

This distinction is fundamental. A classical numerical solver approximates the solution of the governing equation for each new initial condition and parameter choice. An operator-learning model instead attempts to approximate the map

$$(\rho_s, \mu) \longmapsto \rho_{s+\Delta s} \tag{72}$$

over a family of admissible initial states and machine configurations.

The same perspective applies to reduced longitudinal collective dynamics. The induced wake potential is already an integral operator acting on the bunch profile,

$$V_{\parallel}(z) = \int_{-\infty}^{\infty} W_{\parallel}(z - z')\lambda(z')\,\mathrm{d}z'. \tag{73}$$

Thus,

$$\mathcal{W} : \lambda \longmapsto V_{\parallel} \tag{74}$$

is a functional map whose kernel is the wake function.

Likewise, the Haïssinski problem defines a nonlinear self-consistent relation between the bunch density and its induced field. If $\mathcal{H}$ denotes the nonlinear update associated with the stationary equation, the equilibrium profile satisfies [26]

$$\lambda_h = \mathcal{H}[\lambda_h; W, I, \mu]. \tag{75}$$

The equilibrium density is therefore a fixed point of a nonlinear operator. One may consequently seek to learn either the fixed-point solution map

$$\mathcal{G}_{\mathrm{H}}^{\dagger} : (W, I, \mu) \longmapsto \lambda_h, \tag{76}$$

or a finite-step relaxation operator

$$\mathcal{G}_{\Delta k}^{\dagger} : (\lambda^{(k)}, \mu) \longmapsto \lambda^{(k+\Delta k)}. \tag{77}$$

At the full kinetic level, an analogous target is

$$\mathcal{G}_t^\dagger : (\rho_0, \mu) \longmapsto \rho_t, \tag{78}$$

which approximates the evolution generated by a Vlasov–Fokker–Planck-type equation.

From this perspective, it is clear that collective beam dynamics is naturally an operator-learning problem. The target is not merely the value of a scalar observable at one point, but the transformation of an entire density, field, or functional representation under parameter-dependent physical dynamics.

4.3.2 General neural-operator architecture

Neural operators are parameterized models designed to approximate maps between function spaces. Following the general framework described by Kovachki et al.[20], the continuous object of interest is

$$\mathcal{G}^\dagger : \mathcal{A} \longrightarrow \mathcal{U}, \tag{79}$$

where $\mathcal{A}$ and $\mathcal{U}$ are suitable function spaces, typically Banach spaces of scalar- or vector-valued fields [20].

Given an input field $a(x)$, the lifting stage is written as

$$v_0(x) = \mathcal{P}(a(x), x, \mu), \tag{80}$$

where P is generally a pointwise neural network. The explicit inclusion of the coordinate x allows the model to retain information about domain geometry, while μ provides parametric dependence on machine or beam settings.

The latent field is then propagated through T operator layers[4],

$$v_{t+1}(x) = \sigma(W_t v_t(x) + \mathcal{K}_{\theta,t}[v_t](x) + b_t(x)), \quad t = 0, \dots, T-1, \tag{81}$$

where W_t is a local linear transformation, b_t is a bias, σ a pointwise nonlinearity, and $\mathcal{K}_{\theta,t}$ a learned nonlocal operator.

For an integral-kernel parameterization,

$$\mathcal{K}_{\theta,t}[v_t](x) = \int_D \kappa_{\theta,t}(x, y, \mu) v_t(y) \, \mathrm{d}\nu(y), \tag{82}$$

so that

$$v_{t+1}(x) = \sigma\left(W_t v_t(x) + \int_D \kappa_{\theta,t}(x, y, \mu) v_t(y) \, \mathrm{d}\nu(y) + b_t(x)\right). \tag{83}$$

Here, $\kappa_{\theta,t}(x, y, \mu)$ is the learned kernel and ν is the measure associated with the domain discretization.

Finally, a projection map produces the output field,

$$u(x) = \mathcal{Q}(v_T(x)). \tag{84}$$

This formulation is particularly relevant to collective beam physics. The local term $W_t v_t(x)$ can represent pointwise or local corrections, while the kernel term

$$\int_D \kappa_{\theta,t}(x,y) v_t(y)\, \mathrm{d}\nu(y) \tag{85}$$

represents structured nonlocal interactions. Such a decomposition is conceptually compatible with beam dynamics, where local lattice transport coexists with nonlocal collective interactions generated by wakefields and self-consistent fields.

The architecture should not, however, be interpreted as automatically enforcing the governing equations. Unless physical constraints are inserted explicitly, the learned kernel remains data driven. The advantage is instead one of representation as the model class is constructed to approximate operators acting on fields and distributions in functional spaces emerging form underlying parametrized differential equations, rather than only maps between fixed finite-dimensional vectors.

4.3.3 Discretization and approximation

One of the central motivations for neural operators is that the model is conceptually defined at the level of functions rather than at the level of one fixed coordinate vector. In the theoretical neural-operator framework, the parameterization can be constructed so that its number of learned parameters does not depend directly on the number of discretization points [20].

This property is especially relevant in accelerator modelling, where the same underlying density may be represented using different, longitudinal grid resolutions, histogram bin numbers, macroparticle populations, spectral truncations, or observation meshes. A discretization-invariant architecture allows the same learned operator to be applied to different representations of the same underlying physical state.

Nevertheless, cross-resolution transfer should be interpreted carefully. A discretization-invariant architecture does not imply that predictions are automatically identical on arbitrary meshes or that no numerical error is introduced by changing resolution. Rather, the continuous operator and its discrete realization must be distinguished.

Let D_L denote a discretization with resolution parameter L, and let $\mathcal{G}_\theta b$ denote the numerical realization of the continuous learned operator $\mathcal{G}_\theta$. Then

$$\begin{aligned}\left\|\widehat{\mathcal{G}}_\theta(D_L, a|_{D_L}) - \mathcal{G}^\dagger(a)\right\|_{\mathcal{U}} \leq & \underbrace{\left\|\widehat{\mathcal{G}}_\theta(D_L, a|_{D_L}) - \mathcal{G}_\theta(a)\right\|_{\mathcal{U}}}_{\text{discretization error}} \\ & + \underbrace{\left\|\mathcal{G}_\theta(a) - \mathcal{G}^\dagger(a)\right\|_{\mathcal{U}}}_{\text{approximation error}}.\end{aligned} \tag{86}$$

For the present discussion in beam dynamics, the target operator may describe either non-collective transport or evolution including collective self-interactions. This distinction can be made explicit by writing the beam-density equation as

$$\partial_s \rho = \mathcal{L}^\mu_{\text{ext}} \rho + \lambda \mathcal{N}^\mu_{\text{coll}}[\rho], \tag{87}$$

where

$$\mathcal{L}^{\mu}_{\text{ext}}\rho = -\nabla_{\xi} \cdot [\mathbf{F}_{\text{ext}}(\xi, s, \mu)\rho] + \mathcal{D}_{\mu}[\rho], \tag{88}$$

and

$$\mathcal{N}^{\mu}_{\text{coll}}[\rho] = -\nabla_{\xi} \cdot [\mathbf{F}_{\text{coll}}[\rho](\xi, s, \mu)\rho]. \tag{89}$$

Here, λ controls the strength of the collective interaction. For example, it may represent an impedance or wakefield scaling factor.

When $\lambda = 0$, particles evolve under the prescribed external lattice without modifying the force acting on the rest of the beam. In the Hamiltonian limit, this evolution is described by a symplectic single-particle map [27]. The particle trajectories may still be nonlinear functions of the phase-space coordinates because of nonlinear lattice elements. Nevertheless, provided that $\mathcal{D}^{\mu}$ is linear, the corresponding density propagator $\mathcal{G}^{\dagger}_{0}$ remains linear with respect to the input density,

$$\mathcal{G}^{\dagger}_{0}(\alpha\rho_1 + \beta\rho_2; \mu) = \alpha\mathcal{G}^{\dagger}_{0}(\rho_1; \mu) + \beta\mathcal{G}^{\dagger}_{0}(\rho_2; \mu). \tag{90}$$

When $\lambda \neq 0$, the force depends on the evolving beam density through wakefields, impedances, space charge, or other collective mechanisms [2, 3]. The resulting propagator $\mathcal{G}^{\dagger}_{\lambda}$ is therefore generally nonlinear in ρ. Even when the collective force is obtained from a linear convolution of the density with a wake function, the product $F_{\text{coll}}[\rho]\rho$ makes the density evolution nonlinear and typically nonlocal.

This distinction also clarifies the role of the two errors in Eq. (86). In the non-collective regime, discretization error arises principally from representing and transporting the phase-space density on a finite grid or with a finite macroparticle sample. In the collective regime, an inaccurate density representation also perturbs the computed self-consistent force, and this error is fed back into the subsequent evolution. Insufficient resolution of narrow structures, tails, or longitudinal modulations can therefore alter both the represented density and the collective force generated by that density. Cross-resolution transfer is consequently a more demanding test in the collective regime. Increasing the number of bins may reduce this error only when the associated macroparticle sampling noise remains controlled.

A direct comparison between the two regimes may be expressed through the collective correction

$$\Delta\rho_{\text{coll}}(s) = \mathcal{G}^{\dagger}_{\lambda}(\rho_0; \mu) - \mathcal{G}^{\dagger}_{0}(\rho_0; \mu), \tag{91}$$

obtained by propagating the same initial density and external lattice with and without collective interactions. A learned operator conditioned on λ may approximate both regimes, while Eq. (91) isolates the density modification attributable to collective effects.

Under appropriate assumptions, neural-operator classes satisfy universal-approximation results for continuous operators on compact subsets of function spaces [20]. Such results provide theoretical motivation for the framework, but they do not remove the practical requirements of representative data, stable optimization, adequate discretization, and physically meaningful validation.

The central kernel operator

$$\mathcal{K}_\theta[v](x) = \int_D \kappa_\theta(x, y)v(y)\,\mathrm{d}\nu(y) \tag{92}$$

can be parameterized in several ways. The choice of parameterization determines the inductive bias, computational complexity, and geometry of the learned interaction.

A direct kernel parameterization leads to Graph Neural Operator-type models, in which the integral is approximated over sampled points or graph neighborhoods. A low-rank operator imposes a factorized representation of the kernel. Hierarchical variants introduce multiscale interaction structures. On regular grids, a Fourier Neural Operator instead represents the kernel action in spectral space.

These models approximate different operators and act on different representations, but they arise from the same conceptual framework. The FNO parameterizes nonlocal interaction through spectral multipliers, whereas the GNO parameterizes it through a learned kernel evaluated over graph neighborhoods.

This common interpretation will later be important when comparing the GNO with the local-window Transformer. Although attention and graph message passing are different computational mechanisms, both can be interpreted as learned interaction kernels with different support constraints.

4.4 Fourier Neural Operator

For reduced longitudinal problems defined on a regular grid, the Fourier Neural Operator provides a natural specialization of the general architecture. The starting point is the kernel operator

$$\mathcal{K}_\theta[v](x) = \int_D \kappa_\theta(x, y)v(y)\,\mathrm{d}y. \tag{93}$$

If the kernel is translation invariant,

$$\kappa_\theta(x, y) = \kappa_\theta(x - y), \tag{94}$$

the integral becomes a convolution,

$$\mathcal{K}_\theta[v] = \kappa_\theta * v. \tag{95}$$

By the convolution theorem,

$$\mathcal{F}(\mathcal{K}_\theta[v])(k) = \mathcal{F}(\kappa_\theta)(k)\mathcal{F}(v)(k). \tag{96}$$

The FNO replaces the explicit Fourier transform of the kernel by a learned spectral multiplier

$$\mathcal{K}_\theta v = \mathcal{F}^{-1}(R_\theta(k)\mathcal{F}(v)(k)). \tag{97}$$

A complete FNO layer therefore takes the form

$$v_{\ell+1}(x) = \sigma\left(W_\ell v_\ell(x) + \mathcal{F}^{-1}(R_{\theta,\ell}(k)\mathcal{F}(v_\ell)(k))(x) + b_\ell(x)\right). \tag{98}$$

In practice, only a finite set of Fourier modes is retained. If $K_{\max}$ denotes the maximum retained mode,

$$\hat{v}(k) = \mathcal{F}(v)(k), \qquad |k| \leq K_{\max}, \tag{99}$$

then the spectral branch acts as

$$\widehat{\mathcal{K}_\theta v}(k) = R_\theta(k)\hat{v}(k), \qquad |k| \leq K_{\max}. \tag{100}$$

Modes outside the retained range do not pass through the learned spectral multiplier and are either truncated or represented indirectly through the local branch and nonlinear layer composition.

The computational appeal of this construction comes from the use of FFT-based transforms on uniform grids. For a grid containing N_g points, the Fourier transforms scale approximately as

$$\mathcal{O}(N_g \log N_g), \tag{101}$$

up to channel dimensions and spectral-mode multiplication costs.

The FNO is particularly appropriate for the longitudinal collective-effects problem because the underlying physics already possesses a spectral formulation. The induced voltage satisfies

$$V_{\text{ind}}(z) = -\mathcal{F}^{-1}\left[Z_\parallel(\omega)\hat{\lambda}(\omega)\right](z), \tag{102}$$

which has the same mathematical structure as multiplication by a spectral response followed by inverse transformation.

This does not imply that the learned multiplier $R_\theta(k)$ is identical to the physical impedance $Z_\parallel(\omega)$. The FNO is trained to approximate the complete target operator represented in the data, which may include nonlinear relaxation, RF confinement, parameter dependence, discretization effects, and several successive physical transformations. Nevertheless, the spectral architecture introduces an inductive bias consistent with the convolutional and frequency-domain structure of longitudinal collective interactions.

In the present work, the primary reduced FNO task is to learn the evolution of the longitudinal density under collective effects, parametrized by a vector μ containing the physical parameters controlling the simulated collective response. For stationary problems, a second possible formulation is to learn the fixed-point map where the target is the self-consistent Haïssinski equilibrium. A third formulation acts directly on the collective field, where the target is the induced voltage or wake potential, which is the closest learned analogue of the physical wake-convolution operator.

These distinct formulations emphasize an important methodological point in operator learning. It does not prescribe a unique surrogate target, instead the methodology is defined depending on the intended application, one may learn the evolved density, the stationary

solution, the collective field, or reduced observables. The experiments presented later focus on the mappings supported by the available simulation datasets, while the broader formulations define natural extensions of the present work.

4.5 Graph Neural Operator

The Fourier Neural Operator provides an efficient parameterization of nonlocal operators on regular grids. However, its spectral construction becomes less convenient when the underlying domain is irregular or nonuniformly sampled, when fine-scale structures require the retention of many higher Fourier harmonics, or when a high-dimensional domain makes the tensor-product spectral representation computationally expensive. Moreover, some physical systems are more naturally represented as sets of interacting elements rather than as fields on regular grids. In these settings, a Graph Neural Operator (GNO) provides an alternative discretization of the generic kernel operator introduced in Section 4.3. Rather than representing the kernel action in Fourier space, the GNO approximates the integral operator by evaluating learned interactions directly between sampled points or graph nodes [20, 28].

Starting from the continuous operator layer

$$\mathcal{K}_\theta[v](x) = \int_D \kappa_\theta(x, y, \mu) v(y)\, \mathrm{d}\nu(y), \tag{103}$$

let

$$X = \{x_i\}_{i=1}^N \tag{104}$$

be a discrete set of nodes sampling the domain (D). A quadrature approximation of Eq. (103) gives

$$\mathcal{K}_\theta[v](x_i) \approx \sum_{j=1}^N \omega_{ij} \kappa_\theta(x_i, x_j, \mu) v(x_j), \tag{105}$$

where ω_j denotes a quadrature weight associated with node x_j. When interactions are restricted to a neighborhood $\mathcal{N}(i)$ of node i, the discrete operator becomes

$$\mathcal{K}_\theta[v](x_i) \approx \sum_{j \in \mathcal{N}(i)} \omega_{ij} \kappa_\theta(x_i, x_j, \mu) v(x_j). \tag{106}$$

The neighborhood definition therefore determines the support of the learned interaction kernel.

A complete GNO layer may be written as

$$v_i^{(\ell+1)} = \sigma\left(W_\ell v_i^{(\ell)} + \sum_{j \in \mathcal{N}(i)} \omega_{ij} \kappa_{\theta,\ell}(x_i, x_j, v_i^{(\ell)}, v_j^{(\ell)}, \mu) v_j^{(\ell)} + b_\ell \right), \tag{107}$$

where W_ℓ is a local linear transformation, $\kappa_{\theta,\ell}$ is a learned kernel, σ is a nonlinear activation, and b_ℓ is a bias term.

Equivalently, the operator can be expressed in message-passing form. Defining an edge message

$$m_{ij}^{(\ell)} = \varphi_{\theta,\ell}(v_i^{(\ell)}, v_j^{(\ell)}, e_{ij}, \mu), \tag{108}$$

where e_{ij} contains relative geometric or physical information associated with the interaction between nodes i and j, the aggregated message is

$$m_i^{(\ell)} = \sum_{j \in \mathcal{N}(i)} m_{ij}^{(\ell)}, \tag{109}$$

and the node update becomes

$$v_i^{(\ell+1)} = \psi_{\theta,\ell}(v_i^{(\ell)}, m_i^{(\ell)}). \tag{110}$$

This representation makes explicit that graph message passing is a discrete approximation of a learned integral operator rather than merely a generic graph-processing mechanism.

The distinction with the FNO is therefore primarily a distinction in kernel parameterization. The FNO evaluates a global translation-compatible interaction through spectral multiplication,

$$\mathcal{K}_\theta^{\mathrm{FNO}} v = \mathcal{F}^{-1}(R_\theta(k)\mathcal{F}(v)(k)), \tag{111}$$

whereas the GNO evaluates interactions directly between sampled locations,

$$\mathcal{K}_\theta^{\mathrm{GNO}} v(x_i) = \sum_{j \in \mathcal{N}(i)} \omega_{ij} \kappa_\theta(x_i, x_j, \mu) v(x_j). \tag{112}$$

The FNO is therefore particularly natural for regular grids and approximately translationstructured interactions, while the GNO is more flexible when geometry, connectivity, or interaction support vary across the domain.

4.5.1 Directed graph representation of the accelerator lattice

In the present work, the GNO is used not directly on a six-dimensional Cartesian grid, but as a latent propagator along the accelerator lattice. The lattice is represented as a directed graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \tag{113}$$

where each node

$$i \in \mathcal{V} \tag{114}$$

corresponds to a lattice element or observation point and each directed edge

$$(j \to i) \in \mathcal{E} \tag{115}$$

represents an admissible upstream interaction.

Each node is associated with a feature vector

$$h_i = \mathrm{Tok}(\mathrm{type}_i, L_i, p_i, s_i, \mu_i), \tag{116}$$

where type_i denotes the element category, L_i its length, p_i the associated element parameters, s_i its longitudinal coordinate, and μ the global machine and collective-effect parameters.

The directed neighborhood may be defined as

$$\mathcal{N}(i) = \{j : j < i,\ d(i,j) \le r\}, \tag{117}$$

where $d(i,j)$ is a graph or longitudinal distance and r is an interaction radius. An alternative is a fixed upstream neighborhood containing the previous (K) elements,

$$\mathcal{N}_K(i) = \{\max(1, i-K), \dots, i-1\}. \tag{118}$$

Both constructions enforce causal directionality while controlling the spatial support of the learned kernel.

This representation is particularly relevant for beam transport. The evolution through an accelerator is intrinsically ordered, and the state observed downstream results from the accumulated action of upstream lattice elements. The graph construction therefore introduces a structured inductive bias without assuming that every upstream element must interact equally with every downstream state.

4.5.2 Latent-state propagation

The six-dimensional beam distribution is not propagated directly. Instead, the CVAE encoder produces a latent representation

$$z_i = \mathcal{E}_\phi(H(\rho_i), c_i) \in \mathbb{R}^{d_z}, \tag{119}$$

where $\mathcal{H}(\rho_i)$ contains the fifteen pairwise phase-space projections and (c_i) contains physical conditioning information.

The latent state is first lifted to the graph feature dimension,

$$v_i^{(0)} = P_z(z_i) + P_h(h_i), \tag{120}$$

where P_z and P_h are learned projections.

A general latent GNO layer then takes the form

$$v_i^{(\ell+1)} = \sigma\left(W_\ell v_i^{(\ell)} + \sum_{j\in\mathcal{N}(i)} \omega_{ij}\mathcal{K}_{\theta,\ell}(h_i, h_j, v_i^{(\ell)}, v_j^{(\ell)}, \mu) v_j^{(\ell)}\right). \tag{121}$$

After (L) operator layers, the latent increment is predicted through

$$\Delta z_i = \mathcal{Q}_z\left(v_i^{(L)}\right), \tag{122}$$

and the propagated latent state is updated residually,

$$z_{i+1} = z_i + \Delta z_i. \tag{123}$$

The residual formulation is motivated by the fact that each lattice section modifies, rather than reconstructs from scratch, the incoming beam state. It also makes the GNO structurally comparable with the Transformer propagator used later in the model comparison.

For an entire sequence of (N) lattice elements, the accumulated evolution is

$$z_N = z_0 + \sum_{i=0}^{N-1} \Delta z_i, \tag{124}$$

with each increment determined by the local directed graph neighborhood.

4.5.3 Kernel parameterization

The learned kernel may depend on relative node coordinates, element properties, latent features, and global conditioning variables. A general parameterization is

$$\mathcal{K}_\theta(h_i, h_j, v_i, v_j, \mu) = \mathrm{MLP}_\theta(h_i, h_j, h_i - h_j, v_i, v_j, \mu). \tag{125}$$

If the longitudinal positions s_i are explicitly available, the relative coordinate

$$\Delta s_{ij} = s_i - s_j \tag{126}$$

may also be supplied to the kernel,

$$\mathcal{K}_{\theta,ij} = \mathrm{MLP}_\theta(h_i, h_j, \Delta s_{ij}, v_i, v_j, \mu). \tag{127}$$

This allows the learned interaction to depend explicitly on physical separation along the lattice.

In the implemented model, continuous positional information may be further expanded through Fourier features,

$$\gamma(s) = [\sin(\omega_1 s), \cos(\omega_1 s), \dots, \sin(\omega_M s), \cos(\omega_M s)], \tag{128}$$

so that the kernel receives a multi-scale representation of element position. This is useful when relevant interactions span several characteristic length scales.

The resulting kernel should not be interpreted as an exact reconstruction of a physical wake function, transfer matrix, or Green function. Rather, it is a learned effective interaction kernel for the reduced latent dynamics. Its physical relevance arises from the structure of the parameterization and graph support, not from an a priori identification with one specific analytical quantity.

The GNO formulation provides a useful comparison with the Transformer models introduced later[13]. Causal self-attention computes

$$y_i = \sum_{j \le i} \alpha_{ij} V_j, \qquad \alpha_{ij} = \mathrm{softmax}_j \left(\frac{Q_i^\top K_j}{\sqrt{d_k}} \right). \tag{129}$$

The GNO instead computes

$$y_i = \sum_{j \in \mathcal{N}(i)} \mathcal{K}_\theta(h_i, h_j, v_i, v_j, \mu) v_j. \tag{130}$$

Both mechanisms therefore aggregate information from other locations using data-dependent interaction weights. The principal distinction lies in how the interaction kernel and its support are defined. Global causal attention allows every upstream token to influence the current state,

$$\mathcal{N}_{\mathrm{TF}}(i) = \{1, \dots, i\}, \tag{131}$$

whereas the local-window Transformer restricts this support to

$$\mathcal{N}_w(i) = \{j : 0 \le i - j \le w\}. \tag{132}$$

The GNO uses a graph-defined neighborhood $\mathcal{N}_{\mathrm{GNO}}(i)$ that can be selected directly from lattice geometry or interaction assumptions.

This correspondence is central to the experimental comparison developed later in this report. The global Transformer, local-window Transformer, and GNO can be viewed as three different assumptions about the support and parameterization of information transfer. The global Transformer learns dense upstream interactions, the windowed Transformer imposes finite support through masking, and the GNO defines interaction support directly through graph connectivity.

The empirical observation that reducing the Transformer window moves its predictions closer to those of the GNO can therefore be interpreted as evidence that the support of the learned interaction kernel is an important inductive bias for the latent beam-dynamics problem.

The GNO is used in this work to learn the conditional latent evolution

$$\widehat{\mathcal{G}}_\theta^{\mathrm{GNO}} : (z_0, \mathcal{G}_{\mathrm{lat}}, \mu) \longmapsto z_N, \tag{133}$$

where z_0 is the low dimension latent representation of the initial beam, $\mathcal{G}_{\mathrm{lat}}$ is the directed lattice graph, and (μ) contains global physical parameters.

The propagated latent state is then decoded as

$$\widehat{H}(\rho_N) = D_\psi(z_N, c_N), \tag{134}$$

providing reconstructed pairwise phase-space marginals at the downstream location.

In the dual-output formulation, a second head predicts the physical position and extension of the projected distributions,

$$\hat{q}_N = P_{\mathrm{phys}}(z_N, q_0, \mu), \tag{135}$$

where

$$q = \{(\mu_r, \sigma_r)\}_{r=1}^{6} \tag{136}$$

collects the location and scale descriptors associated with the fifteen projected distributions.

The complete GNO surrogate therefore approximates

$$(z_0, q_0, \mathcal{G}_{\mathrm{lat}}, \mu) \longmapsto (z_N, \hat{q}_N) \longmapsto (\widehat{H}(\rho_N), \hat{q}_N). \tag{137}$$

This formulation separates two complementary aspects of beam evolution. The latent state carries information about the shape and correlations of the projected phase-space density, while the auxiliary physical head tracks the position and extension of those distributions. The GNO acts as the structured propagator connecting these representations along the accelerator lattice.

4.6 Variational latent representation

4.6.1 VAE formulation

For the six-dimensional problem, the fifteen projected histograms are high-dimensional objects. A variational autoencoder (VAE) is used to learn a compressed latent representation. Given a beam representation $X = \mathcal{H}(\rho)$, the encoder defines an approximate posterior

$$q_\phi(z \mid X) = \mathcal{N}(\mu_\phi(X), \operatorname{diag} \sigma_\phi^2(X)), \tag{138}$$

while the decoder defines a likelihood $p_\psi(X|z)$. The evidence lower bound is

$$\mathcal{L}_{\mathrm{ELBO}}(\phi, \psi) = \mathbb{E}_{q_\phi(z|X)}[\log p_\psi(X \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid X) \| p(z)), \tag{139}$$

4.6.2 Conditional VAE

In the conditional model, the posterior and decoder also depend on physical conditioning variables c, such as centroids, coordinate-wise RMS spreads, lattice parameters, and collective-effect parameters. The encoder and decoder are therefore

$$q_\phi(z \mid X, c), \qquad p_\psi(X \mid z, c). \tag{140}$$

The corresponding objective is

$$\mathcal{L}_{\mathrm{CVAE}} = \mathbb{E}_{q_\phi(z|X,c)}[\log p_\psi(X \mid z, c)] - \beta D_{\mathrm{KL}}(q_\phi(z \mid X, c) \| p(z \mid c)), \tag{141}$$

where β controls the strength of latent regularization [18]. The conditional formulation is useful because beam distributions are of course not arbitrary. Their scale, position, and evolution depend on physical parameters. Conditioning helps separate geometric variability from dynamical evolution, and it naturally opens the possibility of uncertainty quantification by sampling several latent states at fixed c.

Each particle distribution ρ is mapped to the complete set of 15 pairwise two-dimensional histograms $\mathcal{H}(\rho) \in \mathbb{R}^{15\times64\times64}$. Four residual convolutional downsampling blocks compress this representation to a $256 \times 4 \times 4$ feature tensor. After flattening, the 4096 convolutional

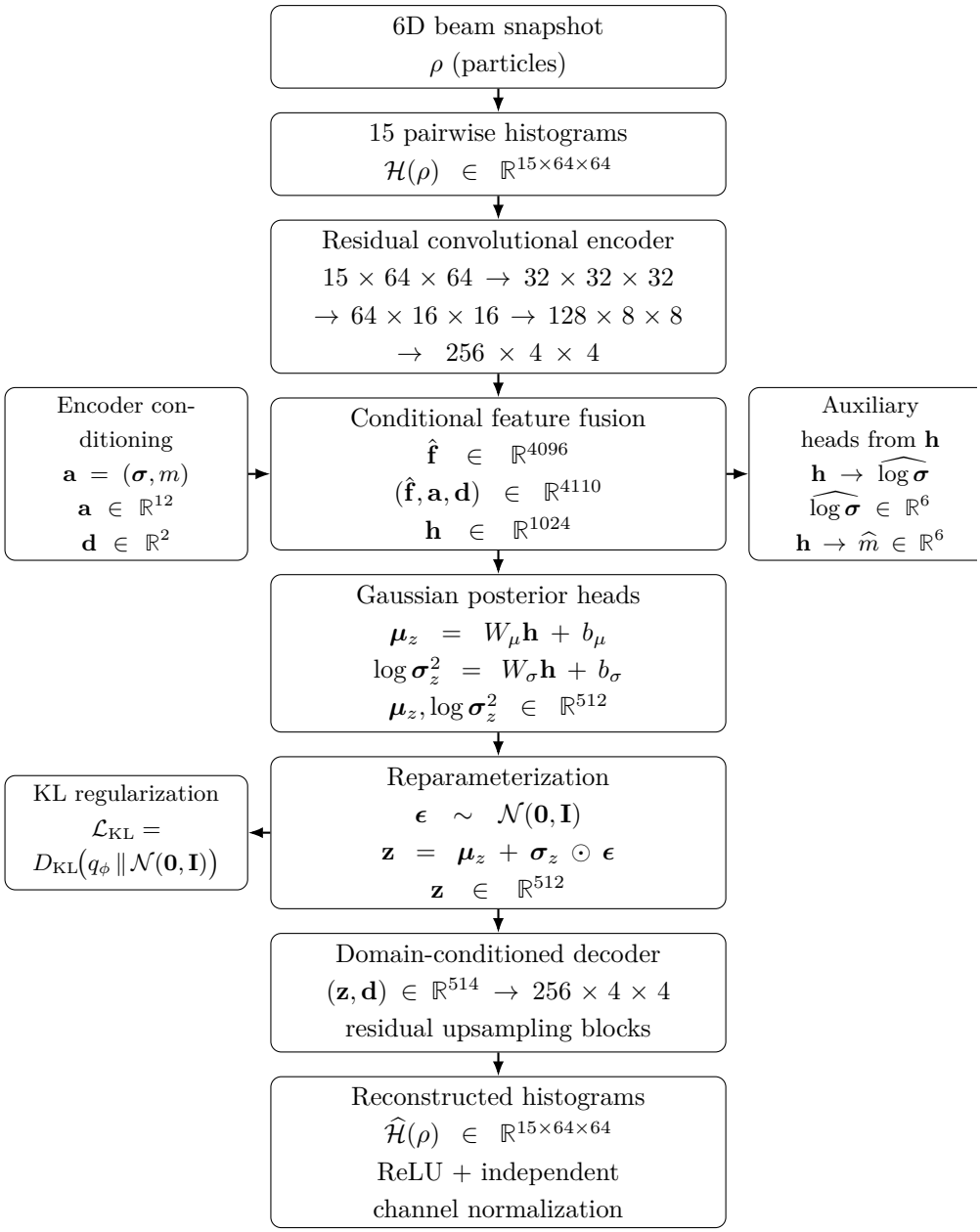


Fig. 3: Implemented conditional variational autoencoder architecture used to construct the reduced representation of six-dimensional beam snapshots.

features are concatenated with the 12 auxiliary beam statistics $\mathbf{a} = (\sigma, m) \in \mathbb{R}^{12}$ and the two-component domain token d, and projected to a shared feature $\mathbf{h} \in \mathbb{R}^{1024}$. Two linear heads parameterize a 512-dimensional Gaussian posterior, from which the latent state $\mathbf{z}$ is sampled through the reparameterization trick. The decoder receives only the latent state and the domain token, (z, d), and reconstructs the 15 pairwise histograms through residual upsampling blocks. In parallel, two auxiliary heads branch directly from the shared encoder feature $\mathbf{h}$ to predict the six logarithmic rms beam sizes and the six phase-space centroids. Each reconstructed histogram channel is constrained to be non-negative and normalized independently over its 64×64 grid.

The implemented CVAE acts on the full collection of pairwise projected beam densities rather than on a single low-dimensional observable. For a six-dimensional phase-space distribution ρ, the preprocessing operator constructs

$$H(\rho) = \{H_i\}_{i=1}^{15}, \qquad H(\rho) \in \mathbb{R}^{15\times 64\times 64},$$

corresponding to the complete set of pairwise two-dimensional projections of the six phase-space coordinates. These 15 histograms are supplied jointly as channels of the convolutional encoder.

The encoder contains four residual downsampling blocks. Starting from the 64×64 spatial grid, successively downsampling the feature map by a factor of two in each dimension produces the sequence until the final 4×4 feature map is reached. The channel dimension is simultaneously increased from 15 to 256 through successive convolutional operations.

Each block first applies a stride-two convolution. The resulting feature map is then transformed through group normalization, GELU activation, and a second convolutional operation, with a residual connection around the transformation. The final feature tensor therefore contains $256 \times 4 \times 4 = 4096$ scalar features.

The convolutional representation is conditioned on explicit low-order beam statistics. The auxiliary vector is defined as $\mathbf{a} = (\sigma, m) \in \mathbb{R}^{12}$,

A two-dimensional domain token $\mathbf{d} \in \mathbb{R}^2$ is additionally concatenated with the encoded histogram features. The input to the fully connected fusion layer is therefore

$$(\hat{f}, \mathbf{a}, \mathbf{d}) \in \mathbb{R}^{4096+12+2} = \mathbb{R}^{4110}.$$

The shared conditioned feature is

$$h = \mathrm{GELU}(W_h(\hat{f}, \mathbf{a}, \mathbf{d}) + b_h), \qquad h \in \mathbb{R}^{1024}. \tag{142}$$

Two independent linear heads parameterize the approximate Gaussian posterior,
$m_z = W_m \mathrm{h} + b_m$, $\log \sigma_z^2 = W_\sigma \mathrm{h} + b_\sigma$,
with m_z, $\log \sigma_z^2 \in \mathbb{R}^{512}$.
The corresponding approximate posterior is

$$q_\phi(z \mid H(\rho), \mathbf{a}, \mathbf{d}) = \mathcal{N}(\mathbf{m}_z, \mathrm{diag}(\sigma_z^2)). \tag{143}$$

Sampling is performed through reparametrization of the latent state, which allows gradients to propagate through the stochastic sampling operation. The reparameterization is given by (224). The reconstructed channels are explicitly constrained to represent discrete two-dimensional densities. First, non-negativity is imposed through

$$\widetilde{H} = \mathrm{ReLU}(\mathrm{Conv}_{1\times1}(f_{\mathrm{dec}})), \tag{144}$$

Each channel is then normalized independently over the spatial grid,

$$\widetilde{H}_i = \mathrm{ReLU}(\mathrm{Conv}_{1\times1}(f_{\mathrm{dec},i})), \qquad i = 1, \dots, 15. \tag{145}$$

Consequently,

$$\sum_{u,v} \widehat{H}_i(u,v) = 1, \qquad i = 1, \dots, 15. \tag{146}$$

If the total intensity of a channel becomes numerically negligible during the early stages of training, the implementation replaces that channel by the uniform density $H_i^b(u,v) = \frac{1}{64\times64}$, which prevents undefined normalization.

In parallel with the variational posterior, two auxiliary regression heads branch directly from the shared conditioned feature $\mathbf{h} \in \mathbb{R}^{1024}$. Each head consists of two linear layers with a GELU nonlinearity between the two linear layers. It is important to distinguish the fact that the auxiliary physical heads are not functions of the sampled latent vector z. They directly supervise the shared encoder representation $\mathbf{h}$ before its projection onto the stochastic latent posterior.

The rms-size head is supervised in logarithmic space,

$$\mathcal{L}_{\sigma} = \mathrm{MSE}(\widehat{\log\sigma}, \log\sigma), \tag{147}$$

while the centroid head is trained using

$$\mathcal{L}_{\mathrm{cent}} = \mathrm{MSE}(\widehat{\mathbf{m}}, \mathbf{m}). \tag{148}$$

The complete snapshot-pretraining objective implemented in the notebook is therefore

$$\mathcal{L}_{\mathrm{CVAE}} = \mathcal{L}_{\mathrm{rec}} + \beta\mathcal{L}_{\mathrm{KL}} + \gamma\mathcal{L}_{\sigma} + \delta\mathcal{L}_{\mathrm{cent}}, \tag{149}$$

where KL regularization term is defined in (227) with β, γ, and δ tuneable parameters in the present implementation.

The encoder combines these projected distributions with the explicit conditioning vector $\mathbf{c} = [\sigma, m, \mathrm{d}]$, containing the six rms beam sizes, the six phase-space centroids, and the domain information. The approximate posterior is parameterized through the latent mean m_z and variance σ_z^2, and a 512-dimensional latent state is sampled using the reparameterization trick. The conditional decoder reconstructs the 15 pairwise histograms. In the dual formulation, an auxiliary physical-dimensions head predicts $\{(\sigma_i b, m_i b)\}_{i=1}^{6}$, providing an explicit representation of physical scale and location in parallel with the histogram-based latent description.

Solid arrows denote forward model evaluation, whereas dashed arrows denote conditioning, regularization, or loss pathways.

The resulting architecture separates two complementary descriptions of the beam. The histogram branch represents the distribution through the set of pairwise projected densities,

$$H(\rho) = \{H_i\}_{i=1}^{15},$$

which retains non-Gaussian structure, inter-variable dependence, tails, asymmetries, and multimodal features that cannot be summarized by a small set of low-order moments. These projected densities are compressed into a latent variable

$\mathbf{z} \sim q_\phi\ (\mathrm{z}\ |\mathcal{H}(\rho),\ \mathrm{c})\ ,$

where the approximate posterior is parameterized as a diagonal Gaussian,

$q_\phi = \mathcal{N} m_z,\ \mathrm{diag}\ \sigma_z^2.$

Sampling is performed through

$\mathbf{z} = m_z + \sigma_z \odot \epsilon, \epsilon \sim \mathcal{N}(0, \mathrm{I}),$

which preserves differentiability of the stochastic latent representation during training.

The explicit conditioning variables are retained outside the histogram representation because the projected densities and the global physical extent of the beam play distinct roles. In the present implementation, the conditioning vector contains the rms beam sizes, the six phase-space centroids, and the domain information,

$\mathbf{c} = [\sigma, \mathbf{m}, \mathbf{d}],$

with

$$\sigma = (\sigma_x, \sigma_y, \sigma_\zeta, \sigma_{p_x}, \sigma_{p_y}, \sigma_\delta),$$

and

$$\mathbf{m} = (\langle x\rangle, \langle y\rangle, \langle \zeta\rangle, \langle p_x\rangle, \langle p_y\rangle, \langle \delta\rangle).$$

This conditioning allows the encoder to account explicitly for global translation and scale. The decoder receives only (z, d), absolute position and extension are therefore predicted separately by the physical-statistics head. The auxiliary physical-dimensions branch extends this separation to the subsequent dynamical surrogate. Rather than requiring the latent state alone to preserve the absolute support of each projected distribution, the model predicts

$$\{(\hat{\sigma}_i, \widehat{m}_i)\}_{i=1}^{6},$$

in parallel with the latent representation. The resulting dual description therefore distinguishes between the evolution of projected distribution shape and the evolution of its physical extent and location. This is particularly relevant for long-horizon tracking, where a latent model may reconstruct the normalized morphology of a distribution while progressively losing information about its absolute scale or displacement.

The CVAE objective combines reconstruction accuracy with regularization of the stochastic latent space and, when the dual branch is used, supervision of the explicit physical variables. The complete loss function is

$\mathcal{L}_{\mathrm{CVAE}} = \mathcal{L}_{\mathrm{rec}} + \beta\mathcal{L}_{\mathrm{KL}} + \lambda_{\mathrm{phys}}\mathcal{L}_{\mathrm{phys}}$.

The Kullback–Leibler contribution,

$\mathcal{L}_{\mathrm{KL}} = D_{\mathrm{KL}}(q_\phi\ (\mathrm{z}\ |\mathcal{H}(\rho),\ \mathrm{c})\ ,\mathcal{N}(0,\ \mathrm{I}))\ ,$

regularizes the learned representation toward a continuous Gaussian latent space, while $\mathcal{L}_{\mathrm{rec}}$ constrains the reconstructed pairwise densities and $\mathcal{L}_{\mathrm{phys}}$ supervises the explicit scale-andlocation branch.

4.7 Tokenized latent dynamics

The latent representation reduces the beam state to a vector $z_i \in \mathbb{R}^{d_z}$ at lattice element or observation index i. The goal is then to learn a propagator

$$z_{i+1} = T_\theta(z_i, h_i, \mu), \tag{150}$$

where h_i is an element token encoding the local lattice element and μ contains global parameters. In residual form,

$$z_{i+1} = z_i + \Delta z_i, \qquad \Delta z_i = f_\theta(z_i, h_i, \mu). \tag{151}$$

This element-indexed formulation describes the computational architecture, not the available supervision. The training archive supplies only the upstream cloud and the final downstream cloud. Consequently, the loss is applied after decoding the final state z_N; no target z_i or particle cloud is available for $0 < i < N$. The intermediate latent states are therefore internal recurrent states and must not be interpreted as validated physical beam distributions. Testing the operator at arbitrary propagation lengths would require simulator checkpoints and CVAE training data at the corresponding intermediate locations.

4.7.1 Element tokenization

Each lattice element is represented by a token containing categorical type information, element length, local optical information, and relevant physical parameters. A general token can be written as

$$h_i = \mathrm{Tok}(\mathrm{type}_i, L_i, k_i, \beta_i, \alpha_i, \mu_i), \tag{152}$$

then projected to the latent model dimension through a learned embedding. Tokenization makes the model aware that beam evolution is not a single black-box map, but a composition of local transformations along a lattice.

4.7.2 Transformer baseline

The Transformer baseline processes the element sequence using causal self-attention [19]. Given token embeddings $H \in \mathbb{R}^{N\times d}$, attention is

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M_{\mathrm{causal}}\right)V, \tag{153}$$

The causal mask prevents dependence on future elements. However, it does not enforce locality, every previous element may still interact with the current one. Thus, the baseline is expressive but may introduce global mixing artifacts when the physical dynamics are dominated by local or near-local transport.

4.7.3 Local-window Transformer

To test the role of locality, a windowed attention mask is introduced by,

$$M_{ij} = \begin{cases} 0, & 0 \le i - j \le w, \\ -\infty, & \text{otherwise}, \end{cases} \tag{154}$$

where w is the attention window. This architecture preserves causality while restricting information flow to the most recent upstream elements. Empirically, this modification is expected to reduce nonphysical global mixing and to improve stability of long-lattice rollouts.

In the full causal mask, token i may attend to every preceding token and to itself, while all future positions are excluded. In the local-window variant, the causal constraint is retained but attention is further restricted to the most recent W positions; the right panel uses $W = 4$ only for visualization. The windowed model therefore changes the temporal connectivity of the Transformer rather than merely reducing its size. In particular, the number and width of the Transformer layers may remain unchanged, while the admissible information pathways are constrained to favor local lattice history.

For a sequence of lattice-conditioned latent tokens indexed by $i = 1, \dots, N$, the full causal Transformer prevents access to future states while retaining the complete available history. Its additive attention mask may be written as

$$M_{ij}^{\text{causal}} = \begin{cases} 0, & j \le i, \\ -\infty, & j > i. \end{cases}$$

Consequently, the representation associated with the $i-$th lattice position may directly aggregate information from every preceding token. This choice is appropriate when long-range dependencies are expected to remain directly relevant throughout the entire sequence.

However, for the beam-dynamics problem considered here, the latent evolution is generated through a sequence of local interactions with lattice elements. The full causal mask therefore allows information to propagate along paths that are not physically admissible, which may introduce nonphysical global mixing artifacts.

The windowed variant preserves causality but restricts the admissible support to a local history of width W. Using the convention that the current token is included in the window, the mask is

$$M_{ij}^{(W)} = \begin{cases} 0, & \max(1, i - W + 1) \le j \le i, \\ -\infty, & \text{otherwise}. \end{cases}$$

The direct attention neighborhood of token i is therefore $\mathcal{N}_W(i) = \{\max(1, i - W + 1), \dots, i\}$. This modification is not equivalent to constructing a smaller Transformer. The

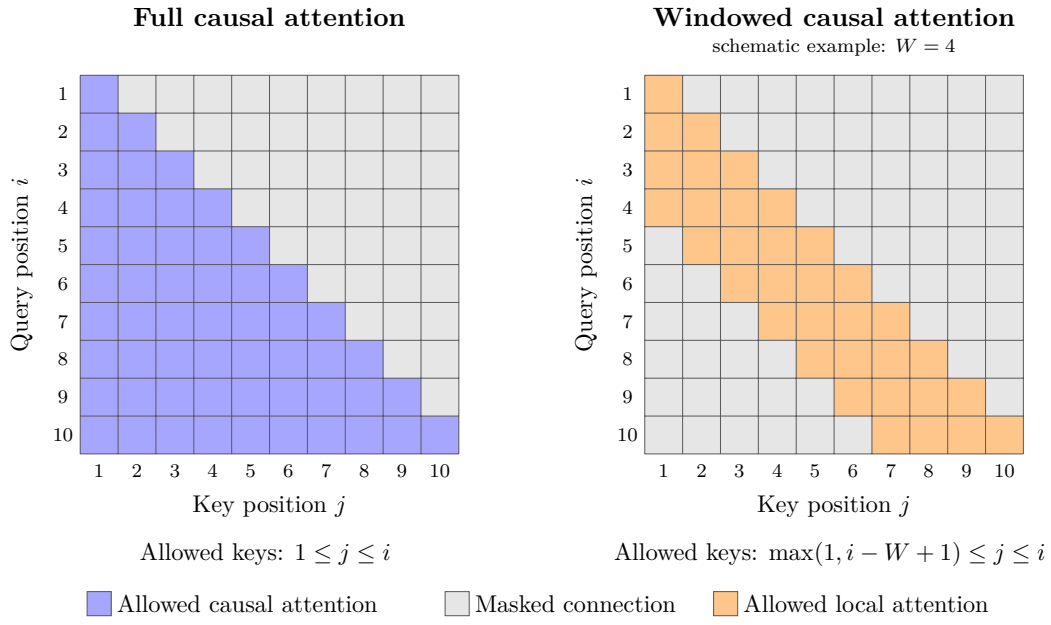


Fig. 4: Comparison of the attention support used by the full causal and windowed Transformer variants. Rows correspond to query positions i and columns to key positions j.

window changes the connectivity pattern through which information can propagate. The distinction is therefore one of inductive bias in the learned latent dynamics. In the full causal architecture, the number of directly accessible keys for query i grows as $\mathcal{N}_i = i$, whereas for the local-window mask it is bounded by W. Thus, the full causal receptive field expands continuously along the lattice, while the windowed attention pattern imposes a fixed local interaction scale at each layer.

The locality constraint does not imply that information older than W positions is completely inaccessible to a multilayer Transformer. Because the representation produced at one layer already summarizes earlier tokens inside its own local neighborhood, successive layers can propagate information indirectly across a broader range. Ignoring boundary effects, after L local-attention layers the maximum backward dependency range can grow approximately as $1+L(W-1)$, while each individual attention operation remains local. The model therefore retains the possibility of building longer-range dependencies hierarchically, but it must do so through successive local transformations rather than through a single unrestricted attention step.

4.7.4 Graph Neural Operator latent propagator

A Graph Neural Operator represents the lattice as a directed graph. Nodes correspond to lattice elements or observation points, and directed edges encode physically admissible interactions. The continuous operator layer

$$v_{\ell+1}(x) = \sigma\left(W v_\ell(x) + \int_\Omega K_\theta(x,y) v_\ell(y)\,\mathrm{d}y\right) \tag{155}$$

is discretized as message passing,

$$v_i^{\ell+1} = \sigma\left(Wv_i^{\ell} + \sum_{j\in\mathcal{N}(i)} K_\theta(h_i, h_j)v_j^{\ell}\right). \tag{156}$$

In the beam-dynamics setting, $\mathcal{N}(i)$ can be chosen to respect directionality and locality along the lattice. This makes the GNO a natural approximation of a localized transport kernel. It also provides a physical interpretation of why its behavior may resemble that of a local-window Transformer while being more explicitly tied to operator learning and message passing.

4.8 Loss functions and validation metrics

The snapshot CVAE is pretrained before the latent propagators. Its implemented objective is

$$\mathcal{L}_{\mathrm{CVAE}} = \mathcal{L}_{\mathrm{rec}} + 10^{-2}\mathcal{L}_{\mathrm{KL}} + 10^{-3}\mathcal{L}_{\sigma} + 10^{-3}\mathcal{L}_{\mathrm{cent}}, \tag{157}$$

where $\mathcal{L}_{\mathrm{rec}}$ is the reconstruction error of the fifteen normalized histograms, $\mathcal{L}_{\sigma}$ is the mean-squared error of the six logarithmic coordinate-wise RMS spreads, and $\mathcal{L}_{\mathrm{cent}}$ is the mean-squared error of the six centroids.

For latent propagation, only the final decoded state is supervised. Let $H_{N,p}^{(s)}$ be target projection p of sample s and let $H_{N,p}^{b(s)} = \mathcal{D}_\psi(z_N^{(s)})$ be the decoded prediction. The endpoint histogram term used in the Transformer window study is

$$\mathcal{L}_{\mathrm{hist},N} = \frac{1}{15B}\sum_{s=1}^{B}\sum_{p=1}^{15}\sum_{a,b}\left(\widehat{H}_{N,p}^{(s)}[a,b] - H_{N,p}^{(s)}[a,b]\right)^2. \tag{158}$$

The complete endpoint objective is

$$\mathcal{L}_{\mathrm{prop}} = \mathcal{L}_{\mathrm{hist},N} + \beta_{\mathrm{KL}}\mathcal{L}_{\mathrm{KL}} + 10\mathcal{L}_{\log\sigma,N} + 10\mathcal{L}_{m,N}, \tag{159}$$

with

$$\mathcal{L}_{\log\sigma,N} = \mathrm{MSE}(\widehat{\log\sigma}_N, \log\sigma_N), \qquad \mathcal{L}_{m,N} = \mathrm{MSE}(\widehat{\mathbf{m}}_N, \mathbf{m}_N). \tag{160}$$

The KL weight is selected independently for each Transformer window and is linearly warmed up during the first ten epochs. The pretrained CVAE decoder remains frozen during this sweep, while the convolutional encoder and posterior heads are trainable so that β_{KL} has a gradient effect. The tokenizer and latent propagator are trained jointly. In the later physical-statistics-head fine-tuning stage, the complete distribution branch is frozen and only the statistics head is optimized. No sum over intermediate physical checkpoints is present in the implemented objective.

Training loss and reported validation metrics are distinct. Each histogram channel is normalized to unit mass, and the reported relative histogram error is the mean relative L^1 discrepancy

$$\epsilon_{\text{rel}} = \frac{1}{15B} \sum_{s=1}^{B} \sum_{p=1}^{15} \frac{\left\| \widehat{H}_{N,p}^{(s)} - H_{N,p}^{(s)} \right\|_1}{\left\| H_{N,p}^{(s)} \right\|_1 + \varepsilon}. \tag{161}$$

Because the target channels have unit mass, the denominator is approximately one; the metric measures total absolute probability-mass discrepancy per projection. The other retained metrics are summarized in Table 5.

Tab. 8: Metrics used for quantitative comparison of surrogate models.

Metric	Symbol	Interpretation
Relative histogram error	ϵ_{rel} in Eq. (161)	Projected mass discrepancy
Mean absolute centroid error	$\|\widehat{\mathbf{m}} - \mathbf{m}\|_1/6$	Centroid accuracy
Mean RMS-spread error	$\|\hat{\sigma} - \sigma\|_1/6$	Coordinate-scale accuracy
Wasserstein distance	$W_1(\hat{\rho}, \rho)$	Geometric mass displacement
Sliced Wasserstein distance	$\mathrm{SW}(\hat{\rho}, \rho)$	Distributional transport discrepancy
Residual bias	$\mathbb{E}[\widehat{H} - H]$	Systematic reconstruction structure
Inference time	t_{inf}	Neural forward-evaluation cost

4.9 Conditional latent uncertainty estimation

The stochastic latent representation of the CVAE can be propagated through the complete global physical surrogate to obtain an ensemble of predictions for a fixed input beam and fixed machine parameters. This construction is used as a diagnostic of the model's conditional latent variability. It is important to distinguish this quantity from a complete Bayesian treatment of the model, which would require marginalization over the model parameters θ in addition to the latent variable z. The present approach instead uses the stochastic latent representation to quantify uncertainty in the predicted beam distribution for a fixed trained model. This model does not include uncertainty associated with finite training data, model misspecification, or parameter uncertainty. It is therefore a conditional uncertainty estimate rather than a complete Bayesian posterior.

4.10 Posterior sampling and propagation

For an input histogram representation $X_0 = \mathcal{H}(\rho_0)$, and conditioning variables c_0, the encoder defines

$$q_\phi(z_0 \mid X_0, c) = \mathcal{N}\left(\mathbf{m}_\phi(X_0, c), \mathrm{diag}[\sigma_\phi^2(X_0, c)]\right). \tag{162}$$

An ensemble of R initial latent states is generated through the reparameterization

$$z_0^{(r)} = \mathbf{m}_\phi(X_0, c) + \sigma_\phi(X_0, c) \odot \epsilon^{(r)}, \quad \epsilon^{(r)} \sim \mathcal{N}(0, I), \quad r = 1, \dots, R. \tag{163}$$

The machine parameters μ, lattice tokens $\{h_n\}_{n=0}^{N-1}$, and physical conditioning variables are held fixed across the ensemble. Each sample is propagated through the same learned latent dynamics,

$$z_N^{(r)} = \mathcal{T}_\theta^N \left(z_0^{(r)}, \{h_n\}_{n=0}^{N-1}, \mu \right), \tag{164}$$

and decoded into the fifteen normalized pairwise histogram masses,

$$\widehat{H}_{N,p}^{(r)} = D_\psi(z_N^{(r)}, c_N)_p, \qquad p = 0, \dots, 14. \tag{165}$$

The global physical model also returns the six predicted RMS extensions and centroids $\sigma_N^{(r)} b$ and $m_N^{(r)} c$ for each sample. These quantities are used to convert the normalized histogram masses into physical densities.

For a pair $p = (a, b)$, the normalized histogram mass is converted into a physical density by assigning bin edges from the predicted centroid and RMS extension. If $\Delta u_{p,i}^{(r)}$ and $\Delta v_{p,j}^{(r)}$ are the resulting physical bin widths, then

$$\hat{\rho}_{N,p,ij}^{(r)} = \frac{\widehat{H}_{N,p,ij}^{(r)}}{\Delta u_{p,i}^{(r)} \Delta v_{p,j}^{(r)}}. \tag{166}$$

Consequently, variability in the physical-density ensemble contains both decoded shape variability and variability introduced by the physical-statistics head.

4.10.1 Monte-Carlo estimators

For any predicted bin quantity $A_{ij}^{(r)}$, representing either normalized mass or physical density, the empirical predictive mean is

$$\overline{A}_{ij} = \frac{1}{R} \sum_{r=1}^{R} A_{ij}^{(r)}. \tag{167}$$

The population-form Monte-Carlo variance and standard deviation used in the implementation are

$$s_{A,ij}^2 = \frac{1}{R} \sum_{r=1}^{R} \left(A_{ij}^{(r)} - \overline{A}_{ij} \right)^2. \tag{168}$$

The signed and absolute discrepancies of the ensemble mean with respect to the reference are

$$b_{A,ij} = \overline{A}_{ij} - A_{ij}^{\text{true}}, \qquad e_{A,ij} = \left| \overline{A}_{ij} - A_{ij}^{\text{true}} \right|. \tag{169}$$

Spatial averages, maxima, and sums of these fields are then used to summarize the uncertainty distribution over a selected 64×64 marginal.

The same estimators are applied to each physical statistic. For coordinate d,

$$\overline{\sigma}_d = \frac{1}{R} \sum_{r=1}^{R} \hat{\sigma}_d^{(r)}, \qquad s_{\sigma_d}^2 = \frac{1}{R} \sum_{r=1}^{R} \left(\hat{\sigma}_d^{(r)} - \overline{\sigma}_d \right)^2, \tag{170}$$

with analogous definitions for m_d and s_{m_d}.

Empirical central intervals may additionally be obtained from ensemble quantiles,

$$I_{1-\alpha}(A_{ij}) = \left[Q_{\alpha/2}(\{A_{ij}^{(r)}\}_{r=1}^R), Q_{1-\alpha/2}(\{A_{ij}^{(r)}\}_{r=1}^R) \right]. \tag{171}$$

5 Results

5.1 Longitudinal operator learning with FNO

5.1.1 Learning reduced longitudinal profiles

The first experimental block evaluates whether a Fourier Neural Operator can learn a reduced longitudinal map

$$\widehat{\mathcal{T}}_\theta : (\lambda_{\mathrm{in}}, \mu) \mapsto \lambda_{\mathrm{out}}. \tag{172}$$

This task is lower-dimensional than full six-dimensional transport, but remains physically meaningful because longitudinal collective effects are naturally expressed through line densities, wake kernels, and impedance spectra. It also provides a controlled setting in which the operator-learning hypothesis can be tested before moving to the latent six-dimensional problem.

5.1.2 Haïssinski-profile reconstruction

For stationary longitudinal problems, the FNO can be used to approximate the equilibrium map

$$(W_\parallel, I, \mu) \mapsto \lambda_h. \tag{173}$$

This formulation is closely connected to the Haïssinski equation because the equilibrium profile is determined by a nonlinear self-consistent potential. The relevant validation is therefore not limited to pointwise profile accuracy, but also concerns whether the surrogate reproduces derived physical observables such as centroid shift, bunch lengthening, and peak displacement.

The title of the representative prediction reports a discrete relative L^2 profile error of approximately 2.90×10^{-1}. This value is not a 29% pointwise error and is not inconsistent with the close visual overlap of the two curves on a linear vertical scale. The global norm accumulates discrepancies over the complete grid and is sensitive to small profile shifts, width differences, and residual structure that are difficult to judge from the superposed curves alone. Because a final distribution-wide table of FNO test metrics was not retained under

the common evaluation protocol, this example is interpreted qualitatively rather than used for cross-model ranking.

5.2 Six-dimensional latent representation

5.2.1 CVAE reconstruction quality

The next experimental block evaluates whether the fifteen-projection representation can be compressed into a latent variable while preserving physically relevant structure of the beam

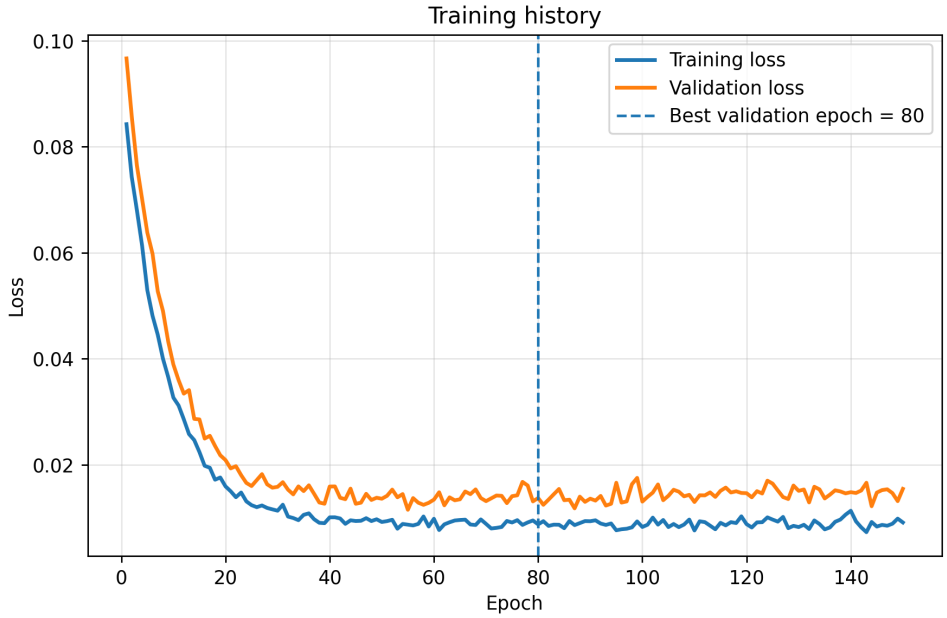


Fig. 5: Training and validation loss histories for the one-dimensional FNO longitudinal surrogate. The curves illustrate the convergence of the optimization procedure over the training epochs and the evolution of the model performance on the validation set.

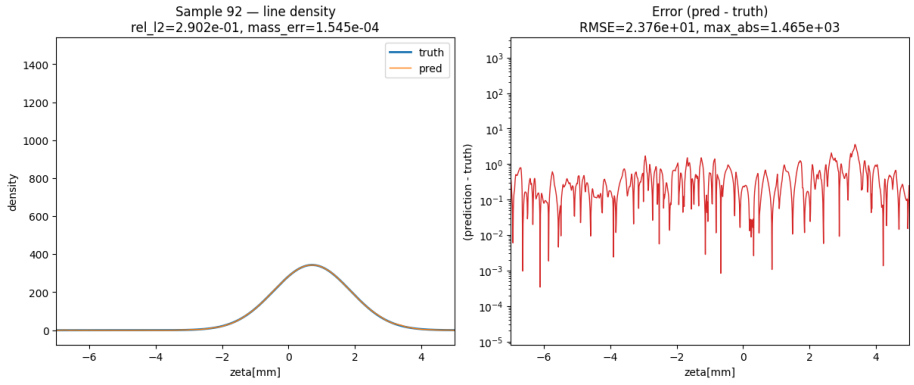


Fig. 6: Representative line density reconstruction showed along logarithmic error. The FNO prediction $\lambda_{\text{out}} b$ is compared with the reference profile λ_{out}. The residual representation, $\lambda^{b}_{\text{out}} - \lambda_{\text{out}}$, provides a complementary view of systematic reconstruction error.

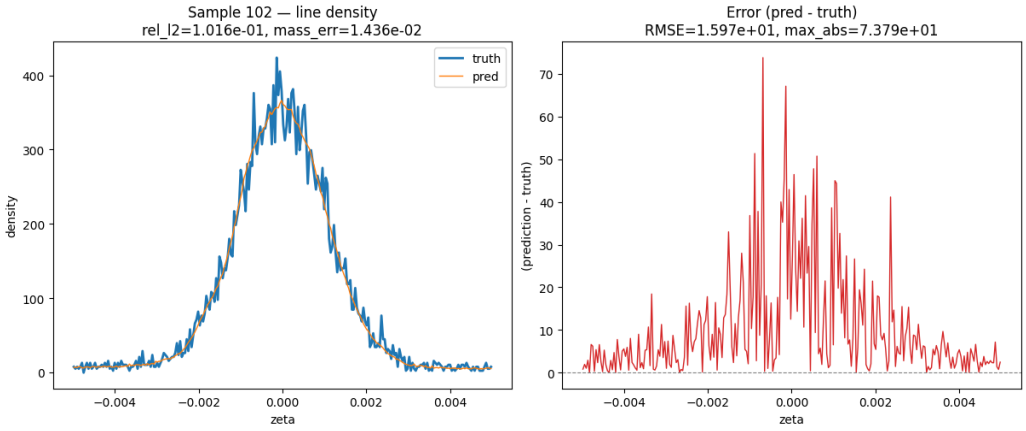


Fig. 7: Representative test-set prediction for the reduced FNO longitudinal surrogate. Randomized noise is added to the sample to illustrate the robustness of the learned operator. The figure compares the reference line density λ_{out} with the FNO prediction $\lambda_{\text{out}}b$ and the residual field $\lambda^b_{\text{out}} - \lambda_{\text{out}}$.

distribution. The CVAE reconstructs the projected histograms from a latent state z and conditioning variables c, while auxiliary physical information may be treated through explicit conditioning or a dedicated physical-dimensions head.

Tab. 9: Quantitative CVAE reconstruction metrics.

Metric	Train	Validation	Test
Histogram relative error	0.362	0.357	0.367
Moment error	0.178	0.243	0.264
KL term	929.275	921.513	920.925
Sliced Wasserstein distance	0.0092	0.0089	0.0096

The standalone CVAE test error of 0.367 quantifies the information lost when an already available beam snapshot is compressed and reconstructed without any lattice propagation. It is therefore a representation baseline for the full pipeline. The downstream CVAE–GNO error of 0.511 contains both this latent-bottleneck contribution and additional endpoint-propagation error; the numerical difference is informative but the two error sources are not algebraically separable.

5.2.2 Latent-space diagnostics

A well-regularized latent space should not simply memorize the training samples. To investigate this point, the latent coordinates are compared with physically interpretable observables. In the present work, Pearson correlations are evaluated between the latent variables and beam statistics including coordinate means and RMS spread extensions. For a physical observable q and latent coordinate z_k, the Pearson correlation coefficient is

$$r_{q,z_k} = \frac{\mathrm{Cov}(q, z_k)}{\sigma_q \sigma_{z_k}}. \tag{174}$$

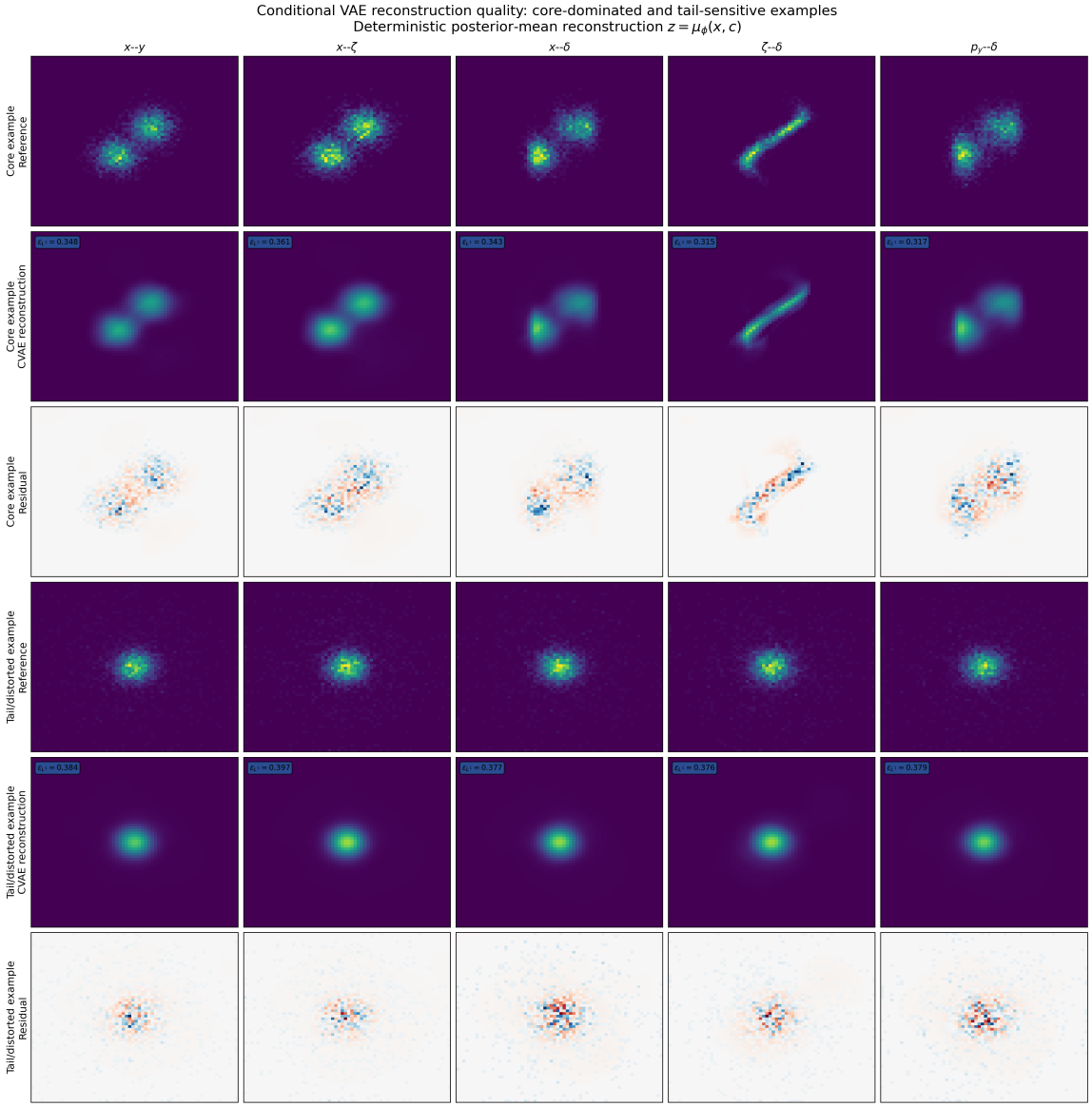


Fig. 8: Conditional VAE reconstruction quality for representative beam distributions. The comparison illustrates both the fidelity of the reconstructed high-density core and the increased difficulty associated with lower-density tails and distorted phase-space structure.

For each physical observable, the twenty latent coordinates with the largest absolute correlations can be displayed,

$$\{z_{k_1}, \dots, z_{k_{20}}\} = \mathrm{TopK}_{20}(|r_{q,z_k}|). \tag{175}$$

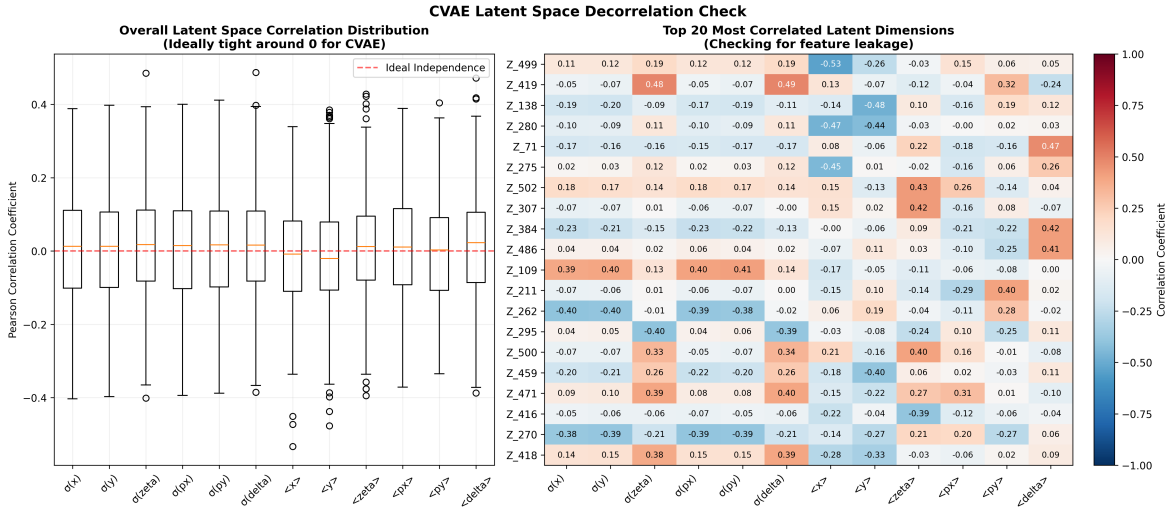


Fig. 9: Pearson-correlation diagnostics between the CVAE latent representation and the auxiliary physical beam statistics. The left panel shows the distribution of signed correlation coefficients across the 512 latent coordinates for each of the 12 physical observables, comprising the six coordinate-wise rms spreads $\sigma = (\sigma_x, \sigma_y, \sigma_\zeta, \sigma_{p_x}, \sigma_{p_y}, \sigma_\delta)$ and the six phase-space centroids $\langle \mathrm{x} \rangle = (\langle x \rangle, \langle y \rangle, \langle \zeta \rangle, \langle p_x \rangle, \langle p_y \rangle, \langle \delta \rangle)$. The right panel displays the full correlation patterns of the 20 latent coordinates attaining the largest absolute correlation with any physical observable. The signed distributions remain concentrated around zero and no single physical statistic produces a systematic displacement of the latent population.

To quantify the behavior shown in Fig. 9, Pearson coefficients were computed between every latent coordinate and every auxiliary physical observable over the validation subset. With $N_{\mathrm{val}} = 51$ samples, a latent dimension $d_z = 512$, and 12 physical observables, the diagnostic contains $512 \times 12 = 6144$ pairwise correlation coefficients. The corresponding global statistics are summarized in Table 7.

The global correlation distribution is tightly concentrated around zero. In particular, the mean and median absolute coefficients are 0.1161 and 0.1006, respectively, while only 18 of the 6144 tested pairs exceed $|r| = 0.4$ and only one exceeds $|r| = 0.5$. The largest individual coefficient,

$$r(z_{499}, \langle x \rangle) = -0.5333,$$

should therefore be interpreted in the context of both the limited validation sample size and the simultaneous inspection of several thousand latent–physical pairs. An isolated extreme

coefficient is not, by itself, evidence that a physical observable is systematically encoded throughout the latent representation.

Tab. 10: Global latent–physical Pearson-correlation statistics for the CVAE. The threshold counts refer to the full set of 6144 latent–physical pairs.

Diagnostic	Value	Fraction of pairs
Mean absolute correlation	0.1161	–
Median absolute correlation	0.1006	–
Standard deviation of $\|r\|$	0.0846	–
Maximum absolute correlation	0.5333	–
$\|r\| \geq 0.20$	1031/6144	16.78%
$\|r\| \geq 0.30$	198/6144	3.22%
$\|r\| \geq 0.40$	18/6144	0.29%
$\|r\| \geq 0.50$	1/6144	0.016%

This interpretation is further supported by the observable-wise statistics in Table 8. The mean absolute correlation remains within the narrow interval

$$0.1098 \leq \frac{1}{d_z} \sum_{j=1}^{d_z} |r(z_j, a_k)| \leq 0.1203$$

for every physical observable a_k. Consequently, no individual beam size or centroid produces a globally enhanced response across the latent coordinates.

Tab. 11: Observable-wise Pearson-correlation summary on the validation set. For each physical statistic, the table reports the mean and median absolute Pearson correlation over all 512 latent coordinates, together with the strongest individual signed coefficient. These statistics quantify linear association only and do not test general nonlinear dependence.

Observable	Mean $\|r\|$	Median $\|r\|$	Max. $\|r\|$	Strongest pair
σ_x	0.1203	0.1078	0.4030	$r(z_{262}, \sigma_x) = -0.4030$
σ_y	0.1199	0.1047	0.3974	$r(z_{262}, \sigma_y) = -0.3974$
σ_ζ	0.1174	0.1013	0.4849	$r(z_{419}, \sigma_\zeta) = +0.4849$
σ_{p_x}	0.1200	0.1062	0.4002	$r(z_{109}, \sigma_{p_x}) = +0.4002$
σ_{p_y}	0.1201	0.1052	0.4112	$r(z_{109}, \sigma_{p_y}) = +0.4112$
σ_δ	0.1172	0.0981	0.4867	$r(z_{419}, \sigma_\delta) = +0.4867$
$\langle x \rangle$	0.1157	0.1003	0.5333	$r(z_{499}, \langle x \rangle) = -0.5333$
$\langle y \rangle$	0.1170	0.0934	0.4767	$r(z_{138}, \langle y \rangle) = -0.4767$
$\langle \zeta \rangle$	0.1098	0.0861	0.4278	$r(z_{502}, \langle \zeta \rangle) = +0.4278$
$\langle p_x \rangle$	0.1143	0.1023	0.3891	$r(z_{381}, \langle p_x \rangle) = +0.3891$
$\langle p_y \rangle$	0.1107	0.0994	0.4038	$r(z_{211}, \langle p_y \rangle) = +0.4038$
$\langle \delta \rangle$	0.1114	0.0954	0.4716	$r(z_{71}, \langle \delta \rangle) = +0.4716$

The strongest Pearson coefficients do not reveal a single physical observable that is uniformly aligned with the latent representation. Instead, the largest coefficients involve different transverse and longitudinal centroids and coordinate-wise RMS spreads, and are distributed across several latent coordinates. For example, the strongest individual associations involve $\langle x \rangle, \sigma_\delta, \langle y \rangle, \langle \delta \rangle, \langle \zeta \rangle$, and σ_{p_y}. This heterogeneity indicates that the largest observed linear

associations are not concentrated on one physical statistic or one latent coordinate. Most importantly, Pearson correlation measures linear association only and therefore cannot rule out nonlinear dependence. The present result should therefore be interpreted only as evidence against a simple, pervasive linear alignment, not as proof of statistical independence or of nonlinear disentanglement.

Overall, the diagnostic provides no evidence of systematic linear leakage of the explicitly supplied beam sizes and centroids into the CVAE latent representation at the resolution allowed by the present validation sample. The result should nevertheless be interpreted as a test of linear dependence rather than a proof of full statistical independence as Pearson coefficients do not exclude nonlinear dependence, and the relatively small value $N_{\mathrm{val}} = 51$ limits the precision of individual correlation estimates. Within these limitations, the observed concentration around zero supports the intended separation between the latent morphological representation and the explicit physical beam statistics.

This result is compatible with the intended role of the CVAE latent representation. The latent coordinates are not required to coincide individually with specific physical observables. Rather, they are expected to provide a distributed reduced representation of the projected density structure. The fact that only a limited subset of latent coordinates carries moderate information about individual observables suggests sparse physical alignment rather than a trivially encoded hidden parameterization.

Complete disentanglement is neither expected nor necessarily desirable. The relevant question is whether the latent state preserves the dynamical degrees of freedom required for propagation while avoiding excessive leakage of scale and position information that can be handled explicitly through physical conditioning variables or the auxiliary physical-dimensions head.

5.3 Transformer latent propagation and the effect of locality

5.3.1 Global-attention baseline

The first sequence-model baseline investigates whether an expressive Transformer can propagate the CVAE latent representation through the ordered lattice description. In this formulation, the causal mask prevents future or downstream tokens from influencing earlier positions, thereby enforcing upstream-to-downstream information flow. A global causal mask does not, however, impose finite interaction support: the state associated with a given position may, in principle, aggregate information from every preceding token.

This distinction is relevant because the beam state results from an ordered composition of lattice transformations. Although long-range dependence may be physically meaningful, unrestricted upstream mixing can introduce an architectural bias that differs from a predominantly local composition of element-wise transport as seen in 10. The no-window Transformer diagnostics provide a qualitative illustration of this effect.

Let H_{ij} denote a reference two-dimensional marginal and $H_{ij}b$ its prediction. The residual field is

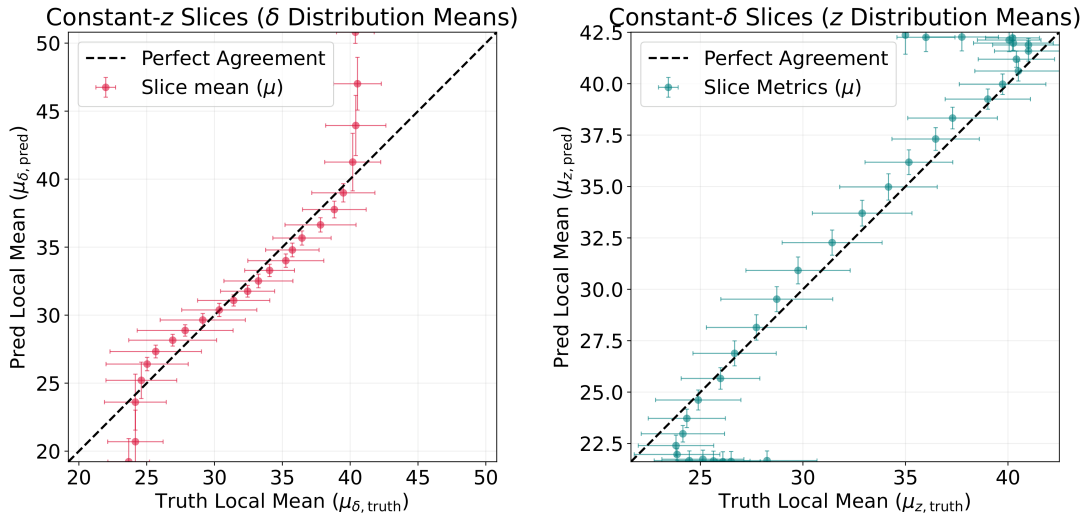


Fig. 10: Conditional-slice reconstruction artifacts obtained with the no-window Transformer baseline for the longitudinal (ζ, δ) marginal. Predicted conditional means for constant $-\,\zeta$ and constant $-\,\delta$ slices are compared with their reference values. Systematic departures from the ideal diagonal and increased scatter indicate coherent distortion of the reconstructed local phase-space structure. The figure is interpreted as a qualitative diagnostic and does not establish global attention as the unique cause of the observed error.

$$\varepsilon_{ij}(u, v) = \widehat{H}_{ij}(u, v) - H_{ij}(u, v). \tag{176}$$

A systematic reconstruction artifact corresponds to a nonzero mean residual

$$b_{ij}(u, v) = \mathbb{E}[\varepsilon_{ij}(u, v)] \neq 0. \tag{177}$$

At the latent level, an accumulated error mechanism can be represented schematically as

$$z_{n+1}^{\text{true}} = z_n + \Delta z_{\theta,n} + \eta_n, \tag{178}$$

which gives, after N propagation steps,

$$z_N^{\text{true}} = z_0 + \sum_{n=0}^{N-1} \Delta z_{\theta,n} + \sum_{n=0}^{N-1} \eta_n. \tag{179}$$

If the learned local error has a systematic component,

$$\mathbb{E}[\eta_{\theta,n}] \neq 0, \tag{180}$$

the decoded rollout can develop a coherent distortion rather than a zero-centered random error. This mechanism does not prove that unrestricted attention is uniquely responsible for the observed artifacts. It nevertheless motivates the controlled locality study introduced next.

5.3.2 Local-window Transformer

To test whether unrestricted upstream mixing contributes to the observed reconstruction error, the Transformer attention support is restricted through the local causal mask introduced in Eq. (154). In the principal Transformer–GNO comparison presented below, the Transformer implementation uses an attention window $w = 8$. Thus, the hidden state at sequence position i can only aggregate information from a finite upstream neighborhood.

The effect of this restriction is evaluated first through the longitudinal marginal (ζ, δ), which is particularly sensitive to the local correlation structure of the beam. The conditional-slice diagnostic introduced in Section 5.3.1 is applied to the local-window Transformer reconstruction. This marginal is central in the study of longitudinal collective effects, which are the primary physical motivation for the present work.

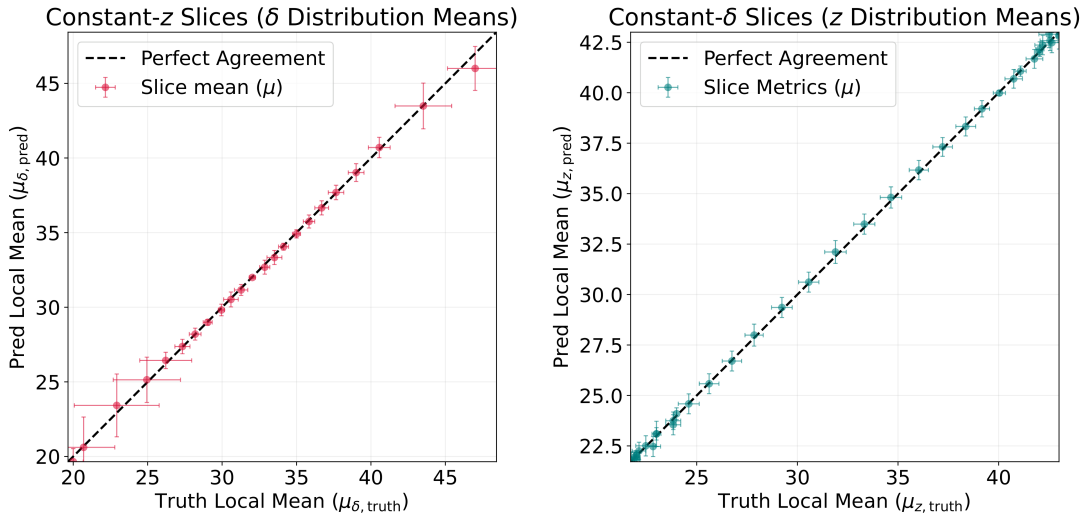


Fig. 11: Conditional-slice consistency of the local-window Transformer for the longitudinal (ζ, δ) marginal. The diagnostic compares predicted and reference conditional means for constant $-\,\zeta$ and constant $-\,\delta$ slices. Error bars encode local conditional spread. Concentration around the ideal parity relation indicates preservation of internal longitudinal phase-space structure beyond agreement of global moments.

The local-window result is important because the architectural modification does not change the underlying CVAE representation or the physical prediction target. Instead, it changes the support of the interaction mechanism used during latent propagation. The resulting comparison therefore isolates, to first order, the role of locality as an inductive bias.

5.4 CVAE with Graph Neural Operator propagation

5.4.1 Consistency in longitudinal dynamics

The Graph Neural Operator provides the second latent propagation mechanism. In contrast with attention-based sequence mixing, the GNO represents information transfer through a learned kernel evaluated over the graph neighborhood introduced in Section 4.5. The same longitudinal conditional diagnostic is applied to the GNO reconstruction.

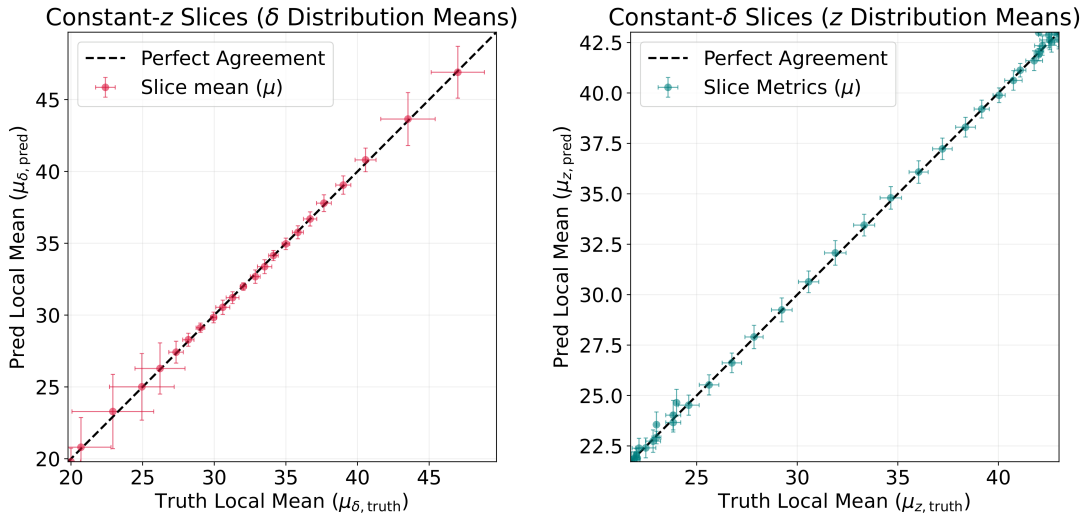


Fig. 12: Conditional-slice consistency of the CVAE–GNO surrogate for the longitudinal (ζ, δ) marginal. The figure evaluates agreement of local conditional means under constant $-\zeta$ and constant $-\delta$ slicing. Comparison with Fig. 11 isolates differences in internal phase-space reconstruction between local-window attention and graph-kernel propagation.

The conditional analysis complements global histogram errors. Two models may attain similar integrated reconstruction losses while distributing their residuals differently across the support of the beam. Conditional means therefore provide a direct probe of whether correlations between longitudinal position and energy deviation are preserved locally.

5.5 Direct comparison of local-window Transformer and GNO

5.5.1 Two-dimensional marginal reconstruction

A direct comparison is performed on the same selected sample, $s = 3$, used in the previous conditional diagnostics. The sample is evaluated at the same four representative pairwise phase-space projections, which are displayed in Figs. 13 and 14. The comparison is performed at the histogram level, with residuals defined as

The residuals are evaluated as

$$\Delta H_{\mathrm{TF}} = \widehat{H}_{\mathrm{TF}} - H, \qquad \Delta H_{\mathrm{GNO}} = \widehat{H}_{\mathrm{GNO}} - H. \tag{181}$$

The selected marginals probe complementary structures. The (x, y) distribution tests coupling between transverse coordinates; (x, ζ) and (x, δ) probe mixed correlations between transverse and longitudinal degrees of freedom; and (ζ, δ) directly tests reconstruction of longitudinal phase-space structure. Consequently, agreement across these projections provides a stronger diagnostic than evaluation of a single preferred plane.

Qualitatively, the comparison should be interpreted through both residual amplitude and residual organization. A model may exhibit a moderate pointwise error while remaining approximately unbiased, whereas coherent positive and negative residual lobes reveal systematic deformation of the predicted distribution. The latter is particularly relevant when evaluat-

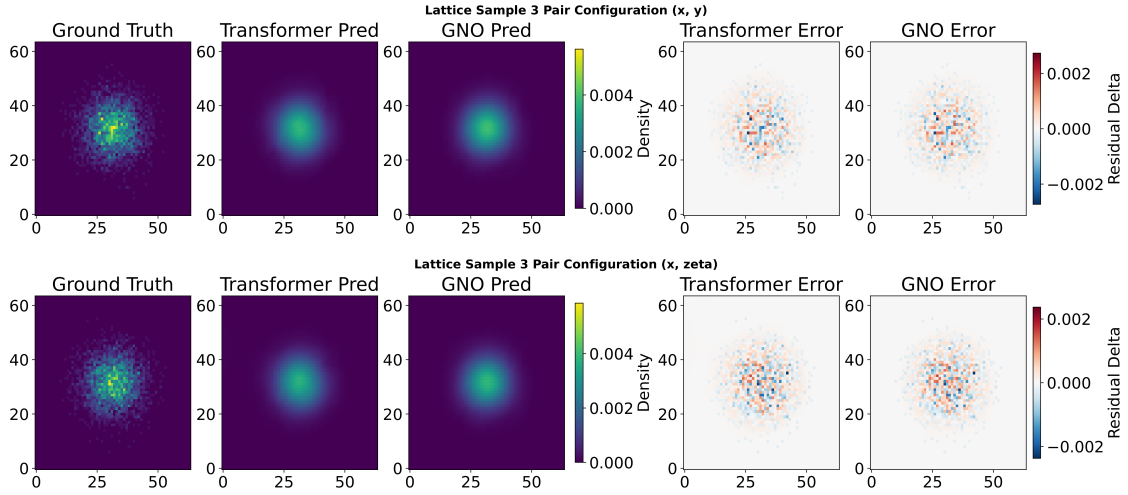


Fig. 13: Direct density-level comparison of the local-window Transformer and GNO for sample $s = 3$. Top: transverse cross-plane marginal (x, y), pair $p = 0$. Bottom: mixed transverse– longitudinal marginal (x, ζ), pair $p = 1$. Each row displays the reference density, both model predictions, and their residual fields.

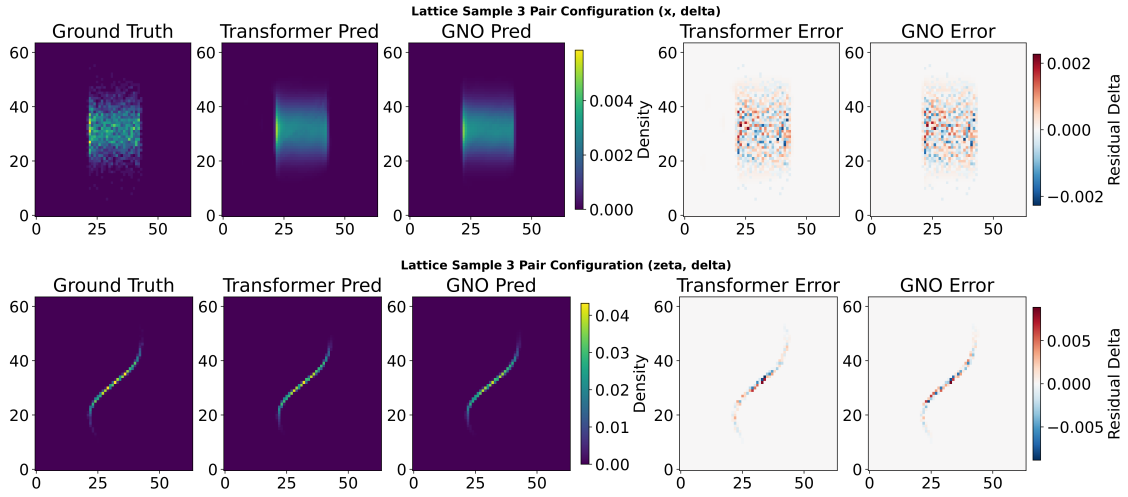


Fig. 14: Direct density-level comparison of the local-window Transformer and GNO for sample $s = 3$. Top: mixed transverse–energy marginal (x, δ), pair $p = 4$. Bottom: longitudinal phase-space marginal (ζ, δ), pair $p = 11$. The common residual representation exposes structured model error that may be obscured by scalar reconstruction metrics.

ing whether the surrogate preserves the location, orientation, and extension of phase-space structures.

The selected marginals probe complementary structures. The (x, y) distribution tests coupling between transverse coordinates, (x, ζ) and (x, δ) probe mixed transverse–longitudinal correlation, and (ζ, δ) directly tests reconstruction of longitudinal phase-space structure. Agreement across these projections therefore provides a more informative qualitative diagnostic than evaluation on a single preferred plane.

Both the local-window Transformer and the CVAE–GNO reproduce the main location, orientation, and spatial extension of the distributions in the examples shown. On the basis of these selected marginals alone, neither model can be declared systematically superior. The comparison should instead distinguish between residual amplitude and residual organization. A model may exhibit moderate pointwise error while remaining approximately unbiased, whereas coherent positive and negative residual lobes indicate a systematic deformation of the reconstructed density.

A quantitative distinction between the two propagators therefore requires evaluation over the complete held-out test set rather than visual inspection of individual examples. The relevant criteria include binwise reconstruction error, transport-based discrepancy, systematic bias, and end-to-end inference time. These quantities are compared globally in Table 10. Under the reported evaluation conditions, the CVAE–GNO provides the lower aggregate distributional errors, the lower measured bias norm, and the shorter inference time, whereas the local-window Transformer remains competitive in its reconstruction of individual marginals. Robustness to perturbations, changes in lattice configuration, or out-of-distribution parameters is not established by the present comparison and would require a dedicated study.

5.5.2 Macroparticle-based distributional diagnostics

Histogram residuals are complemented by a macroparticle-based analysis. Each predicted two-dimensional density is interpreted as a discrete probability distribution and sampled to generate a large synthetic particle ensemble. In the implemented diagnostic, N_{sample} macroparticles are drawn from each of the reference, Transformer, and GNO distributions.

The sampled particle coordinates are transformed back to the corresponding physical range using the reference cloud bounds. Sorted coordinate samples are then compared directly. For a coordinate q, the residuals are

$$r_{\mathrm{TF}}^{(q)} = q_{\mathrm{TF}} - q_{\mathrm{truth}}, \qquad r_{\mathrm{GNO}}^{(q)} = q_{\mathrm{GNO}} - q_{\mathrm{truth}}. \tag{182}$$

The resulting dashboards combine parity plots, residual scatter, and empirical residualdensity summaries.

Histogram-level error does not directly quantify the physical displacement of probability mass, therefore, sampling transforms the reconstructed density into a representation closer to the macroparticle description used by tracking simulations. As a result, it provides an intermediate diagnostic between image-based reconstruction and full six-dimensional particlelevel validation.

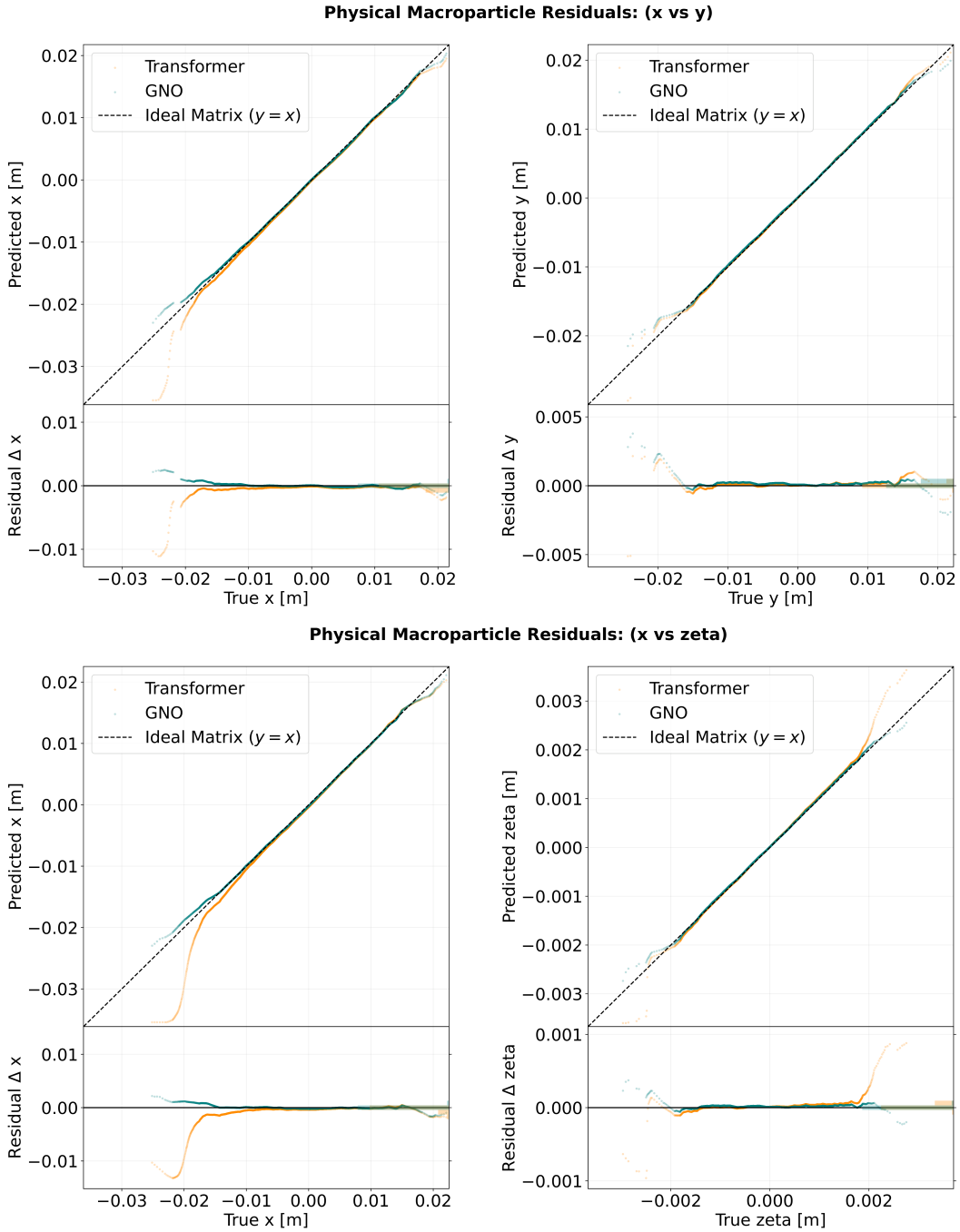


Fig. 15: Macroparticle-based comparison for sample $s = 3$. Top: (x, y) projection. Bottom: (x, ζ) projection. Synthetic particles are sampled from the reference, local-window Transformer, and GNO density maps. Parity and residual diagnostics reveal distributional displacement in physical coordinates.

5.5.3 Windowed attention and graph-kernel correspondence

The comparison between the Transformer and GNO models admits a more precise interpretation than a generic appeal to locality. At the level of a single attention layer, causal windowed self-attention can be written exactly as message passing on a directed graph whose edges are determined by the attention mask. The resulting update is therefore structurally close to a graph-kernel operator acting on the latent token field.

Let $h_i^{(\ell)} \in \mathbb{R}^d$ denote the hidden representation associated with the $i-$th lattice token at layer ℓ. A causal attention window of width w defines the directed graph

$$\mathcal{G}_w = (\mathcal{V}, \mathcal{E}_w), \tag{183}$$

with

$$\mathcal{E}_w = \{(j,i) : 0 \le i - j \le w\}. \tag{184}$$

Equivalently, the incoming neighbourhood of node i is

$$\mathcal{N}_w^-(i) = \{j : \max(1, i-w) \le j \le i\}. \tag{185}$$

Thus, the causal window explicitly replaces the complete causal graph by a directed banded graph, $\mathcal{G}_w$.

For one attention head, define

$$q_i^{(\ell)} = W_Q^{(\ell)} h_i^{(\ell)}, \qquad k_j^{(\ell)} = W_K^{(\ell)} h_j^{(\ell)}, \qquad v_j^{(\ell)} = W_V^{(\ell)} h_j^{(\ell)}. \tag{186}$$

The masked attention coefficients are

$$\alpha_{ij}^{(\ell,w)} = \frac{\mathbf{1}_{j \in \mathcal{N}_w^-(i)} \exp\left(\dfrac{q_i^{(\ell)\top} k_j^{(\ell)}}{\sqrt{d_k}}\right)}{\displaystyle\sum_{r \in \mathcal{N}_w^-(i)} \exp\left(\dfrac{q_i^{(\ell)\top} k_r^{(\ell)}}{\sqrt{d_k}}\right)}. \tag{187}$$

The attention message received by token i is therefore

$$m_i^{(\ell,\mathrm{TF})} = \sum_{j \in \mathcal{N}_w^-(i)} \alpha_{ij}^{(\ell,w)} W_V^{(\ell)} h_j^{(\ell)}. \tag{188}$$

After output projection and residual addition,

$$\tilde{h}_i^{(\ell+1)} = h_i^{(\ell)} + W_O^{(\ell)} m_i^{(\ell,\mathrm{TF})}. \tag{189}$$

Equations (188)–(189) are precisely a message-passing update on $\mathcal{G}_w$, each node aggregates messages only from its incoming graph neighbourhood.

This correspondence becomes explicit by defining a state-dependent matrix kernel

$$K_{ij}^{(\ell,\mathrm{att})}(\mathbf{h}^{(\ell)}) = \alpha_{ij}^{(\ell,w)} W_O^{(\ell)} W_V^{(\ell)}. \tag{190}$$

Then the complete aggregation can be written as

$$\tilde{h}_i^{(\ell+1)} = h_i^{(\ell)} + \sum_{j=1}^{N} K_{ij}^{(\ell,\mathrm{att})}(\mathbf{h}^{(\ell)}) h_j^{(\ell)}. \tag{191}$$

Because

$$K_{ij}^{(\ell,\mathrm{att})} = 0 \qquad \text{for } j \notin \mathcal{N}_w^-(i), \tag{192}$$

the kernel has compact support on the directed latent graph induced by the causal window. For H attention heads, the same construction gives

$$K_{ij}^{(\ell,\mathrm{MHA})}(\mathbf{h}^{(\ell)}) = \mathbf{1}_{j\in\mathcal{N}_w^-(i)} \sum_{a=1}^{H} \alpha_{ij}^{(\ell,a)} W_O^{(\ell,a)} W_V^{(\ell,a)}, \tag{193}$$

so that multi-head attention is naturally interpreted as a sum of learned graph kernels,

$$\tilde{h}_i^{(\ell+1)} = h_i^{(\ell)} + \sum_{j=1}^{N} K_{ij}^{(\ell,\mathrm{MHA})}(\mathbf{h}^{(\ell)}) h_j^{(\ell)}. \tag{194}$$

The GNO admits an analogous representation at the message passing level. A generic graph-kernel layer may be written as

$$\tilde{h}_i^{(\ell+1)} = h_i^{(\ell)} + \sum_{j\in\mathcal{N}_{\mathrm{GNO}}(i)} \omega_{ij} K_\theta^{(\ell)}(\xi_i, \xi_j, h_i^{(\ell)}, h_j^{(\ell)}, \mu) h_j^{(\ell)}, \tag{195}$$

where ξ_i and ξ_j denote node or lattice coordinates, μ contains global conditioning variables, and ω_{ij} may represent graph or quadrature weights. The GNO therefore performs message passing through an explicitly parameterized kernel, whereas the Transformer constructs its kernel implicitly from query–key compatibility.

Both architectures can consequently be embedded in the common discrete operator form

$$(\mathcal{T}_\Theta \mathbf{h})_i = h_i + \sum_{j=1}^{N} A_{ij} K_\Theta(i, j, h_i, h_j, \mu) h_j, \tag{196}$$

where A_{ij} specifies the support of information transfer. For the causal windowed Transformer,

$$A_{ij}^{\mathrm{TF}} = \mathbf{1}_{0\le i-j\le w}, \tag{197}$$

while for the GNO,

$$A_{ij}^{\mathrm{GNO}} = \mathbf{1}_{j\in\mathcal{N}_{\mathrm{GNO}}(i)}. \tag{198}$$

The principal architectural difference is therefore not whether message passing occurs, but how the kernel and graph support are parameterized.

For the Transformer, the effective kernel is a normalized, state-dependent kernel,

$$K_{ij}^{\mathrm{TF}} \propto \frac{\exp\left(\frac{q_i^\top k_j}{\sqrt{d_k}}\right)}{\sum_{r\in\mathcal{N}_w^-(i)} \exp\left(\frac{q_i^\top k_r}{\sqrt{d_k}}\right)}, \tag{199}$$

whereas the GNO learns a more general kernel directly,

$$K_{ij}^{\mathrm{GNO}} = K_\theta(\xi_i, \xi_j, h_i, h_j, \mu). \tag{200}$$

Hence, attention may be viewed as a constrained graph-kernel parameterization at its core. The interaction weights are generated by query–key similarity and are normalized over the local neighbourhood, while the GNO kernel can depend explicitly on geometry, edge information, latent states, and physical conditioning variables.

The same relation appears in a continuum interpretation. Let the ordered lattice coordinate be s, let $h(s)$ denote the corresponding latent field, and let

$$r_w = w\Delta s \tag{201}$$

be the physical extent of the attention window. A one-sided local attention operator formally approaches

$$(\mathcal{K}_{\mathrm{att}}[h])(s) = \frac{\displaystyle\int_{s-r_w}^{s} \exp\left(\frac{q(h(s))^\top k(h(\sigma))}{\sqrt{d_k}}\right) Vh(\sigma)\,\mathrm{d}\sigma}{\displaystyle\int_{s-r_w}^{s} \exp\left(\frac{q(h(s))^\top k(h(\sigma))}{\sqrt{d_k}}\right) \mathrm{d}\sigma}. \tag{202}$$

This is a nonlinear integral operator with compact, causal support. By comparison, a local graph neural operator has the form

$$(\mathcal{K}_{\mathrm{GNO}}[h])(s) = \int_{B_r^-(s)} \kappa_\theta(s, \sigma, h(s), h(\sigma), \mu) h(\sigma)\,\mathrm{d}\nu(\sigma), \tag{203}$$

where $\mathcal{B}_r^-(s)$ denotes a local causal neighbourhood. The operators in Equations (202) and (203) differ primarily in kernel parameterization and normalization, rather than in their basic operator structure.

An additional consequence concerns depth. Although a single windowed layer only receives information from $\mathcal{N}_w^-(i)$, repeated layers propagate information through successive graph hops. In the absence of other long-range connections, after L layers the maximal causal receptive field satisfies

$$\mathcal{N}_w^{-,(L)}(i) \subseteq \{j : 0 \le i - j \le Lw\}. \tag{204}$$

Thus, depth in the windowed Transformer plays the same qualitative role as repeated message-passing iterations in a graph network. In this case, local interactions compose to generate longer-range transport without introducing direct all-to-all coupling at every layer.

The empirical tendency of the local-window Transformer to approach the GNO prediction is therefore consistent with a sharper structural interpretation. The dense causal Transformer operates on an almost complete directed upstream graph, whereas the windowed Transformer replaces this graph by a sparse banded one. The resulting latent dynamics become closer to repeated local message passing, which is the basic computational mechanism of the GNO. In this sense, the observed agreement between the two models may indicate that the relevant beam evolution is better represented by composition of local latent interactions than by unrestricted upstream mixing.

The correspondence is not an exact equivalence of the full networks. Transformer kernels are softmax-normalized and state-dependent, the GNO may employ unnormalized matrix-valued kernels and explicit geometric information, and the Transformer additionally contains residual paths, normalization layers, and pointwise feed-forward blocks. Nevertheless, at the level of the non-local interaction operator, the windowed Transformer and the GNO belong to closely related message-passing constructions. This motivates the interpretation of the attention window not merely as a reduction in model size, but as a change in the graph topology and therefore in the inductive bias imposed on latent beam transport.

5.6 Window-size dependence

To investigate the locality hypothesis directly, Transformer models are trained over attention windows $w \in [1, 32]$. The smallest values enforce strongly local information transfer, while sufficiently large windows approach dense causal attention over the available sequence length.

For each window, the same validation protocol was applied. Let

$$E_p(w) = \frac{\left\| \widehat{H}_p^{(w)} - H_p \right\|_1}{\|H_p\|_1 + \varepsilon} \tag{205}$$

denote the relative error for projection p. A global window-dependent score may be defined as

$$\overline{E}(w) = \frac{1}{15} \sum_{p=1}^{15} E_p(w), \tag{206}$$

with a worst-projection score

$$E_{\max}(w) = \max_p E_p(w). \tag{207}$$

The window study provides a validation-based model-selection test of the locality interpretation developed above. Lower errors for intermediate windows than for very narrow or broad windows are consistent with a finite effective interaction scale, while the absence of repeated-seed estimates prevents a statistical claim about the exact minimizing window.

Figure 16 and Table 9 show that the dependence on the attention window is nonmonotonic. The smallest window, $w = 1$, yields the largest errors, indicating that strictly one-step latent interactions do not provide sufficient context to reconstruct the downstream phase-space structure. Increasing the receptive field to $w = 2$ and $w = 4$ improves both the average

reconstruction and the most difficult projection, with the lowest observed errors obtained at $w = 8$ in this run. This behaviour suggests that the latent dynamics require information to propagate across several neighbouring lattice tokens rather than through isolated element-wise updates.

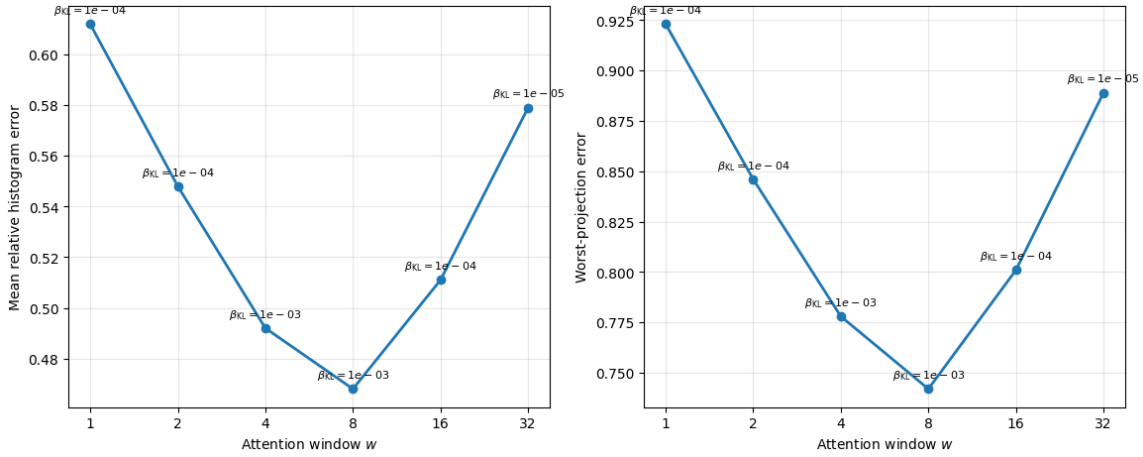


Fig. 16: Transformer window-size study after independent selection of the KL regularization weight β_{KL} for each causal attention window w. The left panel reports the mean relative L^1 histogram error across the reconstructed phase-space projections, while the right panel shows the corresponding worst-projection error. In this single-seed sweep, the lowest observed values occur at $w = 8$; no seed-to-seed error bars are available.

Tab. 12: Selected Transformer configurations for the window-size study. For each attention window w, the KL weight β_{KL} was selected independently according to validation performance. Values correspond to one seeded training sweep; $w = 8$ has the lowest observed validation errors, but the ranking is not accompanied by repeated-seed uncertainty.

w	β_{KL}	Selected epoch	Mean rel. error	Worst-proj. error	Mean KL
1	10^{-4}	18	0.612	0.923	0.418
2	10^{-4}	24	0.548	0.846	0.356
4	10^{-3}	31	0.492	0.778	0.238
8	10^{-3}	34	0.468	0.742	0.211
16	10^{-4}	27	0.511	0.801	0.304
32	10^{-5}	21	0.579	0.889	0.447

The trend reverses for broader windows. At $w = 16$ and $w = 32$, both error metrics increase again despite the larger amount of information available to the attention mechanism. This observation is consistent with the interpretation developed above: although a Transformer can exploit long-range and eventually near-global mixing, unrestricted access to increasingly distant upstream tokens is not necessarily advantageous for the present transport problem. Beyond the intermediate interaction scale, distant latent states may contribute weakly relevant or competing information to the attention aggregation, thereby reducing the locality bias that was beneficial at smaller windows.

The optimum near $w = 8$ therefore reflects a balance between two limiting regimes.

A window that is too narrow prevents the model from representing dependencies extending across multiple lattice elements, whereas a window that is too broad progressively approaches global upstream mixing and weakens the structured local propagation identified previously. In view of the graph-kernel interpretation of windowed attention, the result is consistent with latent beam transport being most effectively represented by repeated finite-range interactions whose composition generates longer-range dependencies through depth, rather than by direct coupling to the entire causal history at every layer.

5.7 Physics-statistics prediction

Distributional reconstruction should be complemented by direct prediction of physically interpretable beam statistics. For the six-dimensional state, ξ, the principal first- and second-order statistics are the mean m_d and standard deviation σ_d of each coordinate d. A dedicated statistics head can be evaluated through parity relations $m_d b \approx m_d$ and $\sigma_d b \approx \sigma_d$, together with coordinate-wise biases

$$B_{m_d} = \mathbb{E}[\widehat{m}_d - m_d], \qquad B_{\sigma_d} = \mathbb{E}[\hat{\sigma}_d - \sigma_d]. \tag{208}$$

Therefore the statistics predictor provides a direct test of whether the latent propagation preserves physically relevant first- and second-order beam observables.

The statistics predictor therefore probes information complementary to the histogram decoder and directly tests whether latent propagation preserves beam-scale observables.

5.8 Global comparison of latent propagators

The latent propagators were evaluated on the same held-out endpoint test set under the reported lattice-token generation protocol. The comparison distinguishes the no-window Transformer, local-window Transformer, and GNO.

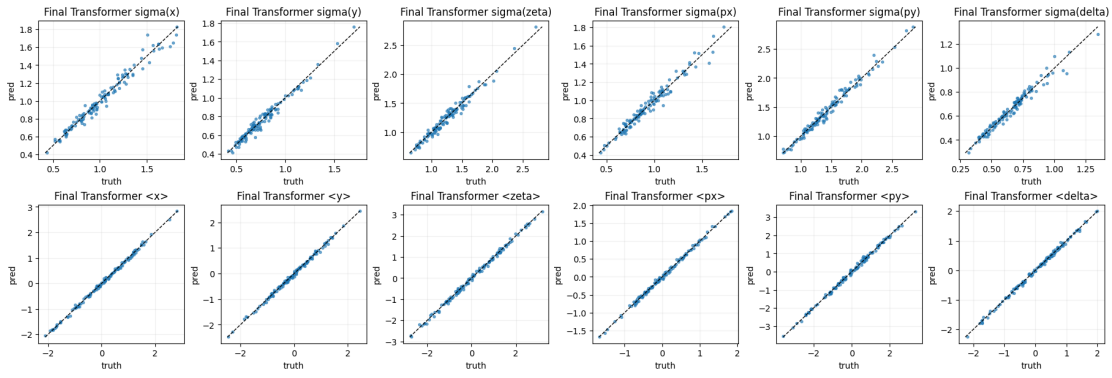


Fig. 17: Parity comparison for the physics-statistics prediction head. Predicted beam centroids and RMS spread extensions are compared with their reference values across the six phase-space coordinates. Agreement with the diagonal indicates accurate reconstruction of the corresponding first- and second-order beam statistics, while deviations from the $y = x$ line quantify systematic bias and prediction error.

Tab. 13: Comparison of the standalone representation and latent propagators. The CVAE-only row is a snapshot-reconstruction baseline and is not a propagation model. The propagator rows use the common held-out endpoint test set. The local-window Transformer uses the selected $w = 8$ checkpoint. Values are selected single-seed results and do not include seed-to-seed error bars.

Model	ϵ_{rel}	SW	Bias norm	τ [ms/sample]
CVAE only, no propagation	0.367	0.0096	–	–
Transformer, no window	0.681	0.087	17.337	18.74
Transformer, $w = 1$	0.650	0.026	52.389	14.35
Windowed Transformer ($w = 8$)	0.534	0.022	50.414	12.01
CVAE–GNO	0.511	0.014	0.376	2.338

The quantitative comparison is consistent with the locality-based interpretation developed above, within the variability of the selected runs. Relative to the no-window Transformer, restricting causal attention to $w = 8$ reduces the mean relative histogram error from 0.681 to 0.534 and the sliced-Wasserstein discrepancy from 0.0867 to 0.022. The improvement is therefore observed both at the binwise density level and in a transport-based comparison of the reconstructed distributions.

Among the selected propagator checkpoints, the CVAE–GNO has the lowest observed distribution errors, with a relative histogram error of 0.511 and a sliced-Wasserstein discrepancy of 0.014. The CVAE-only row shows that 0.367 relative error is already introduced by snapshot compression and reconstruction; the full-pipeline value should therefore not be attributed to the propagator alone. This ordering is consistent with the earlier structural interpretation of the models. Dense causal attention allows broad upstream mixing at every layer, whereas the local-window Transformer restricts the support of direct information exchange. The GNO goes further by encoding the interaction through an explicit graph-kernel construction. The results therefore suggest that constraining the geometry of latent information transfer provides a useful inductive bias for the present beam-transport problem.

The comparison should not be interpreted as evidence that the underlying collective beam dynamics are strictly local. Rather, the improvement obtained by the windowed Transformer indicates that unrestricted direct mixing is not required to represent the relevant effective dynamics. Longer-range dependencies may instead emerge through the composition of successive local interactions across depth, in close analogy with repeated graph message passing.

The model table reports one selected checkpoint per configuration. Since the training sweep was not repeated over several independent random seeds, no uncertainty interval is available for the differences between architectures or attention windows. The numerical ordering is therefore descriptive and should not be interpreted as a statistically resolved ranking.

The timing measurements also reveal a substantial computational difference. The Transformer variants require approximately 12–19 ms per sample under the present neural evaluation protocol, while the GNO requires approximately 2.34 ms per sample. Within that common neural benchmark, the graph-based propagator is about eight times faster than

the global Transformer. Comparing 2.338 ms with the exploratory Xsuite generation time of 1.41 s gives a nominal ratio of approximately 603, but this is not a controlled simulator speed-up because the simulator and network measurements were not obtained under identical hardware, batching, and numerical-precision conditions.

5.9 Conditional latent uncertainty of the CVAE–GNO surrogate

The final experimental block propagates stochastic samples from the conditional CVAE posterior through the complete global physical CVAE–GNO model. Unlike the deterministic comparisons presented above, this analysis keeps the input beam, machine parameters, lattice tokens, and trained network weights fixed while repeatedly sampling the initial latent state. The resulting ensemble therefore measures conditional latent variability, as defined in Section 4.9, rather than the full epistemic uncertainty of the surrogate.

5.9.1 Fixed-condition ensemble propagation

A first experiment was performed for one validation sample using $R = 100$ stochastic forward passes. Each realization produced a 64×64 normalized-mass histogram, a corresponding physical-density histogram, six RMS spread extensions, and six centroids. The complete ensemble shapes were therefore $100 \times 64 \times 64$ for each selected marginal and 100×6 for each physical-statistics branch.

Table 11 summarizes the per-bin ensemble spread and the discrepancy between the ensemble mean and the reference distribution. In normalized-mass units, the mean predictive

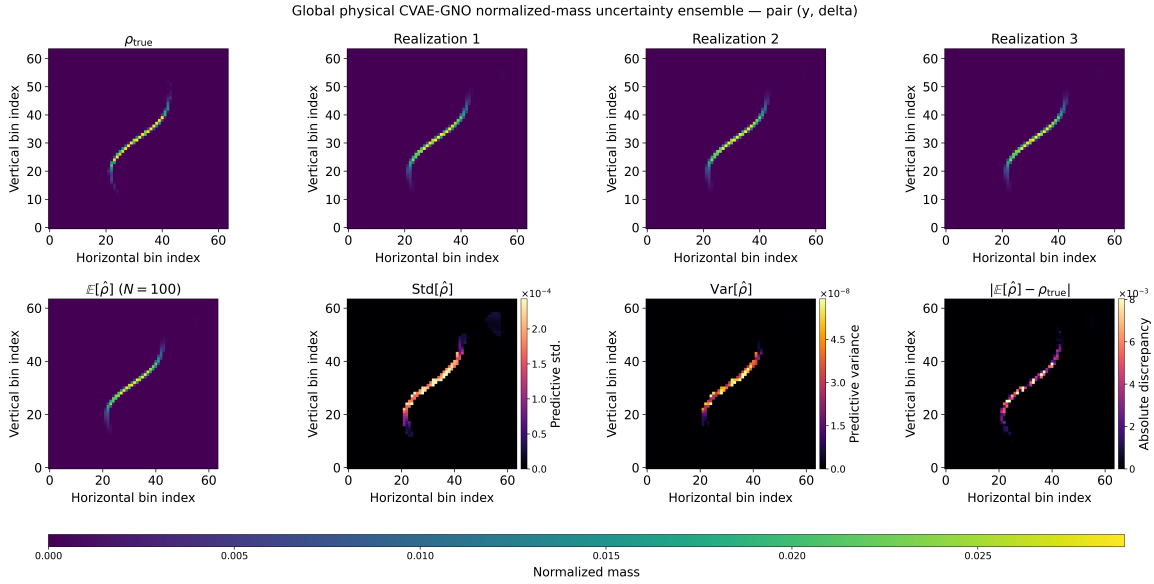


Fig. 18: Conditional latent uncertainty for the normalized (y, δ) histogram mass of validation sample $s = 3$, obtained from $R = 100$ stochastic CVAE samples propagated through the complete GNO surrogate. The panels compare the reference distribution, representative posterior draws, the ensemble mean, predictive standard deviation, predictive variance, and absolute ensemble-mean discrepancy. Because the histogram is normalized, this figure isolates variability of the decoded distributional shape before physical rescaling.

standard deviation is 3.749×10^{-6}, whereas the mean absolute discrepancy of the ensemble mean is 7.364×10^{-5}. The discrepancy is therefore approximately 19.6 times larger than the stochastic spread produced by latent sampling.

This separation indicates that the sampled predictions are tightly clustered, but that their common mean remains displaced from the reference distribution. For this example, the dominant prediction error is therefore not explained by variability under the sampled conditional latent distribution. Instead, it is consistent with systematic component arising from the learned surrogate, including possible reconstruction, propagation, or decoder bias. Consequently, the conditional latent ensemble is under-dispersed relative to the observed error and should not be interpreted as a calibrated predictive uncertainty interval.

After conversion to physical-density units, the mean predictive standard deviation is 4.003× 102, while the mean absolute ensemble-mean discrepancy is 2.222× 104, giving a ratio of approximately 55.5. The larger ratio does not represent an independent increase in model uncertainty. The conversion from normalized bin mass to density weights each bin by the inverse of its physical area and therefore amplifies discrepancies in narrow or high-density regions differently from the normalized-mass representation. The physical-density values should thus be interpreted as the same underlying mismatch expressed in physically scaled units.

Overall, this diagnostic shows that posterior latent sampling captures only a limited component of the total surrogate error for the selected sample. A calibrated uncertainty

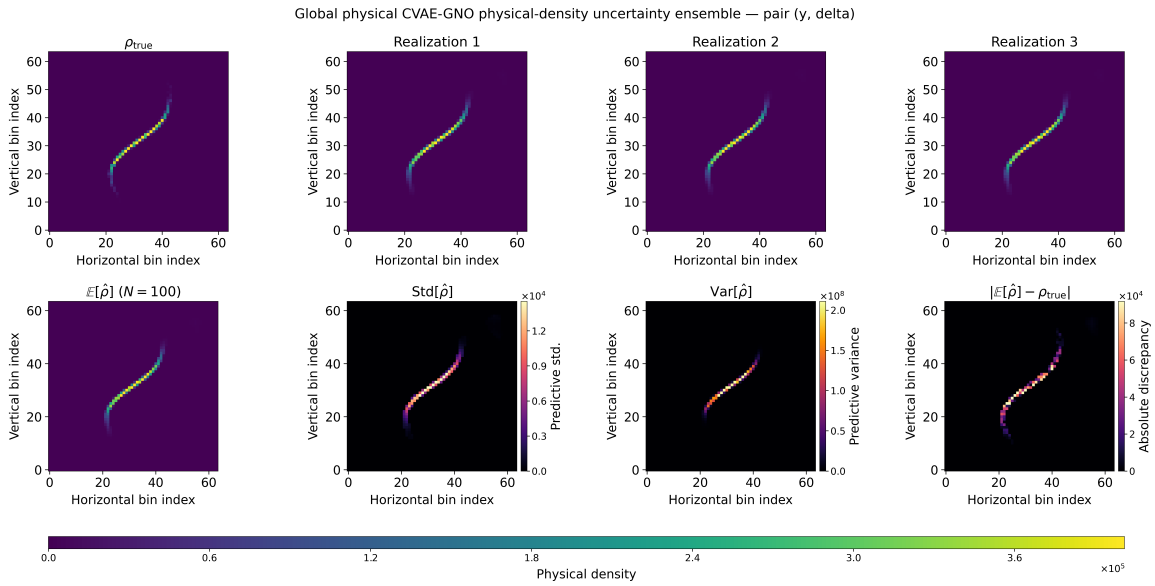


Fig. 19: Conditional latent uncertainty for the physical-density reconstruction of the same (y, δ) marginal and validation condition. Physical bin edges are generated from the predicted centroids and RMS spread extensions. The density ensemble therefore combines uncertainty in the decoded normalized shape with variability of the physical-statistics head. The comparison is displayed in bin-index coordinates and should not be interpreted as interpolation onto a single common physical grid.

estimate would also need to account for model-parameter uncertainty, finite-training-data effects, and systematic approximation error. Since the present result concerns one validation

sample and one phase-space projection, it should be treated as a local diagnostic rather than a general calibration statement.

Tab. 14: Per-bin conditional latent spread and ensemble-mean error for validation sample $s = 3$, the (y, δ) projection, and $R = 100$ stochastic forward passes. The standard deviation and variance quantify variability induced by conditional latent sampling, whereas the absolute discrepancy is computed between the ensemble mean and the reference histogram. Physical-density values are obtained by rescaling normalized bin masses using the corresponding physical bin areas.

Representation	Mean std.	Maximum std.	Mean variance	Maximum variance	Mean abs. discrepancy	Maximum abs. discrepancy
Normalized mass	3.748986×10^{-6}	3.138230×10^{-4}	6.816308×10^{-10}	9.848490×10^{-8}	7.364131×10^{-5}	1.159624×10^{-2}
Physical density	4.003429×10^{2}	3.244647×10^{4}	7.850748×10^{6}	1.052774×10^{9}	2.222052×10^{4}	3.394383×10^{6}

The posterior draws are therefore tightly concentrated relative to the remaining prediction error. The ensemble captures where the decoder and physical head are sensitive to latent perturbations, but its spread is not yet a calibrated estimate of the total surrogate error. In particular, the deterministic model bias is not removed by increasing the number of latent samples.

5.9.2 Uncertainty of the physical-statistics head

The physical-statistics ensemble is shown in Fig. 20. Since the six coordinates have different units and characteristic scales, the physical relevance of the prediction errors is assessed using dimensionless quantities rather than raw coordinate differences alone.

For each coordinate d, define the relative RMS-spread error and its posterior-sampling spread as

$$e_{\sigma,d} = \frac{\overline{\sigma}_d - \sigma_d^{\text{true}}}{\sigma_d^{\text{true}}}, \qquad u_{\sigma,d} = \frac{s_{\sigma_d}}{\sigma_d^{\text{true}}}. \tag{209}$$

The centroid error and its posterior-sampling spread are normalized by the reference RMS extent in the same coordinate,

$$b_{m,d} = \frac{\overline{m}_d - m_d^{\text{true}}}{\sigma_d^{\text{true}}}, \qquad u_{m,d} = \frac{s_{m_d}}{\sigma_d^{\text{true}}}. \tag{210}$$

Thus, $e_{\sigma,d}$ quantifies the fractional error in the predicted coordinate-wise RMS spread, while $b_{m,d}$ expresses the centroid displacement in units of the corresponding reference beam extent.

For this sample, the coordinate-wise RMS spreads in x, p_x, and y are overestimated by approximately $2.94\%, 2.68\%$, and 3.36%, respectively. The predicted p_y and ζ spreads agree with their reference values to better than 0.04%. The corresponding discrepancies are therefore small relative to the physical extent of the beam in these coordinates, even though they remain larger than the variability produced by conditional latent sampling.

The dominant physical error occurs in the energy-deviation coordinate. The predictedto-reference spread ratio is 0.118. The physical-statistics head therefore underestimates the energy spread by approximately 88.2%. The latent-sampling standard deviation is only approximately 0.026% of the reference energy spread, and is therefore far too small to account

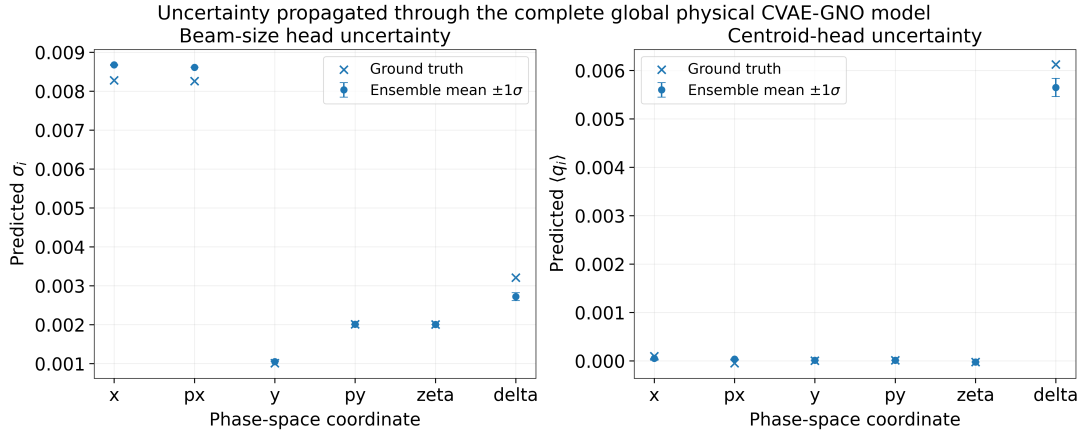


Fig. 20: Conditional latent variability of the six predicted coordinate-wise RMS spreads and six centroids for validation sample $s = 3$. Circles and error bars show the ensemble mean and one posterior-sampling standard deviation over $R = 100$ stochastic forward passes, crosses show the corresponding reference values. Raw coordinate units are shown in the figure, while the dimensionless physical errors are summarized in Table 12. The intervals quantify sensitivity to conditional latent sampling only and are not calibrated confidence intervals.

Tab. 15: Dimensionless physical-statistics diagnostic for validation sample $s = 3$, based on $R = 100$ stochastic forward passes. The RMS-spread bias and sampling spread are expressed as percentages of the reference RMS-spread. Centroid quantities are also expressed as percentages of the reference RMS extent in the corresponding coordinate.

Coordinate	$100e_{\sigma,d}$ [%]	$100u_{\sigma,d}$ [%]	$100b_{m,d}$ [% of RMS]	$100u_{m,d}$ [% of RMS]
x	2.939	0.262	-0.240	0.251
p_x	2.683	0.220	2.011	0.180
y	3.357	0.060	-3.399	2.499
p_y	0.005	0.027	-1.420	0.824
ζ	-0.033	0.033	0.432	0.981
δ	-88.195	0.026	-20.918	4.884

for this systematic discrepancy. For this sample, the predicted distribution would consequently appear substantially too narrow in the δ direction.

The centroid results should likewise be interpreted relative to the beam extent rather than solely in units of posterior-sampling standard deviations. The centroid biases in x, p_x, y, p_y, and ζ correspond to approximately $-0.24\%, +2.01\%, -3.40\%, -1.42\%$, and $+0.43\%$ of their respective reference RMS spreads. These offsets are small relative to the characteristic extent of the beam. In particular, although the p_x centroid error corresponds to approximately 11.2 posterior-sampling standard deviations, it represents only about 2.0% of the reference p_x spread. Counting posterior standard deviations alone would therefore exaggerate its physical importance because the latent ensemble is itself extremely narrow.

The energy-deviation centroid shows a more significant physical bias. The predicted centroid is displaced by approximately 21% of the reference energy spread. Together with the strong underestimation of σ_δ, this indicates that the physical-statistics head does not correctly reconstruct either the location or the extent of the distribution along the energy-deviation coordinate for this sample.

Overall, most transverse and longitudinal-position statistics are reconstructed to within a few percent of the relevant beam scale, whereas the energy-deviation statistics exhibit a clear systematic failure. In all coordinates, the posterior-sampling spread measures only sensitivity to the sampled conditional latent variable. Its small magnitude relative to several observed errors confirms that it does not constitute a calibrated estimate of the total physical prediction uncertainty.

5.9.3 Held-out-set per-bin diagnostic

A second experiment repeated the complete stochastic forward pass $R = 64$ times for every sample in the held-out loader. The loader contained 51 samples, processed in batches of 16, 16, 16, and 3. Figure 21 shows the physical-density variance and discrepancy fields for the representative held-out sample $s = 7$ and the (y, δ) projection.

The large numerical values in the physical-density panels arise from the conversion of dimensionless bin probability mass into density. For a uniform (y, δ) grid,

$$\rho_{ij} = \frac{p_{ij}}{\Delta y \, \Delta \delta}, \tag{211}$$

where p_{ij} is the probability mass in bin (i, j). Because the energy-deviation bin width $\Delta\delta$ is small, the inverse bin area can be large. The absolute values of the density, variance, and standard deviation therefore depend on the selected coordinate units and histogram resolution. Their magnitude should not by itself be interpreted as evidence of a large physical uncertainty.

A dimensionless comparison is obtained by relating the ensemble-mean discrepancy to the variability induced by latent sampling. For the selected sample,

$$\mathcal{R}_{\text{global}} = \frac{3.487909 \times 10^3}{4.686791 \times 10^2} \simeq 7.44.$$

Thus, the average absolute discrepancy between the ensemble mean and the reference

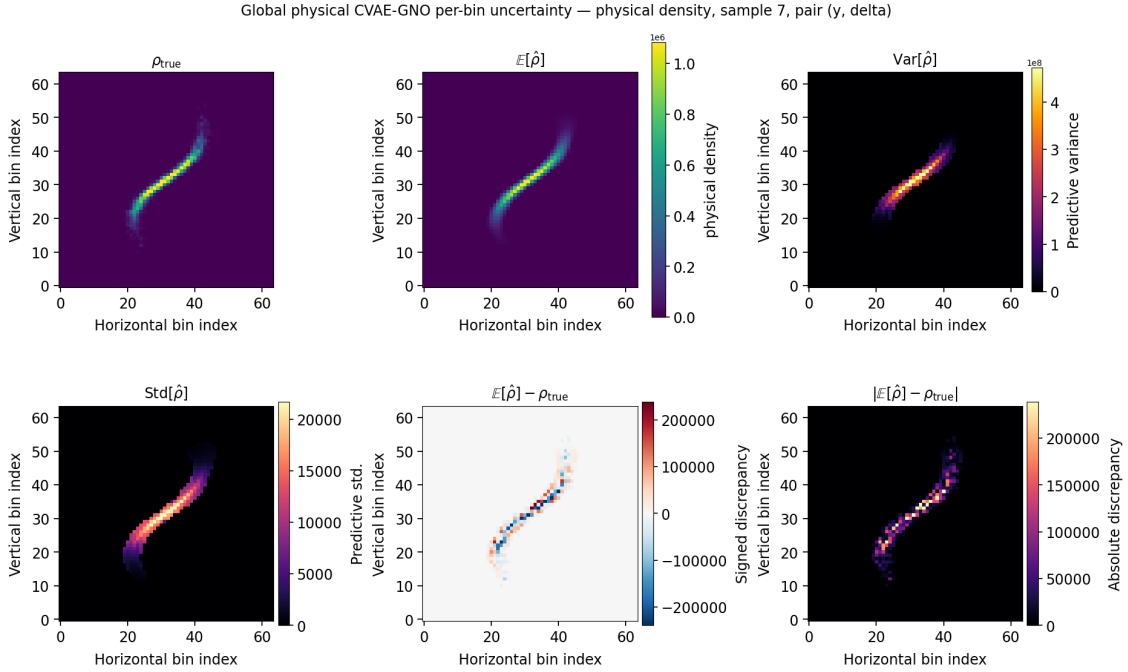


Fig. 21: Per-bin physical-density diagnostic for held-out sample $s = 7$, the (y, δ) projection, and $R = 64$ stochastic forward passes. The panels show the reference density, ensemble mean, predictive variance, predictive standard deviation, signed ensemble-mean discrepancy, and absolute discrepancy. The raw density scale depends on the physical coordinate units and bin area, the dimensionless discrepancy-to-spread ratios discussed in the text provide the relevant comparison between prediction error and conditional latent variability.

distribution is more than seven times the average variation generated by conditional latent sampling. The ensemble is therefore under-dispersed relative to the observed reconstruction error as most of the discrepancy cannot be explained by variation of the sampled CVAE latent variable alone.

The largest absolute variance occurs at bin (32, 32), close to the high-density core. The relative ensemble-mean bias at this bin is

$$\frac{\overline{\rho}_{32,32} - \rho^{\text{true}}_{32,32}}{\rho^{\text{true}}_{32,32}} \simeq 0.107,$$

corresponding to an overprediction of approximately 10.7%. The posterior-sampling standard deviation is only 0.0238 or approximately 2.38% of the reference density. Consequently, the ensemble mean is therefore approximately 4.5 latent-sampling standard deviations away from the reference value at the highest-variance bin. Even in the region where the ensemble exhibits its largest absolute variability, latent sampling does not explain the local reconstruction error.

The concentration of absolute variance near the histogram core should also be interpreted cautiously. Since the density itself is largest in this region, a fixed relative fluctuation naturally produces a larger absolute variance. The observed maximum does not by itself prove that the decoder is intrinsically more sensitive in the core. Establishing such enhanced sensitivity would require examination of a relative uncertainty field, for example

$$u^{\text{rel}}_{ij} = \frac{s_{\rho,ij}}{\rho^{\text{true}}_{ij} + \varepsilon}, \tag{212}$$

with a small regularizing constant ε to avoid unstable ratios in nearly empty bins.

From a beam-physics perspective, coherent residual structures in the (y, δ) projection can indicate errors in the predicted energy spread, energy centroid, or transverse–longitudinal correlation. Residual lobes aligned primarily along δ would be associated with an incorrect energy-deviation location or extension, whereas diagonally organized residuals would indicate an error in the learned y–δ dependence. The present diagnostic therefore provides more information than a single global density norm, because it identifies where and how the predicted phase-space structure differs from the reference.

Overall, the held-out example shows that conditional latent sampling captures only a limited component of the total prediction error. The ensemble variability is concentrated in the high-density region, but both the global discrepancy-to-spread ratio and the normalized core-bin analysis reveal substantial systematic error beyond that variability. The resulting intervals should therefore be interpreted as local sensitivity to CVAE latent sampling rather than as calibrated predictive uncertainty.

6 Discussion and perspectives

6.1 Exploratory research and architectural inductive bias

The modeling architecture was not fixed a priori. Instead, several surrogate formulations were investigated to determine which representations and inductive biases are appropriate for the evolution of high-dimensional beam distributions. The FNO, Transformer, windowed Transformer, and GNO should therefore be interpreted as different hypotheses about the structure of the learned evolution operator rather than as unrelated competing models.

The comparison indicates that the organization of information transfer is central to predictive performance. The dense causal Transformer permits direct interaction with the complete upstream sequence, whereas the windowed Transformer restricts this interaction to a finite causal neighbourhood. The GNO imposes a related but more explicit structure through graph-based message passing and a learned kernel. The improved performance of the windowed Transformer and GNO suggests that unrestricted global mixing is not required to represent the dominant latent dynamics and may introduce weakly constrained couplings during rollout.

This result is physically plausible because beam transport is generated through the composition of lattice transformations and structured collective interactions. Wakefield effects may be nonlocal, but their dependence is directional and constrained rather than arbitrary. A restricted interaction support can therefore provide a useful inductive bias without implying that the complete dynamics are strictly local. Repeated interactions across network depth can still transmit information over longer distances.

The lowest observed validation value at an intermediate attention window is consistent with this interpretation, although it is not statistically resolved across training seeds. A very narrow window may omit relevant multi-element dependencies, while an excessively broad one gradually recovers unrestricted mixing. The results therefore identify the support and geometry of latent information transfer as important modeling choices. They do not establish the universal superiority of one architecture, but show that architectures aligned with meaningful structure in the physical problem can provide more accurate and efficient surrogates.

6.2 Conditional latent uncertainty

Sampling from $q_\phi(z_0|X_0, c)$ can be propagated through the complete CVAE–GNO pipeline, producing spatially resolved variability maps and uncertainty estimates for the predicted physical statistics. The inferred variability is structured and concentrated mainly in regions containing significant probability mass, rather than appearing as spatially uniform noise.

The principal result, however, is that the ensemble is under-dispersed relative to the observed prediction error. In the fixed-sample experiment, the mean ensemble-mean discrepancy was approximately 19.6 times the mean latent-sampling standard deviation in normalized-mass units. The corresponding ratio in physical-density units was 55.5, although this value is affected by inverse-bin-area rescaling. In the independent held-out example, the discrep-

ancy remained approximately 7.44 times larger than the sampled spread. Latent sampling therefore explains only a limited fraction of the total surrogate error.

The physical-statistics head shows the same limitation. For one validation example, the predicted energy-deviation spread was only approximately 11.8% of its reference value, while its latent-sampling variability was negligible in comparison. Since the model weights remain fixed throughout the ensemble, the experiment captures sensitivity to the conditional latent state but excludes parameter uncertainty, finite-data effects, architecture uncertainty, model discrepancy, and out-of-distribution behaviour. The resulting maps may therefore be used as relative sensitivity indicators, but not as calibrated confidence intervals. More complete uncertainty quantification would require additional model variability and empirical coverage tests, for example through independently trained ensembles or post-hoc calibration.

6.3 Toward accelerator digital twins and optimization

Fast differentiable surrogates can support tasks beyond direct reproduction of simulation outputs, including sensitivity analysis, parameter scans, model calibration, and optimization. In this sense, the present work contributes to the broader development of accelerator digital twins: reduced models that complement high-fidelity simulation and experimental measurements.

A natural extension is to combine the surrogate with Bayesian optimization. The learned model could provide inexpensive evaluations of beam-quality objectives, while a higher-level optimizer explores machine parameters under operational constraints [5]. Such a workflow is particularly relevant when direct simulation or machine evaluation is costly. Its reliability would nevertheless depend on improved uncertainty calibration and explicit detection of extrapolative conditions.

6.4 Limitations

Several limitations define the scope of the conclusions. First, the models are validated only within the sampled parameter domain, and generalization outside this region cannot be assumed. Second, the fifteen pairwise projections do not uniquely determine the complete six-dimensional density: three- and higher-coordinate correlations are not identifiable from the chosen representation, and no direct numerical estimate of this representation error was performed. CVAE compression may additionally smooth fine filamentary or weak tail structures. Third, the six-dimensional data contain only upstream and final downstream snapshots. The propagators are supervised at the endpoint, not at intermediate lattice positions, so the internal latent rollout is not a validated physical trajectory and the general operator $\mathcal{G}_{\Delta s,\mu}$ has not been tested for arbitrary Δs. Fourth, the attention-window and architecture comparisons use selected single-seed checkpoints; without repeated training seeds, the observed ordering has no statistical uncertainty margin. Fifth, the present architectures do not exactly enforce symplecticity, charge conservation, Panofsky–Wenzel consistency, or the Haïssinski equilibrium relation.

The uncertainty ensemble varies only the conditional latent state and is explicitly under-

dispersed relative to the observed error. The mean discrepancy-to-sampling-spread ratios are approximately 19.6 for normalized mass, 55.5 after physical-density rescaling in the fixed sample, and 7.44 in the held-out per-bin example. These ensembles are therefore sensitivity diagnostics, not calibrated confidence intervals or predictive coverage. Further work should combine intermediate simulator checkpoints, repeated-seed or deep-ensemble studies, direct six-dimensional representation benchmarks such as point-cloud or normalizing-flow models, broader simulation domains, physics-constrained architectures, and empirical uncertainty calibration. These extensions are necessary before the surrogate can be considered for reliable optimization or operational decision making.

7 Conclusion

This internship investigated machine-learning surrogate models for collective-efefcts studies in the FCC-ee High-Energy Booster. Starting from the formulation of beam dynamics as distribution evolution under external and collective forces, the work developed a hierarchy of learned approximations: Fourier Neural Operators for reduced longitudinal profiles, a conditional variational representation for fifteen pairwise projections of six-dimensional beam clouds, Transformer-based latent dynamics, local-window attention, and Graph Neural Operator propagation.

The principal empirical result is that structured support is beneficial for the evaluated full-lattice endpoint task. Relative to dense causal attention, local-window attention reduces the observed reconstruction error, while the selected GNO checkpoint gives the lowest relative histogram error and neural inference time among the tested propagators. This conclusion concerns selected single-seed models and does not establish that the physical dynamics are strictly local. It also does not validate the internal element-indexed latent states: the available six-dimensional data supervise only the upstream and final downstream clouds, so arbitrarystep evolution and intermediate physical trajectories remain untested.

The standalone CVAE test error of 0.367 shows that the fifteen-marginal latent bottleneck already introduces non-negligible reconstruction loss before propagation. The full CVAE–GNO error of 0.511 therefore combines representation and endpoint-propagation effects. Moreover, pairwise projections do not identify three- and higher-coordinate correlations of a general non-Gaussian six-dimensional distribution. The present results support this representation as a practical reduced description, not as a unique reconstruction of the complete joint density.

The conditional-latent uncertainty study provides a further qualification. Posterior sampling produces spatially structured variability, but the ensemble spread is substantially smaller than the actual discrepancy: representative discrepancy-to-spread ratios are approximately $19.6, 55.5$, and 7.44 under the reported diagnostics. These intervals are under-dispersed sensitivity indicators and must not be interpreted as calibrated predictive coverage.

The FNO results connect longitudinal density evolution with the wake-convolution and Haïssinski formulations, while the six-dimensional study identifies local attention and graph-kernel message passing as promising architectural biases. Future work should add simulator

checkpoints along the lattice, repeat model comparisons over independent seeds, benchmark the pairwise representation against joint point-cloud or normalizing-flow models, impose relevant physical constraints explicitly, and calibrate predictive uncertainty. With these extensions, structured latent operators could support faster parameter scans, design-space exploration, and accelerator digital-twin workflows. Preliminary results from this research were published in the JACoW proceedings of IPAC [1].

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A Auxiliary theoretical results

A.1 Machine-learning preliminaries

This appendix collects standard machine-learning definitions used in the methodology. They are included for completeness but are not specific to the proposed accelerator-surrogate architectures.

A.1.1 Loss minimization and gradient-based training

For regression problems, a common objective is the mean-squared error,

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N}\sum_{s=1}^{N} \left\| \widehat{y}^{(s)} - y^{(s)} \right\|_2^2 . \tag{213}$$

The parameters are updated iteratively using an estimate of the loss gradient,

$$\theta_{k+1} = \theta_k - \eta_k \widehat{\nabla_\theta \mathcal{L}}, \tag{214}$$

where η_k denotes the learning rate and the gradient is generally estimated from a mini-batch of training examples.

A.1.2 Dense neural networks

Let $h^{(0)} = x$ denote the input vector. A dense layer computes

$$h^{(\ell+1)} = \sigma\left(W^{(\ell)} h^{(\ell)} + b^{(\ell)}\right), \tag{215}$$

where $W^{(\ell)}$ and $b^{(\ell)}$ are trainable parameters and σ is a nonlinear activation. Examples used in modern neural networks include

$$\mathrm{ReLU}(x) = \max(0, x), \tag{216}$$

and

$$\mathrm{GELU}(x) = \frac{x}{2}\left[1 + \mathrm{erf}\left(\frac{x}{\sqrt{2}}\right)\right]. \tag{217}$$

A multilayer perceptron is obtained by composing several dense transformations. The nonlinear activations are essential because a composition of affine maps without nonlinearities reduces to a single affine transformation.

A.1.3 Convolutional neural networks

For an input feature tensor $h^{(\ell)}$, a two-dimensional convolutional layer computes

$$h^{(\ell+1)}_{c_{\mathrm{out}}}(i,j) = \sigma\left[b_{c_{\mathrm{out}}} + \sum_{c_{\mathrm{in}}}\sum_{r}\sum_{q} K_{c_{\mathrm{out}},c_{\mathrm{in}}}(r,q)\, h^{(\ell)}_{c_{\mathrm{in}}}(i-r, j-q)\right], \tag{218}$$

where $K_{c_{\text{out}},c_{\text{in}}}$ is a learned kernel. Parameter sharing allows the same local feature to be detected at different spatial positions. Successive layers increase the effective receptive field, while downsampling and upsampling operations permit encoder–decoder constructions.

A.1.4 Autoencoders

An autoencoder consists of an encoder

$$\mathcal{E}_\phi : X \longmapsto z \tag{219}$$

and a decoder

$$\mathcal{D}_\psi : z \longmapsto \widehat{X}. \tag{220}$$

The reconstruction is

$$\widehat{X} = \mathcal{D}_\psi(\mathcal{E}_\phi(X)), \tag{221}$$

and a standard deterministic autoencoder can be trained by minimizing

$$\mathcal{L}_{\text{AE}} = \|\widehat{X} - X\|_2^2. \tag{222}$$

A.1.5 Variational and conditional autoencoders

A variational autoencoder represents each input through an approximate latent posterior. For a diagonal Gaussian posterior,

$$q_\phi(z \mid X) = \mathcal{N}\left(\mu_\phi(X), \operatorname{diag}\sigma_\phi^2(X)\right). \tag{223}$$

Sampling is implemented using the reparameterization

$$z = \mu_\phi(X) + \sigma_\phi(X) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \tag{224}$$

The VAE objective combines reconstruction and latent-prior regularization,

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{rec}} + \beta D_{\text{KL}}(q_\phi(z \mid X)\|p(z)), \tag{225}$$

where the prior is commonly

$$p(z) = \mathcal{N}(0, I). \tag{226}$$

For a diagonal Gaussian posterior, the non-negative divergence from the unit Gaussian is

$$D_{\text{KL}}(q_\phi(z \mid X)\|p(z)) = \frac{1}{2}\sum_{j=1}^{d_z}\left[\mu_{\phi,j}^2(X) + \sigma_{\phi,j}^2(X) - 1 - \log\sigma_{\phi,j}^2(X)\right]. \tag{227}$$

A conditional variational autoencoder extends the posterior and decoder by introducing

$$q_\phi(z \mid X, c), \qquad p_\psi(X \mid z, c). \tag{228}$$

Conditioning allows known physical or machine variables to contribute explicitly to the latent representation and reconstruction.

A.2 Key theoretical guarantees

The neural-operator results used to motivate the methodology are approximation statements for maps between function spaces, not guarantees about a particular trained checkpoint. Under the hypotheses of Theorem 8 in [20], a fixed continuum neural operator has finite-dimensional realisations that converge under mesh refinement, uniformly on compact subsets of the admissible input space. Schematically, for fixed parameters θ,

$$\sup_{a\in K}\left\|\widehat{\mathcal{G}}_{L,\theta}(D_L, a|_{D_L}) - \mathcal{G}_\theta(a)\right\|_{\mathcal{U}} \longrightarrow 0 \quad (L\to\infty). \tag{229}$$

This is a statement about consistency of the discretised architecture. It does not establish robustness to changing the number of Monte Carlo macroparticles used to estimate an input density.

Likewise, Theorems 11–14 in [20] provide density results for neural operators under specified domain, function-space, continuity, activation, and measurability hypotheses. On a compact admissible set K, the uniform-approximation implication has the form

$$\forall\varepsilon>0, \quad \exists\mathcal{G}_\theta : \ \sup_{a\in K}\left\|\mathcal{G}_\theta(a) - \mathcal{G}^\dagger(a)\right\|_{\mathcal{U}} < \varepsilon. \tag{230}$$

Related FNO-specific approximation results are given in [30]. These statements are existential: they do not prove that the finite width, retained Fourier modes, optimiser, sample size, or training budget used in this report attain a prescribed error. Because the reported FNO experiment was evaluated on its training-grid resolution, this thesis makes no empirical claim of cross-resolution generalisation.

The architectures considered in this work are motivated not only by empirical performance but also by several approximation, consistency, and structure-preservation results. These guarantees apply at different levels and should therefore be distinguished carefully. Neural-operator theory provides approximation and discretization-consistency results for maps between function spaces; the variational formulation provides a lower-bound identity for the conditional latent model; masked attention imposes exact support constraints on information propagation; and H'enon-type maps preserve symplectic structure by construction. None of these results, however, constitutes a general guarantee of successful optimization, finite-sample generalization, or stable long-horizon autoregressive rollout.

A.2.1 Discretization invariance of neural operators

The first key property is that the continuum neural-operator architecture is not intrinsically tied to one fixed discretization. The relevant result is more precise than generic resolution invariance.

Theorem 1 (Discretization invariance; adapted from Theorem 8 of [20]). *Let $D\subset\mathbb{R}^d$ and $D'\subset\mathbb{R}^{d'}$ be domains, and let A and U be real-valued Banach function spaces on D and*

D', *respectively. Assume that*

$$\mathcal{A} \hookrightarrow C(D), \qquad \mathcal{U} \hookrightarrow C(D') \tag{231}$$

continuously, and let the activation functions be continuous. Then, for any fixed neural-operator depth and fixed parameter value θ, *the corresponding neural-operator class is discretization-invariant.*

More explicitly, if

$$D_1 \subset D_2 \subset \cdots \subset D \tag{232}$$

is a discrete refinement and $G_{L,\theta}b$ denotes a finite-dimensional realization of the continuum operator G_θ, then for every compact subset $K \subset A$,

$$\lim_{L\to\infty} \sup_{a\in K} \left\| \widehat{\mathcal{G}}_{L,\theta}(D_L, a|_{D_L}) - \mathcal{G}_\theta(a) \right\|_{\mathcal{U}} = 0. \tag{233}$$

The important point is that the parameter vector θ remains fixed while the discretization is refined. Thus, the statement is stronger than merely accepting arrays of different lengths: the discrete realizations converge to a common continuum operator. This distinction is essential because a finite-dimensional network may be resolution-flexible without possessing a consistent continuum limit.

For the present work, this result provides the theoretical motivation for evaluating operator surrogates under changes in density-grid resolution. It does not, by itself, imply invariance with respect to the number of Monte Carlo macroparticles used to estimate a beam distribution. Particle-number robustness additionally depends on the statistical consistency of the empirical histogram or density estimator used to construct the network input.

A.2.2 Uniform approximation of continuous operators

The second central result concerns expressivity in function space. Under the hypotheses established in [20], neural operators are dense in appropriate classes of continuous nonlinear operators.

Theorem 2 (Uniform approximation on compact sets; adapted from Theorem 11 of [20]). *Let* A *and* U *satisfy the function-space assumptions of [20]; in particular, the admissible spaces include appropriate* L^p, *Sobolev, and continuous-function spaces on Lipschitz domains. Let*

$$\mathcal{G}^\dagger : \mathcal{A} \to \mathcal{U} \tag{234}$$

be continuous, and let $K \subset A$ *be compact. For every* $\varepsilon > 0$, *there exists a neural operator* G_θ *such that*

$$\sup_{a\in K} \left\| \mathcal{G}_\theta(a) - \mathcal{G}^\dagger(a) \right\|_{\mathcal{U}} < \varepsilon. \tag{235}$$

This theorem is the function-space analogue of finite-dimensional universal approximation. The target is not a vector-valued function

$$f : \mathbb{R}^m \to \mathbb{R}^n, \tag{236}$$

but an operator

$$\mathcal{G}^\dagger : \mathcal{A} \to \mathcal{U} \tag{237}$$

between infinite-dimensional spaces. Consequently, it provides the appropriate theoretical setting for learning maps such as

$$\lambda_s \longmapsto \lambda_{s+\Delta s}, \tag{238}$$

or, more generally,

$$\rho_s \longmapsto \rho_{s+\Delta s}, \tag{239}$$

when these maps are represented as continuous operators on suitable subsets of the corresponding function spaces.

The theorem is existential. It states that a sufficiently expressive neural operator exists, but it does not specify that a particular finite architecture, fixed channel width, fixed number of Fourier modes, or fixed graph radius attains the desired tolerance.

A.2.3 Approximation in expected risk

Uniform approximation on compact sets is not the only relevant topology. Since model training minimizes an empirical average over samples, an L^2_μ approximation result is particularly pertinent.

Theorem 3 (L^2_μ density; adapted from Theorem 13 of [20]). *Let μ be a probability measure on the input function space A, let the output be an appropriate separable Hilbert–Sobolev space, and suppose that*

$$\mathcal{G}^\dagger \in L^2_\mu(\mathcal{A}; \mathcal{U}). \tag{240}$$

Under the activation and function-space assumptions of [20], for every $\varepsilon > 0$ there exists a neural operator G_θ such that

$$\left\| \mathcal{G}_\theta - \mathcal{G}^\dagger \right\|_{L^2_\mu(\mathcal{A};\mathcal{U})} < \varepsilon. \tag{241}$$

Equivalently,

$$\left(\int_{\mathcal{A}} \left\| \mathcal{G}_\theta(a) - \mathcal{G}^\dagger(a) \right\|^2_{\mathcal{U}} \, \mathrm{d}\mu(a) \right)^{1/2} < \varepsilon. \tag{242}$$

This result is conceptually aligned with the population-risk objective

$$\mathcal{R}(\theta) = \mathbb{E}_{a\sim\mu} \left[\left\| \mathcal{G}_\theta(a) - \mathcal{G}^\dagger(a) \right\|^2_{\mathcal{U}} \right], \tag{243}$$

which is approximated in practice by the empirical risk

$$\widehat{\mathcal{R}}_N(\theta) = \frac{1}{N}\sum_{i=1}^{N} \left\| u^{(i)} - \mathcal{G}_\theta(a^{(i)}) \right\|_{\mathcal{U}}^2 . \tag{244}$$

The distinction is important. Density in $L^2_\mu(A;U)$ establishes representational capability with respect to the population measure μ; it does not prove that finite-sample empirical-risk minimization recovers the corresponding optimal parameters.

A.2.4 Fourier neural operator universality

The longitudinal surrogate uses a Fourier neural operator. In addition to the general neural-operator approximation theory, FNOs possess architecture-specific universality results.

Theorem 4 (Universal approximation by FNOs, informal; [20]). *Let $G^\dagger$ be a continuous operator between suitable function spaces on a compact or Lipschitz domain, and let K be a compact subset of the input space. Then, under the hypotheses of [20], for every $\varepsilon > 0$ there exists a Fourier neural operator G_θ^{FNO} such that*

$$\sup_{a\in K} \left\| \mathcal{G}_\theta^{\mathrm{FNO}}(a) - \mathcal{G}^\dagger(a) \right\|_{\mathcal{U}} < \varepsilon. \tag{245}$$

The same work derives explicit approximation-rate results for particular PDE solution operators, including Darcy-type elliptic problems and incompressible Navier–Stokes equations. These problem-specific rates are significant because they show that universality need not always require an impractically large architecture. They should not, however, be transferred directly to the longitudinal collective-effects problem considered here: no corresponding complexity bound is established in this work for the beam-dynamics operator.

A.2.5 Conditional variational lower bound

The CVAE component has a probabilistic guarantee based on an exact decomposition of the conditional log evidence. Let x denote the beam representation, c the conditioning information, and z the latent variable. For an approximate posterior

$$q_\phi(z \mid x, c), \tag{246}$$

the conditional log likelihood satisfies

$$\begin{aligned} \log p_\theta(x \mid c) = {} & \mathcal{L}_{\mathrm{ELBO}}(x, c; \theta, \phi) \\ & + D_{\mathrm{KL}}(q_\phi(z \mid x, c) \| p_\theta(z \mid x, c)), \end{aligned} \tag{247}$$

where

$$\begin{aligned} \mathcal{L}_{\mathrm{ELBO}}(x, c; \theta, \phi) = {} & \mathbb{E}_{q_\phi(z|x,c)}[\log p_\theta(x \mid z, c)] \\ & - D_{\mathrm{KL}}(q_\phi(z \mid x, c) \| p_\theta(z \mid c)). \end{aligned} \tag{248}$$

Since the Kullback–Leibler divergence is non-negative,

$$\mathcal{L}_{\mathrm{ELBO}}(x, c; \theta, \phi) \le \log p_\theta(x \mid c). \tag{249}$$

[Conditional ELBO identity] For any approximate posterior $q_\phi(z|x,c)$, maximizing the conditional ELBO maximizes a lower bound on the conditional log evidence. For a given pair (x,c), the bound is tight if and only if

$$q_\phi(z \mid x,c) = p_\theta(z \mid x,c) \tag{250}$$

almost everywhere with respect to the latent-space base measure.

This identity follows from the standard variational argument [17, 18] and provides the probabilistic justification for the reconstruction–KL objective used by the CVAE. It does not guarantee disentanglement, identifiability of the latent coordinates, absence of posterior collapse, or optimality of the chosen finite-dimensional latent representation. These properties must therefore be assessed empirically. When the KL contribution is weighted by a coefficient $\beta \neq 1$, the resulting objective is a $\beta-$CVAE objective rather than the standard ELBO itself.

A.2.6 Exact causal and local-support guarantees

The causal and windowed attention masks introduce deterministic architectural constraints that hold independently of training.

Let

$$\mathcal{N}_w^-(i) = \{j : \max(1, i-w) \leq j \leq i\}. \tag{251}$$

For one masked attention layer,

$$h_i^{(\ell+1)} = F_i^{(\ell)}\left(h_i^{(\ell)}, \{h_j^{(\ell)}\}_{j\in\mathcal{N}_w^-(i)}\right). \tag{252}$$

Therefore, in the absence of an additional global bypass,

$$\frac{\partial h_i^{(\ell+1)}}{\partial h_j^{(\ell)}} = 0, \qquad j \notin \mathcal{N}_w^-(i). \tag{253}$$

This yields the following structural statement.

[Finite one-layer interaction support] A causal attention window of width w guarantees that a token cannot directly depend on latent states lying outside its directed upstream neighbourhood $\mathcal{N}_w^-(i)$ in a single attention layer.

After L stacked local layers, dependencies may propagate through successive graph hops. Consequently,

$$\mathcal{N}_w^{-,(L)}(i) \subseteq \{j : 0 \leq i-j \leq Lw\}, \tag{254}$$

up to boundary effects and assuming no other global communication path.

Thus, local attention does not imply a globally local end-to-end model. Instead, it guarantees local support per interaction layer while permitting longer-range dependence through composition.

A.2.7 Graph support and message-passing locality

An analogous exact support statement holds for the GNO discretization. A graph-kernel layer may be written as

$$h_i^{(\ell+1)} = \Psi_i^{(\ell)} \left(h_i^{(\ell)}, \sum_{j \in \mathcal{N}_{\mathrm{GNO}}(i)} \omega_{ij} K_\theta^{(\ell)}(\xi_i, \xi_j, h_i^{(\ell)}, h_j^{(\ell)}, \mu) h_j^{(\ell)} \right). \tag{255}$$

If

$$j \notin \mathcal{N}_{\mathrm{GNO}}(i), \tag{256}$$

then node j contributes no direct message to node i in that layer. Consequently, the graph topology gives an exact sparsity pattern for the interaction operator. As with the windowed Transformer, repeated GNO layers enlarge the effective receptive field through successive graph hops.

This establishes the precise sense in which the two architectures share a locality bias: the support restriction is exact, while the learned values of the interaction kernel remain data-dependent.

A.2.8 Combined discretization–approximation error decomposition

The neural-operator framework yields a direct decomposition of the continuum prediction error. Let $G^\dagger$ denote the target physical operator, G_θ the learned continuum neural operator, and $G_{L,\theta} b$ its realization on an $L-$ point discretization. Then, by the triangle inequality,

$$\begin{aligned} \left\| \widehat{\mathcal{G}}_{L,\theta}(D_L, a|_{D_L}) - \mathcal{G}^\dagger(a) \right\|_{\mathcal{U}} \leq & \left\| \widehat{\mathcal{G}}_{L,\theta}(D_L, a|_{D_L}) - \mathcal{G}_\theta(a) \right\|_{\mathcal{U}} \\ & + \left\| \mathcal{G}_\theta(a) - \mathcal{G}^\dagger(a) \right\|_{\mathcal{U}}. \end{aligned}$$

The two theoretical results act on different terms. Discretization invariance implies that, for fixed θ and under the assumptions of the theorem,

$$\sup_{a \in K} \left\| \widehat{\mathcal{G}}_{L,\theta}(D_L, a|_{D_L}) - \mathcal{G}_\theta(a) \right\|_{\mathcal{U}} \to 0 \tag{257}$$

as $L \to \infty$. Universal approximation implies that the second term can be made arbitrarily small over compact subsets by increasing the expressive capacity of the operator class.

A.3 Prospective wake reconstruction from Haïssinski profiles

The following inverse problem is a proposed extension and was not implemented or evaluated in the reported experiments. Under a chosen nondimensionalisation and sign convention, let $q = z/\sigma_{z0}$ and let $\lambda(q) > 0$ be a normalised Haïssinski profile satisfying

$$\frac{\mathrm{d}\lambda}{\mathrm{d}q} + [q - F(q;\lambda)]\lambda(q) = 0, \qquad F(q;\lambda) = I \int_{-\infty}^{\infty} W(q - q')\lambda(q')\,\mathrm{d}q'. \tag{258}$$

Then

$$F(q) = \frac{\mathrm{d}}{\mathrm{d}q} \log \lambda(q) + q. \tag{259}$$

Profiles measured or simulated at several currents lead to an inverse convolution problem [7, 26]. On a uniform grid, one possible regularised estimator is

$$\widehat{w} = \arg\min_{w} \left\{ \frac{1}{M} \sum_{n=1}^{M} \left\| \widehat{F}^{(n)}(w) - F^{(n)} \right\|_2^2 + \alpha \|Dw\|_2^2 + \beta \|w\|_2^2 \right\}. \tag{260}$$

Because differentiating $\log \lambda$ amplifies sampling and measurement noise, any future study must specify smoothing, support, causality, current calibration, and regularisation before interpreting wb as a reconstructed physical wake.

B Declaration of originality and absence of plagiarism

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